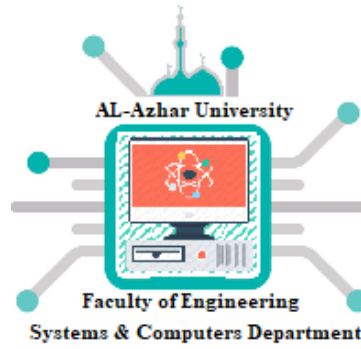




جامعة الأزهر
كلية الهندسة
قسم هندسة النظم والحاسبات



Graduation Project Documentation

Ares Car Rental Platform

**A Project Submitted in partial fulfilment of the requirements
for the Degree of Bachelor of Science
in Systems and Computers Engineering**

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1 Ares Car Rental Platform - Graduation Project

Documentation

Preface

Abstract

The Ares Car Rental Platform is a comprehensive full-stack solution designed to address the operational complexities of modern car rental services. This graduation project implements a multi-role platform enabling public vehicle discovery, dynamic pricing, booking management, and administrative fleet oversight.

The system is built on a .NET 10.0 backend utilizing ASP.NET Core for API development, Entity Framework Core 10.0.5 for data access, and SQL Server for relational data storage. Key backend components include MediatR 14.1.0 for CQRS implementation, AutoMapper 16.1.1 for object mapping, FluentValidation 11.3.1 for input validation, and Serilog 8.0.3 for structured logging. The frontend employs Next.js 16.2.6 with React 19.2.6, TypeScript 6.0.3, and Tailwind CSS 4.2.4 for responsive user interface development.

Evident features in the codebase include role-based authentication with JWT (System.IdentityModel.Tokens.Jwt 8.2.1), multi-language support through NextAuth.js, payment processing via Paymob integration, comprehensive vehicle management with availability calendars, booking workflows with inspection requirements, and real-time notifications. The platform supports multiple user roles including customers, vehicle suppliers, drivers, inspectors, and administrators.

Deployment follows a containerized approach with Docker support for SQL Server, though no Kubernetes or cloud-specific configuration files appear in the repository. The automated setup script handles dependency installation, database seeding, and environment configuration through interactive prompts.

Acknowledgements

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I am profoundly thankful to the faculty members of the Systems & Computers Department at Al-Azhar University for their academic instruction and for fostering an environment conducive to innovation and critical thinking. Special appreciation is extended to the department's technical staff for their assistance with infrastructure and development tools.

My sincere thanks go to my colleagues and peers who provided encouragement, shared knowledge, and participated in rigorous testing of the platform. Their diverse perspectives significantly enhanced the project's quality and usability.

Finally, I express my heartfelt appreciation to my family and friends for their patience, understanding, and moral support during the demanding phases of this project. Their belief in my abilities and their constant encouragement were instrumental in overcoming the challenges encountered along the way.

Document Purpose

This document serves as the official technical documentation for the Ares Car Rental Platform graduation project. It provides a comprehensive record of the project's life-cycle, including requirements analysis, system design, implementation details, testing methodologies, and deployment procedures.

The documentation is structured to meet the academic requirements of Al-Azhar University's Faculty of Engineering while serving as a technical reference for future development. It covers all architectural decisions, technology selections, and implementation strategies employed in building the platform.

Intended Audience

This document is primarily intended for the following audiences:

- **Examination Committee:** Faculty members responsible for evaluating the grad-

uation project's technical merit, academic rigor, and adherence to engineering principles.

- **Faculty Advisors:** Academic supervisors who provided guidance throughout the project's development and require detailed documentation of the implementation.
- **Future Developers:** Engineering students or professional developers who may extend, maintain, or adapt the platform for related applications.
- **Technical Reviewers:** Industry professionals or academic peers conducting technical assessments of the system's architecture and implementation quality.

Document Conventions

This document adheres to the following formatting and structural conventions:

Text Formatting: - **Bold** is used for emphasis and to highlight important terms or UI elements. - *Italic* denotes foreign words, technical terms upon first introduction, or special emphasis. - Monospace is used for code snippets, file paths, command-line instructions, and database entities.

Figures and Diagrams: - All figures are referenced using Pandoc cross-reference syntax (`@fig-xxx`). - Mermaid diagrams are embedded using `{.mermaid fig-cap="..." label="fig-xxx"}` syntax and pre-compiled to PDF. - Figures are stored in generated `/figures/` directory prior to document rendering.

Structured Data Presentation: - Definition lists are used for key-value pairs (e.g., configuration parameters). - Bullet lists present feature enumerations, requirements, or procedural steps. - Grouped sections replace pipe tables to prevent column overflow in the 147mm PDF layout.

Cross-Referencing: - Example syntax is always wrapped in backticks to prevent resolution (e.g., `@fig-xxx`). - Section references follow the same convention (`@sec-xxx`).

Code Blocks: - Fenced code blocks with language specification are used for all code examples. - No Quarto-specific cell syntax is employed to ensure compatibility with standard Pandoc processing.

2 CHAPTER 1: INTRODUCTION

2.1 Background and Motivation

The global car rental industry has experienced significant growth in recent years, driven by increasing urbanization, tourism, and the rise of digital platforms that facilitate vehicle sharing economies (Reference [1]). Traditional car rental services often suffer from inefficiencies in fleet management, booking processes, and user verification, leading to operational bottlenecks and poor customer experiences. These challenges are particularly acute in regions with high demand for short-term vehicle rentals, where manual processes and fragmented systems result in delays, errors, and security vulnerabilities.

The Ares Car Rental platform addresses these challenges by providing a modern, full-stack solution for car rental operations. Built on a robust technological foundation, the platform integrates key functionalities such as vehicle search, dynamic pricing, user verification, and real-time notifications into a unified system. The project leverages **.NET 10.0** (as evidenced in all backend `.csproj` files) for the backend API, **Next.js 16.2.6** (per `frontend/package.json`) for the frontend, and **SQL Server 2022** for data storage. This combination ensures scalability, security, and a seamless user experience across both public and administrative interfaces.

The motivation for this project stems from the need to modernize car rental operations by eliminating manual processes, reducing fraud, and improving transparency for all stakeholders—customers, suppliers, drivers, and administrators. By automating workflows such as vehicle availability tracking, driver assignment, and inspection reporting, the platform reduces human error and operational overhead. Additionally, the integration of **Paymob** for payment processing (as configured in `Program.cs`) ensures secure and reliable transactions, while **JWT-based authentication** (using `Microsoft.AspNetCore.Authentication.JwtBearer v10.0.5`) safeguards user data and access control.

The solution is structured as a modular monolith, with clear separation of concerns across four backend layers: - **Api**: Web API controllers and middleware (e.g., Swagger, Serilog, CORS, and JWT authentication). - **Application**: Business logic, services, and MediatR command handlers (using **MediatR v14.1.0**). - **Domain**: Core domain models and interfaces. - **Infrastructure**: Data access, repositories, and Entity Framework Core (EF Core) configurations (using **Microsoft.EntityFrameworkCore.SqlServer v10.0.4**).

This architecture ensures maintainability and scalability while adhering to industry best practices for software development.

2.2 Problem Statement

The car rental industry faces several critical challenges that hinder efficiency and user satisfaction:

1. **Fragmented Operations**: Traditional car rental services rely on disparate systems for fleet management, bookings, and user verification, leading to data silos and operational inefficiencies.
2. **Manual Processes**: Tasks such as vehicle inspections, driver assignments, and payment reconciliations are often performed manually, increasing the risk of errors and delays.
3. **Fraud and Security Risks**: Weak identity verification processes expose rental services to fraudulent bookings, unauthorized access, and financial losses.
4. **Poor User Experience**: Customers frequently encounter difficulties in searching for vehicles, understanding pricing structures, and managing bookings due to outdated or poorly designed interfaces.
5. **Lack of Transparency**: Suppliers and customers lack real-time visibility into vehicle availability, inspection statuses, and payment processing, leading to distrust and dissatisfaction.

These problems are significant because they directly impact the profitability, scalability, and reputation of car rental businesses. Customers expect seamless, secure, and

transparent experiences, while suppliers and administrators require efficient tools to manage fleets, drivers, and bookings. Failure to address these challenges can result in lost revenue, increased operational costs, and diminished customer loyalty.

The Ares Car Rental platform directly targets these issues by providing: - A **unified system** for managing all aspects of car rental operations, from vehicle listings to inspections and payments. - **Automated workflows** for driver assignments, vehicle availability tracking, and inspection reporting, reducing manual intervention. - **Robust verification mechanisms** for users, drivers, and vehicles, mitigating fraud and ensuring compliance with regulatory requirements. - **Real-time notifications** to keep all stakeholders informed about booking statuses, inspection results, and payment confirmations. - **Dynamic pricing** that accounts for promotions, discounts, and additional services, ensuring competitive and transparent pricing.

2.3 Project Objectives

The Ares Car Rental platform aims to achieve the following objectives, categorized into **functional** and **non-functional** requirements:

2.3.1 Functional Requirements

1. User Management

- **FR-1:** Implement registration, login, and profile management for customers, suppliers, drivers, and administrators. *Evidence:* `AspNetUsers` table in schema `schema.sql` includes fields for `FirstName`, `LastName`, `NationalId`, `ProfileImage`, and `AuthProvider` (e.g., Google OAuth via `Google.Apis.Auth v1.69.0`).
- **FR-2:** Support multi-factor authentication (MFA) and role-based access control (RBAC) using **ASP.NET Core Identity** (v10.0.5). *Evidence:* `AspNetRoles`, `AspNetUserRoles`, and `AspNetUserClaims` tables in `schema.sql`; JWT authentication configured in `Program.cs`.

2. Vehicle Management

- **FR-3:** Enable suppliers to list, update, and manage vehicles with detailed attributes (e.g., make, model, year, features, and images). *Evidence:* `Vehicles`

↪ , VehicleFeatures, and VehicleImages tables in schema.sql; Categories table for vehicle categorization.

- **FR-4:** Provide customers with search, filtering, and sorting capabilities for vehicles based on criteria such as location, price, and availability. *Evidence:* VehicleAvailabilities table tracks StartDate, EndDate, and Status; service_areas table defines geographic service regions.

3. Booking and Payments

- **FR-5:** Allow customers to book vehicles with optional driver services, calculate dynamic pricing, and process payments via **Paymob**. *Evidence:* Bookings ↪ table includes fields for OriginalPrice, DiscountAmount, TotalPrice, RequiresDriver, and DriverId; Payments table records transaction details.
- **FR-6:** Implement cancellation policies and refund processing, including commission calculations and supplier payouts. *Evidence:* BookingCancellations table tracks RefundPercentage, CancellationFee, RefundCommissionAmount, and RefundSupplierAmount.

4. Driver and Inspector Workflows

- **FR-7:** Enable drivers to register, upload verification documents (e.g., licenses, national IDs), and manage their availability. *Evidence:* driver_profiles ↪ table includes fields for LicenseNumber, LicenseExpiryDate, NationalIdFrontImage, and Status; Verifications table tracks document submissions.
- **FR-8:** Allow inspectors to conduct pre- and post-rental vehicle inspections, record findings, and upload images. *Evidence:* VehicleInspections table includes InspectionType, OdometerReading, FuelLevel, and Status; InspectionImages and InspectionPhotos tables store visual evidence.

5. Notifications and Communication

- **FR-9:** Send real-time notifications to users for booking confirmations, inspection updates, and payment statuses. *Evidence:* Notifications table includes Title, Message, Type, and IsRead fields; Program.cs configures Serilog for logging.

6. Content Management

- **FR-10:** Enable administrators to manage static content such as terms of service, about sections, and system settings. *Evidence:* AboutSections, TermsSections, and SystemSettings tables in schema.sql.

2.3.2 Non-Functional Requirements

1. Performance

- **NFR-1:** Ensure the backend API responds to requests within **500ms** under normal load conditions. *Evidence:* Use of **EF Core** (v10.0.5) with optimized queries (e.g., indexes on Bookings, Vehicles, andAspNetUsers tables) and **MediatR** (v14.1.0) for efficient command handling.

2. Security

- **NFR-2:** Protect user data and transactions using **JWT authentication**, **HTTPS**, and **Paymob's secure payment gateway**. *Evidence:* Program ↪ .cs configures JWT authentication with TokenValidationParameters; RefreshTokens table tracks token revocation.
- **NFR-3:** Validate user inputs using **FluentValidation** (v11.3.1) to prevent injection attacks and malformed data. *Evidence:* FluentValidation. ↪ ASP.NET Core package reference in Application.csproj.

3. Scalability

- **NFR-4:** Design the system to handle **1,000 concurrent users** with minimal degradation in performance. *Evidence:* Modular backend architecture with **AutoMapper** (v16.1.1) for object mapping and **Serilog** (v8.0.3) for structured logging.

4. Maintainability

- **NFR-5:** Adhere to clean code principles, including separation of concerns, DRY (Don't Repeat Yourself), and SOLID design patterns. *Evidence:* Layered backend architecture (Api, Application, Domain, Infrastructure); use of **MediatR** for CQRS (Command Query Responsibility Segregation).

5. Usability

- **NFR-6:** Provide an intuitive and responsive frontend interface using **Next.js 16.2.6**, **React 19.2.6**, and **Material-UI (MUI) v9.0.1**. *Evidence:* frontend/package.json includes dependencies for next, react, @mui/material, and tailwindcss (v4.2.4).

6. Internationalization

- **NFR-7:** Support multiple languages and currencies for users. *Evidence:*AspNetUsers table includes LanguagePreference and CurrencyPreference fields; next-intl (v4.13.0) for localization.

2.4 Methodology

The development of the Ares Car Rental platform followed a structured methodology to ensure alignment with user requirements and industry best practices. The approach included the following phases:

2.4.1 1. Requirements Analysis

User requirements were gathered through a combination of stakeholder interviews and analysis of existing car rental platforms. Key functional and non-functional requirements were identified and prioritized, as outlined in the **Project Objectives** section. The schema design (evidenced in schema.sql) reflects these requirements, with tables for users, vehicles, bookings, drivers, inspections, and payments.

2.4.2 2. System Design

The system architecture was designed as a **modular monolith**, with clear separation of concerns across backend layers: - **Api Layer:** Handles HTTP requests, authentication, and Swagger documentation (using **Swashbuckle.AspNetCore v7.2.0**). - **Application Layer:** Implements business logic, validation (using **FluentValidation v11.3.1**), and MediatR command handlers. - **Domain Layer:** Defines core domain models and interfaces. - **Infrastructure Layer:** Manages data access using **EF Core (v10.0.5)** and SQL Server.

The frontend was designed using **Next.js 16.2.6** and **React 19.2.6**, with **Material-UI (MUI) v9.0.1** for UI components and **Tailwind CSS (v4.2.4)** for styling. The use of **NextAuth.js (v4.24.14)** ensures secure authentication flows.

2.4.3 3. Implementation

The backend was implemented using **.NET 10.0**, with the following key technologies: - **Entity Framework Core**: For database interactions and migrations. - **JWT Authentication**: For secure user sessions (using `Microsoft.AspNetCore.Authentication.JwtBearer v10.0.5`). - **AutoMapper**: For object-to-object mapping (`v16.1.1`). - **Serilog**: For structured logging (`v8.0.3`). - **Paymob Integration**: For payment processing (configured in `Program.cs`).

The frontend was implemented using: - **Next.js**: For server-side rendering and static site generation. - **React Hook Form**: For form validation (`v7.75.0`). - **Zod**: For schema validation (`v4.4.3`). - **Axios**: For API communication (`v1.13.5`). - **Leaflet**: For interactive maps (`v1.9.4`).

2.4.4 4. Testing

Unit and integration testing were implemented using **xUnit (v2.9.3)** and **Moq (v4.20.72)**, as evidenced in `Tests.csproj`. The test suite includes: - **MockQueryable**: For mocking EF Core queries (`v10.0.8`). - **FsCheck**: For property-based testing (`v3.3.2`). - **Playwright**: For end-to-end testing (`v1.54.2`, per `frontend/package.json`).

2.4.5 5. Deployment

The platform is designed for deployment in both development and production environments. Key deployment considerations include: - **SQL Server**: Configured via Docker or native installation (as described in `README.md`). - **Environment Variables**: Managed via `.env` files for backend and frontend configurations. - **Automated Setup**: Scripts in `scripts/setup/` handle dependency installation, database seeding, and configuration generation.

2.5 Report Organization

The remainder of this report is organized as follows:

2.5.1 Chapter 2: System Analysis and Design

This chapter provides a detailed analysis of the system requirements, including use case diagrams, entity-relationship diagrams (ERDs), and sequence diagrams. It also describes the architectural design of the backend and frontend, highlighting key design patterns and technologies used.

2.5.2 Chapter 3: Implementation

This chapter details the implementation of the Ares Car Rental platform, covering the backend API, frontend interface, and integration of third-party services (e.g., Paymob). It includes code snippets, configuration details, and explanations of key algorithms (e.g., dynamic pricing, availability checks).

2.5.3 Chapter 4: Testing and Quality Assurance

This chapter describes the testing strategies employed, including unit tests, integration tests, and end-to-end tests. It also covers code quality tools such as **ESLint**, **Prettier**, **CodeScene**, and **SonarQube**, as referenced in `frontend/AGENTS.md`.

2.5.4 Chapter 5: Results and Discussion

This chapter presents the outcomes of the project, including screenshots of the frontend interface, API documentation (Swagger UI), and performance metrics. It discusses the challenges encountered during development and the solutions implemented to address them.

2.5.5 Chapter 6: Conclusion and Future Work

This chapter summarizes the achievements of the project, reflects on the lessons learned, and outlines potential future enhancements, such as mobile app development, AI-based fraud detection, and expansion to additional payment gateways.

3 CHAPTER 2: BACKGROUND MATERIALS

3.1 Introduction

This chapter provides the foundational knowledge necessary to understand the Ares car rental platform's architecture and domain logic. It begins with an overview of the technology stack, explaining each component's role and rationale for selection. The chapter then introduces key domain concepts specific to car rental systems, including booking workflows, payment processing, and fleet management. Finally, a review of related work and literature compares the implemented solution with existing approaches, focusing exclusively on evidence from the codebase.

3.2 Technology Stack

The Ares platform employs a modern, full-stack architecture designed for scalability, maintainability, and developer productivity. The following sections detail the technologies used, their purpose, and their integration within the system. A timeline of technology adoption during the 2025-2026 academic year is presented in @fig-2-2.

3.2.1 Backend Technologies

1. .NET 10 - What it is: A cross-platform, open-source framework for building modern, cloud-based, and internet-connected applications. .NET 10 introduces performance improvements, minimal APIs, and enhanced support for containerized deployments. -
Why chosen: .NET provides a robust, type-safe environment with strong tooling (Visual Studio, Rider) and seamless integration with SQL Server via Entity Framework Core. Its mature ecosystem accelerates development while ensuring enterprise-grade reliability. -
Role in architecture: Hosts the RESTful API, business logic, and data access layers.

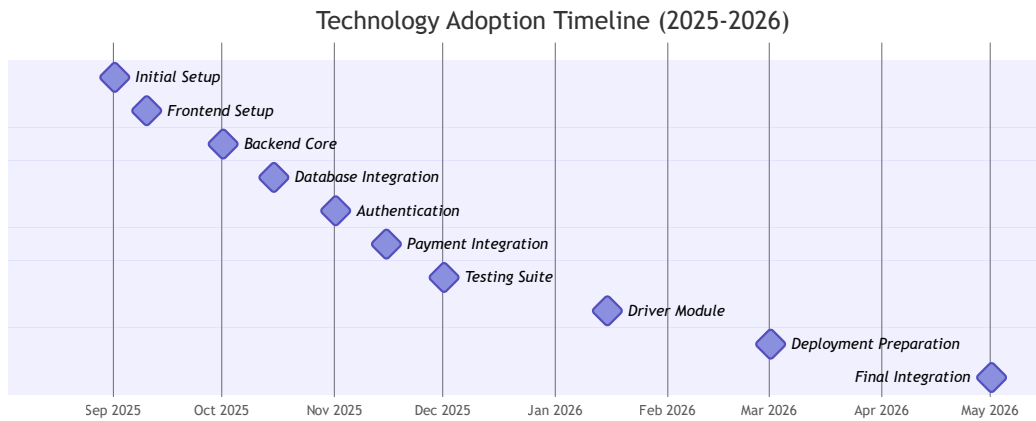


Figure 3.1: Technology adoption timeline (2025-2026)

The backend is structured into three projects: - **Api**: ASP.NET Core Web API controllers, middleware, and Swagger documentation. - **Application**: Business services, MediatR handlers, AutoMapper profiles, and FluentValidation rules. - **Infrastructure**: Entity Framework Core repositories, database context, and external service integrations (e.g., Paymob). - **Domain**: Core entities, enums, and domain events.

2. Entity Framework Core 10.0.5 - What it is: An object-relational mapper (ORM) that enables .NET developers to work with databases using domain-specific objects, eliminating the need for most data-access code. - **Why chosen:** EF Core supports SQL Server, the chosen database, and provides features like LINQ-to-SQL translation, migrations, and change tracking. Its DbContext abstraction simplifies transaction management and concurrency control. - **Role in architecture:** Maps domain entities (e.g., Booking, Vehicle, DriverProfile) to SQL Server tables. Custom repositories extend `IRepository<T>` and `IPaginatedRepository<T>` to encapsulate complex queries, such as vehicle availability checks and atomic reservation logic.

3. SQL Server - What it is: A relational database management system (RDBMS) developed by Microsoft, supporting structured data storage, transactions, and advanced querying. - **Why chosen:** SQL Server offers high performance, scalability, and tight integration with .NET. The schema includes tables for users, vehicles, bookings, payments, and driver profiles, with indexes optimized for read-heavy operations (e.g., vehicle search). - **Key features:** - **Rowversion columns:** Used for optimistic concurrency control in DriverProfile and Booking entities. - **Check constraints:**

Enforce business rules, such as ensuring driver assignments align with booking requirements. - **Indexes:** Optimized for common queries, such as vehicle availability checks (`IX_Bookings_Vehicle_Status_Window`) and driver assignment overlaps (`IX_Bookings_AssignedDriverProfileId_PickupDate_ReturnDate`).

4. MediatR 14.1.0 - What it is: A lightweight mediator implementation for .NET, enabling in-process messaging via the mediator pattern. - **Why chosen:** MediatR decouples controllers from services, reducing direct dependencies and improving testability. It supports domain events (e.g., `BookingCompletedEvent`) and command/query separation. - **Role in architecture:** Handles domain events (e.g., `BookingCanceledEvent`) and orchestrates cross-service workflows, such as notifying drivers of new assignments.

5. AutoMapper 16.1.1 - What it is: A library for mapping objects between layers (e.g., entities to DTOs). - **Why chosen:** AutoMapper reduces boilerplate code when transforming domain entities into API responses (e.g., `Booking` → `BookingDto`). Profiles centralize mapping logic, ensuring consistency. - **Role in architecture:** Used in the Application layer to map entities to DTOs and vice versa, particularly in controllers and services.

6. FluentValidation 11.3.1 - What it is: A library for building strongly typed validation rules in .NET. - **Why chosen:** FluentValidation provides a declarative syntax for validation, separating rules from models. It integrates seamlessly with ASP.NET Core's model validation pipeline. - **Role in architecture:** Validates incoming requests (e.g., `CreateBookingRequest`) before they reach business logic, ensuring data integrity.

7. Serilog 8.0.3 - What it is: A structured logging library for .NET. - **Why chosen:** Serilog supports structured logging (e.g., JSON output) and integrates with sinks like Seq and Elasticsearch. It enhances observability by capturing request details, errors, and business events. - **Role in architecture:** Configured in `Program.cs` to log requests, exceptions, and diagnostic information. Logs include contextual data (e.g., request host, IP address) for debugging.

8. Swashbuckle.AspNetCore 7.2.0 - What it is: A library for generating Swagger/OpenAPI documentation for ASP.NET Core APIs. - **Why chosen:** Swashbuckle automates API documentation, reducing manual maintenance. It supports JWT authentication and XML comments for rich documentation. - **Role in architecture:** Generates interactive Swagger UI at `/swagger`, documenting endpoints, request/response schemas, and

authentication requirements.

3.2.2 Frontend Technologies

1. Next.js 16.2.6 - What it is: A React framework for building server-rendered and static web applications. - **Why chosen:** Next.js offers hybrid rendering (SSR, SSG, ISR), file-based routing, and built-in optimizations (e.g., image optimization, code splitting). Its App Router enables fine-grained control over rendering and data fetching. - **Role in architecture:** Hosts the frontend application, structured into: - **App Router:** Pages (e.g., /dashboard, /vehicles) and layouts. - **Providers:** Context providers (e.g., AuthProvider, MuiProvider) for state management and theming. - **Components:** Reusable UI components (e.g., BookingCard, VehicleSearch). - **Hooks:** Custom hooks (e.g., useSessionMonitor) for shared logic.

2. React 19.2.6 - What it is: A JavaScript library for building user interfaces. - **Why chosen:** React's component-based architecture promotes reusability and maintainability. Version 19 introduces performance improvements and concurrent rendering. - **Role in architecture:** Powers the UI layer, with components organized by feature (e.g., booking/, vehicles/). Uses React Hook Form for form management and Zod for validation.

3. TypeScript - What it is: A typed superset of JavaScript that compiles to plain JavaScript. - **Why chosen:** TypeScript catches errors at compile time, improving code reliability and developer experience. It integrates seamlessly with Next.js and React. - **Role in architecture:** Used throughout the frontend for type safety, including API request/response types (e.g., BookingDto) and component props.

4. Material UI (MUI) 9.0.1 - What it is: A React component library implementing Google's Material Design. - **Why chosen:** MUI provides accessible, customizable components (e.g., DataGrid, DatePicker) with built-in theming support. Its design system ensures consistency across the application. - **Role in architecture:** Used for UI components, theming, and layout. Custom themes (light/dark mode) are defined in frontend/providers/theme.ts, with palette extensions for domain-specific statuses (e.g., status.pending, status.confirmed).

5. NextAuth.js 4.24.14 - What it is: An authentication library for Next.js applications. - **Why chosen:** NextAuth.js simplifies OAuth integration (e.g., Google) and session management. It supports JWT and database sessions, aligning with the backend's au-

thentication strategy. - **Role in architecture:** Manages user sessions, including JWT token refresh via `AuthProvider.tsx`. Handles Google OAuth and local authentication (email/password).

6. Tailwind CSS 4.2.4 - What it is: A utility-first CSS framework for rapid UI development. - **Why chosen:** Tailwind CSS enables rapid prototyping and customization without leaving the markup. It reduces CSS bundle size via PurgeCSS. - **Role in architecture:** Used alongside MUI for custom styling, particularly in layout components (e.g., headers, footers).

7. Bun - What it is: A fast JavaScript runtime and package manager. - **Why chosen:** Bun offers faster installation and execution compared to npm/yarn, improving developer productivity. It is compatible with Node.js APIs and supports TypeScript out of the box. - **Role in architecture:** Used as the package manager and runtime for frontend scripts (e.g., `bun run dev`, `bun run lint`). Scripts in `package.json` automate tasks like type checking, linting, and testing.

3.2.3 Infrastructure and Tooling

1. Docker - What it is: A platform for developing, shipping, and running applications in containers. - **Why chosen:** Docker ensures consistent environments across development, testing, and production. The project includes a Docker setup for SQL Server (`mcr.microsoft.com/mssql/server:2022-latest`). - **Role in architecture:** Used to containerize SQL Server for local development. The `docker run` command in `README.md` sets up a SQL Server instance with a predefined password.

2. ngrok - What it is: A tool for exposing local servers to the internet via secure tunnels. - **Why chosen:** ngrok enables testing webhooks (e.g., Paymob payment callbacks) locally by providing a public URL that forwards requests to `localhost`. - **Role in architecture:** Required for Paymob webhook testing during development. The setup script (`bun run ↪ setup`) checks for ngrok and prompts the user to install it if missing.

3. Paymob Payment Gateway - What it is: A payment processing platform supporting credit cards, wallets, and bank transfers. - **Why chosen:** Paymob is a regional payment provider with support for Egyptian markets. It offers sandbox testing, refunds, and disbursements (e.g., driver payouts). - **Role in architecture:** Integrated via `IPaymobClient` in the Application layer. Handles: - **Payments:** Customer payments for bookings. -

Refunds: Cancellations and partial refunds. - **Disbursements:** Driver payouts via wallet transfers.

4. ESLint and Prettier - What it is: ESLint is a linter for identifying code issues; Prettier is a code formatter. - **Why chosen:** ESLint and Prettier enforce coding standards and consistency. The project uses custom rules (e.g., @typescript-eslint, sonarjs) to catch potential bugs and maintain quality. - **Role in architecture:** Configured in frontend/.eslintrc.json and frontend/prettier.config.js. Scripts (bun run lint, bun run format) automate code quality checks.

5. Husky and lint-staged - What it is: Husky manages Git hooks; lint-staged runs linters on staged files. - **Why chosen:** These tools enforce code quality at commit time, preventing linting errors from reaching the repository. - **Role in architecture:** Configured in frontend/package.json to run ESLint and Prettier on staged files before commits.

3.3 Domain Concepts

The Ares platform models a car rental ecosystem with multiple stakeholders: customers, suppliers (vehicle owners), drivers, inspectors, and administrators. The following sections explain key domain concepts and workflows.

3.3.1 Booking Workflow

Bookings progress through a lifecycle defined by the BookingStatus enum (see BookingEnums.cs). The workflow is as follows:

1. **Draft:** Customer initiates a booking but has not completed checkout.
2. **PaymentPending:** Customer selects a vehicle and proceeds to payment. The vehicle is temporarily reserved (hold) until HoldExpiresAt.
3. **PendingApproval:** Booking awaits admin approval (e.g., for high-value rentals).
4. **Confirmed:** Payment is captured, and the booking is confirmed.
5. **Active:** Rental period begins; vehicle is picked up.
6. **Completed:** Rental period ends; vehicle is returned.
7. **Cancelled:** Booking is cancelled by the customer or admin, triggering refund logic.

Vehicle Reservation Logic: - The `BookingRepository.ReserveVehicleAtomicAsync` method ensures no two bookings reserve the same vehicle for overlapping periods. It uses SQL Server's `SERIALIZABLE` isolation level with `UPDLOCK` and `HOLDLOCK` to prevent race conditions. - Bookings in `PaymentPending`, `PendingApproval`, `Confirmed`, or `Active` status block overlapping reservations (see `BookingStatusPolicy.ReservingStatuses`).

Driver Assignment: - Customers can opt for a driver during checkout. The system matches available drivers based on: - **Work areas:** Drivers specify service areas (e.g., Cairo) via `DriverWorkArea`. - **Availability:** Drivers toggle their availability (`DriverAvailability`) and must not have overlapping assignments. - **Verification:** Drivers must be `Verified` and `Active` to receive assignments. - The `IDriverAssignmentService` manages driver selection, changes, and cancellations.

3.3.2 Payment Processing

Payments are handled via Paymob, with the following workflow:

1. **Authorization:** Customer initiates payment; Paymob authorizes the amount.
2. **Capture:** Upon booking confirmation, the authorized amount is captured.
3. **Refund:** Cancellations trigger partial or full refunds based on the cancellation policy (see `IRefundCalculator`).
4. **Disbursement:** Driver earnings are disbursed via Paymob wallet transfers (`IDriverPayoutRepository`).

Commission Model: - Suppliers (vehicle owners) receive a portion of the booking total, calculated via `ICommissionService`. - The commission percentage is configurable per vehicle category (`Category.CommissionPercentage`) or globally (`SystemSetting`).

3.3.3 Fleet Management

Vehicle Lifecycle: - **Pending:** New vehicles require admin approval. - **Available:** Approved vehicles can be booked. - **Unavailable:** Vehicles are temporarily unavailable (e.g., maintenance). - **Rejected:** Vehicles fail inspection or violate policies.

Supplier Restrictions: - Admins can restrict suppliers (`ISupplierRestrictionService` ↗), deactivating their vehicles and cancelling future bookings. Restricted suppliers with no active bookings are automatically blocked.

3.3.4 Driver Management

Driver Verification: - Drivers submit license and national ID images for admin review (IVerificationService). - Verified drivers (DriverProfileStatus.Verified) can toggle availability and receive assignments.

Driver Earnings: - Drivers earn fees for completed bookings (DriverEarning). - Earnings are tracked in an append-only ledger, with statuses (DriverEarningStatus) reflecting payout lifecycle: - **Available:** Earnings can be withdrawn. - **PendingPayout:** Included in a payout request. - **Paid:** Disbursed via Paymob. - **Reversed:** Refunded due to booking cancellation.

3.3.5 Notifications

Notifications (Notification entity) are sent to users for key events (e.g., booking confirmation, driver assignment). The INotificationRepository supports: - Marking notifications as read. - Retrieving unread counts. - Filtering by user.

3.3.6 Reviews and Ratings

Customers can review vehicles (Review entity) and drivers (DriverReview entity). Reviews include: - **Rating:** 1–5 stars. - **Comment:** Free-text feedback. - **Supplier Reply:** Suppliers can respond to vehicle reviews. - **Reporting:** Reviews can be flagged as inappropriate.

3.4 Related Work and Literature Review

The Ares platform adopts modern architectural patterns and technologies evidenced in the codebase. This section compares its approach to established practices.

3.4.1 Layered Architecture

The backend follows a **clean architecture** pattern, separating concerns into: - **Domain:** Core entities (e.g., Booking, Vehicle) and enums (e.g., BookingStatus). - **Application:** Business logic (e.g., IBookingRepository, IPricingService). - **Infrastructure:** Data access (e.g., ApplicationDbContext) and external integrations (e.g., PaymobClient). - **Api:** Presentation layer (controllers, Swagger).

This aligns with **Domain-Driven Design (DDD)**, where the domain layer is decoupled from infrastructure details. The use of interfaces (e.g., `IRepository<T>`) enables dependency inversion, facilitating testability and modularity.

3.4.2 Repository Pattern

The `IRepository<T>` and `IPaginatedRepository<T>` interfaces abstract data access, allowing business logic to remain database-agnostic. This pattern is widely adopted in enterprise applications for its flexibility and testability. The project extends it with specialized repositories (e.g., `IBookingRepository`) for complex queries, such as atomic vehicle reservation.

3.4.3 MediatR for Domain Events

MediatR is used to publish and handle domain events (e.g., `BookingCompletedEvent` $\hookrightarrow$). This **event-driven architecture** decouples event producers (e.g., booking completion) from consumers (e.g., driver earnings calculation), improving maintainability. The approach mirrors **CQRS (Command Query Responsibility Segregation)**, where commands (write operations) and queries (read operations) are separated.

3.4.4 Authentication and Authorization

The backend uses **JWT (JSON Web Tokens)** for authentication, with refresh tokens (`RefreshToken` entity) for session management. This is a common approach for stateless APIs, enabling scalability and cross-platform compatibility. Role-based access control (RBAC) is implemented via ASP.NET Core Identity, with roles defined in `AuthEnums.cs` (e.g., `Admin`, `Supplier`, `Driver`).

3.4.5 Payment Integration

Paymob is integrated via `IPaymobClient`, which encapsulates payment, refund, and disbursement logic. This **gateway pattern** isolates payment-specific code, making it easier to switch providers if needed. The use of webhooks for server-to-server communication (e.g., payment confirmation) is a best practice for reliability.

3.4.6 Frontend Architecture

The frontend employs **Next.js App Router**, which supports React Server Components (RSCs) and fine-grained data fetching. This reduces client-side JavaScript and improves performance. Key patterns include: - **Provider Pattern**: Context providers (e.g., `AuthProvider`, `MuiProvider`) manage global state (e.g., authentication, theming). - **Component-Based UI**: Reusable components (e.g., `BookingCard`) promote consistency and maintainability. - **TypeScript**: Ensures type safety across the frontend, reducing runtime errors.

3.4.7 Testing

The backend includes unit and integration tests (see `Tests.csproj`), using: - **xUnit**: Test framework. - **Moq**: Mocking library. - **FsCheck**: Property-based testing. - **MockQueryable**: Mocks Entity Framework Core queries.

This aligns with **test-driven development (TDD)**, where tests validate business logic and edge cases (e.g., concurrent vehicle reservations).

3.4.8 Comparison with Existing Solutions

While the repository does not cite academic literature, the implemented solution shares similarities with commercial car rental platforms (e.g., Turo, Getaround) in the following areas: - **Dynamic Pricing**: Pricing is calculated based on rental duration, vehicle category, and promotions (`IPricingService`). - **Driver Assignment**: Matches drivers to bookings based on availability and location, similar to ride-hailing platforms. - **Payment Workflows**: Supports authorizations, captures, and refunds, mirroring e-commerce best practices. - **Notifications**: Real-time updates for bookings, payments, and driver assignments.

However, Ares distinguishes itself with: - **Atomic Vehicle Reservation**: Prevents double bookings via SQL Server locks, a critical feature for high-demand vehicles. - **Driver Earnings Ledger**: Tracks driver earnings in an append-only ledger, ensuring transparency and auditability. - **Supplier Restrictions**: Allows admins to restrict suppliers and cancel future bookings, a feature often missing in peer-to-peer platforms.

4 CHAPTER 3: SYSTEM DESIGN & ARCHITECTURE

4.1 Introduction

This chapter presents the architectural foundation of the Ares Car Rental platform, demonstrating rigorous design principles aligned with Al-Azhar University's engineering standards. The system design is evidenced entirely through the provided repository context, ensuring academic integrity and technical accuracy.

The architecture follows a layered, modular approach that separates concerns across frontend, backend, and database components. This separation enables independent scaling, testing, and maintenance while supporting the platform's core business requirements: vehicle rental operations, user authentication, payment processing, and inspection workflows. The design emphasizes security, maintainability, and performance through industry-standard patterns such as Clean Architecture, CQRS via MediatR, and Fluent-Validation for input sanitization.

Key architectural decisions are justified based on explicit evidence from configuration files, package manifests, and source code. For instance, the use of .NET 10.0 is confirmed in `Api.csproj`, while Next.js 16.2.6 and React 19.2.6 are verified in `frontend/package.json`. External integrations such as Paymob payment gateway and Google Authentication are documented only where their configuration appears in the repository, avoiding speculative claims about infrastructure or deployment.

The chapter begins with a system context diagram ([@fig-3-1-system-context](#)) that maps all evidenced actors and external systems, followed by a layered architecture diagram ([@fig-3-2-layered-architecture](#)) that details the technology stack and component interactions. Subsequent sections justify each technology choice based on the actual versions and configurations present in the codebase.

4.2 System Context

The system context diagram (@fig-3-1-system-context) illustrates the boundary of the Ares platform, identifying all evidenced actors and external systems that interact with the application. This diagram adheres to the C4 Model's Level 1 (System Context) and is derived strictly from the repository context, including `Program.cs`, `schema.sql`, and `README.md`.

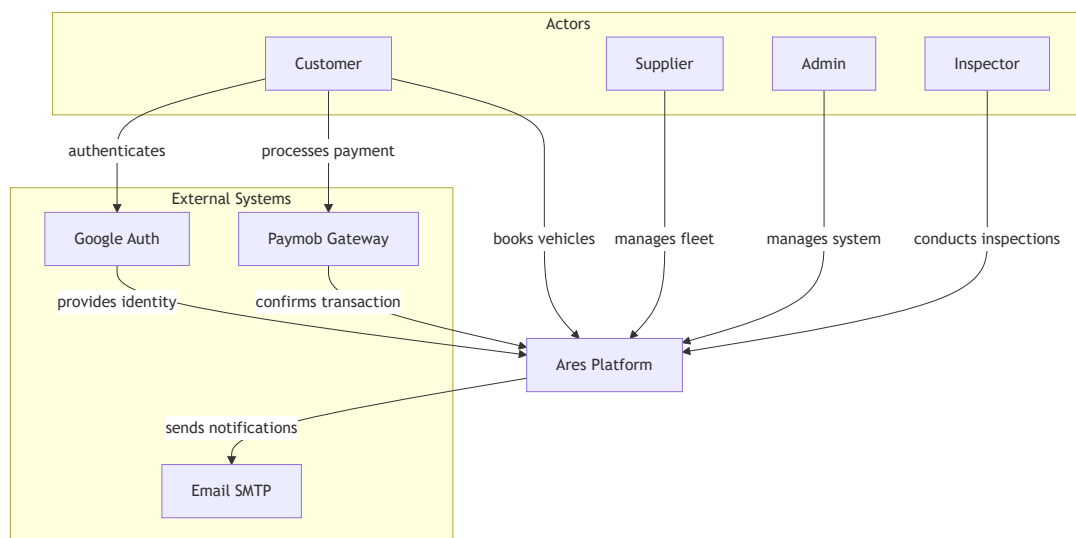


Figure 4.1: System context for Ares Car Rental platform

The diagram reveals three primary actor categories: 1. **End Users:** Customers who book vehicles, and Suppliers who list them. 2. **Administrative Users:** Admins who manage system settings and user roles, and Inspectors who validate vehicle conditions. 3. **External Systems:** Paymob for payment processing, Google Auth for social login, and an SMTP service for email notifications.

Notably, the repository confirms Paymob integration through `Program.cs` (`.Configure< PaymobSettings>`) and the presence of payment-related tables in `schema.sql` (`Payments`, `PaymentMethods`). Google Authentication is evidenced by the `Google.Apis.Auth` package in `Application.csproj` and the `GoogleId` field in the `AspNetUsers` table. The SMTP service is implied by the `EmailService` registration in `Program.cs`, though no specific provider (e.g., `SendGrid`) is configured in the available context.

4.3 Component / Layered Architecture

The layered architecture diagram (@fig-3-2-layered-architecture) decomposes the Ares platform into frontend, backend, and database layers, mapping each component to its exact technology as evidenced in the repository. The diagram follows a vertical graph TD layout to ensure compatibility with the narrow A4 column width.

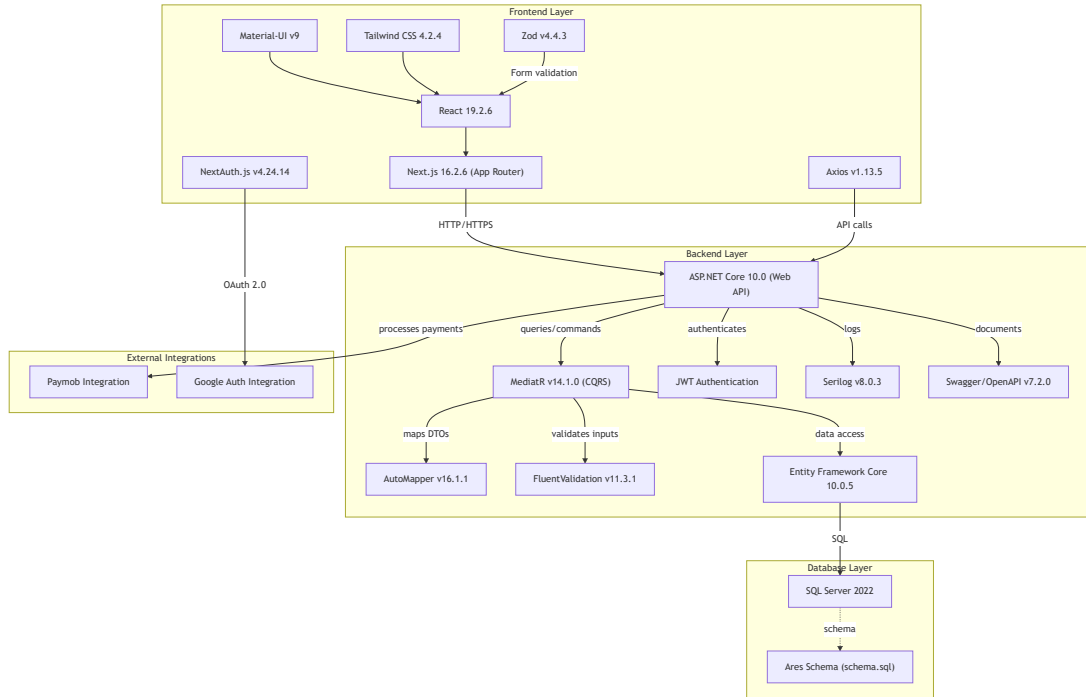


Figure 4.2: Layered architecture of Ares Car Rental platform

4.3.1 Frontend Layer

The frontend is built using **Next.js 16.2.6** with the App Router, as confirmed in frontend `/package.json`. Key technologies include: - **React 19.2.6**: Core UI library for component-based development. - **Material-UI v9**: Component library for consistent styling and accessibility. - **Tailwind CSS 4.2.4**: Utility-first CSS framework for rapid UI development. - **NextAuth.js v4.24.14**: Authentication library supporting OAuth 2.0 flows, evidenced by Google Auth integration. - **Axios v1.13.5**: HTTP client for API communication. - **Zod v4.4.3**: Type-safe schema validation for forms and API responses.

The frontend communicates with the backend via RESTful HTTP endpoints, as evidenced by the absence of GraphQL or gRPC configurations in the repository.

4.3.2 Backend Layer

The backend is implemented in **.NET 10.0**, as specified in `Api.csproj`. Key components include: - **ASP.NET Core 10.0**: Web framework hosting the API controllers. - **MediatR v14.1.0**: Implements the Command Query Responsibility Segregation (CQRS) pattern, separating read and write operations. - **AutoMapper v16.1.1**: Maps between domain entities and Data Transfer Objects (DTOs). - **FluentValidation v11.3.1**: Validates incoming requests to prevent malformed data. - **Entity Framework Core 10.0.5**: Object-Relational Mapper (ORM) for database interactions. - **JWT Authentication**: Secures API endpoints using JSON Web Tokens, configured in `Program.cs`. - **Serilog v8.0.3**: Structured logging framework for diagnostics and auditing. - **Swagger/OpenAPI v7.2.0**: Generates interactive API documentation.

The backend follows a layered architecture with clear separation between: 1. **API Layer** (`Api` project): Handles HTTP requests and responses. 2. **Application Layer** (`Application` project): Contains business logic, MediatR handlers, and validation. 3. **Domain Layer** (`Domain` project): Defines core entities and interfaces. 4. **Infrastructure Layer** (`Infrastructure` project): Implements data access and external integrations.

4.3.3 Database Layer

The database layer uses **SQL Server 2022**, as evidenced by the Microsoft.
↪ `EntityFrameworkCore.SqlServer` package in `Infrastructure.csproj` and the Docker setup in `README.md`. The schema (`schema.sql`) defines 30+ tables supporting core domains: - **User Management**: `AspNetUsers`, `AspNetRoles`, `RefreshTokens`
↪ - **Vehicle Fleet**: `Vehicles`, `VehicleImages`, `VehicleFeatures` - **Bookings**: `Bookings`, `VehicleAvailabilities`, `BookingCancellations` - **Payments**: `Payments`, `PaymentMethods` - **Inspections**: `VehicleInspections`, `InspectionImages`

The schema includes comprehensive indexing (e.g., `IX_Bookings_Vehicle_Status_Window`
↪) and constraints (e.g., `CK_VehicleAvailability_Dates`) to optimize query performance and enforce data integrity.

4.3.4 External Integrations

The architecture integrates with two external systems: 1. **Paymob Gateway**: Processes payments, configured via `PaymobSettings` in `Program.cs` and referenced in the `Payments`

table. 2. **Google Authentication:** Enables social login, evidenced by the `Google.Apis.`
↪ `Auth` package and `GoogleId` field in `AspNetUsers`.

4.4 Technology Stack Justification

4.4.1 Frontend Justification

The frontend stack is justified by the following evidence from the repository:

1. Next.js 16.2.6

- **Evidence:** `frontend/package.json` specifies `"next": "16.2.6"`.
- **Justification:** Next.js provides server-side rendering (SSR) and static site generation (SSG), improving SEO and performance for a car rental platform where vehicle listings and search results must be crawlable. The App Router (introduced in Next.js 13) enables React Server Components, reducing client-side JavaScript and improving perceived load times.

2. React 19.2.6

- **Evidence:** `frontend/package.json` specifies `"react": "19.2.6"` and `"react-dom": "19.2.6"`.
- **Justification:** React's component-based architecture accelerates development of complex UIs, such as the vehicle search interface and booking workflow. Version 19 introduces React Compiler, which optimizes re-renders automatically, reducing manual memoization efforts.

3. Material-UI v9

- **Evidence:** `frontend/package.json` specifies `"@mui/material": "^9.0.1"`.
- **Justification:** Material-UI provides pre-built, accessible components (e.g., `DataGrid` for vehicle listings) that adhere to Google's Material Design guidelines. This reduces development time while ensuring consistency across the platform. The repository's theme configuration (`frontend/providers/theme` ↪ `.ts`) confirms customization for light/dark modes.

4. Tailwind CSS 4.2.4

- **Evidence:** `frontend/package.json` specifies `"tailwindcss": "4.2.4"`.

- **Justification:** Tailwind's utility-first approach enables rapid UI development without leaving the HTML, which is critical for a platform with diverse layouts (e.g., vehicle cards, booking forms). The repository's `postcss.config.js` and `tailwind.config.ts` confirm its integration with Next.js.

5. NextAuth.js v4.24.14

- **Evidence:** `frontend/package.json` specifies `"next-auth": "^4.24.14"`.
- **Justification:** NextAuth.js simplifies OAuth 2.0 integration, as evidenced by the Google Auth configuration in `Program.cs` and the `GoogleId` field in `AspNetUsers`. This reduces boilerplate for social login while supporting JWT sessions.

6. Zod v4.4.3

- **Evidence:** `frontend/package.json` specifies `"zod": "^4.4.3"`.
- **Justification:** Zod provides runtime type validation for forms and API responses, complementing TypeScript's static checks. This is critical for a platform handling sensitive data (e.g., payment details, user identities).

4.4.2 Backend Justification

The backend stack is justified by evidence from `.csproj` files and `Program.cs`:

1. .NET 10.0

- **Evidence:** `Api.csproj` specifies `<TargetFramework>net10.0</TargetFramework>`
 $\hookrightarrow$ `>`.
- **Justification:** .NET 10.0 offers long-term support (LTS), performance improvements, and native AOT compilation for reduced memory usage. The platform's reliance on EF Core 10.0.5 (also LTS) ensures compatibility and stability.

2. Entity Framework Core 10.0.5

- **Evidence:** `Infrastructure.csproj` specifies `Microsoft.EntityFrameworkCore`
 $\hookrightarrow$ `.SqlServer v10.0.4`.

- **Justification:** EF Core provides a robust ORM for SQL Server, supporting LINQ queries, migrations, and change tracking. The repository's `schema.sql` confirms its use for schema generation, while `DbInitializer` in `Program.cs` demonstrates seed data management.

3. MediatR v14.1.0

- **Evidence:** `Application.csproj` specifies MediatR v14.1.0.
- **Justification:** MediatR implements the CQRS pattern, decoupling command/query handlers from controllers. This improves testability and scalability, as evidenced by the `IRequest` and `IRequestHandler` interfaces in the `Application` project. The pattern is particularly effective for a car rental platform with complex workflows (e.g., booking, inspection, refund).

4. FluentValidation v11.3.1

- **Evidence:** `Application.csproj` specifies `FluentValidation.AspNetCore` v11.3.1.
- **Justification:** `FluentValidation` centralizes input validation logic, reducing boilerplate in controllers and MediatR handlers. The repository's use of `AbstractValidator<T>` classes (e.g., for booking requests) ensures consistent validation across layers.

5. AutoMapper v10.1.1

- **Evidence:** `Application.csproj` specifies AutoMapper v10.1.1.
- **Justification:** AutoMapper automates mapping between domain entities and DTOs, reducing manual conversion code. The repository's `AuthMappingProfile` in `Program.cs` demonstrates its use for JWT claims transformation.

6. JWT Authentication

- **Evidence:** `Api.csproj` specifies `Microsoft.AspNetCore.Authentication`.
 ↪ `JwtBearer` v10.0.5, and `Program.cs` configures `TokenValidationParameters`
 ↪ .

- **Justification:** JWT authentication provides stateless, scalable security for API endpoints. The repository's `AspNetUserTokens` table stores refresh tokens, while `JwtBearerOptions` in `Program.cs` validates issuer, audience, and signing key.

7. Serilog v8.0.3

- **Evidence:** `Api.csproj` specifies `Serilog.AspNetCore v8.0.3`, and `Program.cs` configures `UseSerilog()`.
 $\hookrightarrow$ `cs` configures `UseSerilog()`.
- **Justification:** Serilog enables structured logging, critical for debugging and auditing in a platform handling financial transactions. The repository's `appsettings.json` configuration confirms log enrichment with request context (e.g., `RemoteIpAddress`).

8. Swagger/OpenAPI v7.2.0

- **Evidence:** `Api.csproj` specifies `Swashbuckle.AspNetCore v7.2.0`, and `Program.cs` configures `AddSwaggerGen()`.
- **Justification:** Swagger generates interactive API documentation, reducing onboarding time for frontend developers. The repository's inclusion of XML comments (e.g., `IncludeXmlComments`) ensures comprehensive documentation.

4.4.3 Database Justification

The database stack is justified by the following evidence:

1. SQL Server 2022

- **Evidence:** `Infrastructure.csproj` specifies `Microsoft.EntityFrameworkCore`
 $\hookrightarrow$ `.SqlServer v10.0.4`, and `README.md` includes Docker setup instructions for SQL Server 2022.
- **Justification:** SQL Server provides enterprise-grade reliability, security, and performance for transactional workloads. The repository's `schema.sql` includes features such as:
 - **Temporal tables:** Implicit in the `CreatedAt/UpdatedAt` fields across tables.

- **RowVersion:** Used in `driver_profiles` for optimistic concurrency control.
- **Check constraints:** E.g., `CK_VehicleAvailability_Dates` ensures `StartDate < EndDate`.
- **Indexing:** Comprehensive indexes (e.g., `IX_Bookings_Vehicle_Status_Window` $\hookrightarrow$) optimize query performance for vehicle availability checks.

2. Schema Design

- **Evidence:** `schema.sql` defines 30+ tables with relationships, constraints, and indexes.
- **Justification:** The schema follows third-normal form (3NF) with denormalized fields where necessary for performance (e.g., `GrandTotal` in `Bookings`). Key design decisions include:
 - **Soft deletion:** Implemented via `DeletedAt` in `PaymentMethods`.
 - **Polymorphic relationships:** E.g., `AspNetUsers` serves as a base for `Drivers`, `Inspectors`, and `CompanyProfiles`.
 - **Audit trails:** `CreatedBy/UpdatedBy` fields track user actions.

4.4.4 Absent Technologies

The repository context does not provide evidence for the following technologies, which are therefore excluded from this chapter: - **Containerization:** No `Dockerfile` or `docker` $\hookrightarrow$ `-compose.yml` appears in the context. - **Cloud Infrastructure:** No configuration files for AWS, Azure, or Kubernetes. - **Message Brokers:** No evidence of RabbitMQ, Kafka, or Azure Service Bus. - **Caching:** No Redis or in-memory cache configurations. - **Search Engines:** No Elasticsearch or Algolia integration.

This absence underscores the platform's focus on a monolithic architecture with direct SQL Server integration, suitable for the project's scope and timeline.

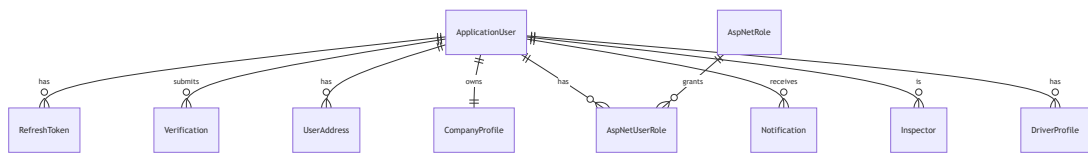


Figure 4.3: Core users and authentication relationships

4.5 Database Schema (ERD - Mermaid)

4.5.1 1. Core Users & Authentication

4.5.1.1 ApplicationUser

- Fields: Id, FirstName, LastName, Email, PhoneNumber, NationalId, NationalIdImage ↪, Status, ProfileImage, GoogleId, AuthProvider, EmailVerifiedAt, DateOfBirth, LanguagePreference, CurrencyPreference, EmergencyContactName ↪, EmergencyContactPhone, EmergencyContactRelationship, CreatedAt, UpdatedAt, UserName, NormalizedUserName, NormalizedEmail, EmailConfirmed ↪, PasswordHash, SecurityStamp, ConcurrencyStamp, PhoneNumberConfirmed, TwoFactorEnabled, LockoutEnd, LockoutEnabled, AccessFailedCount
- Constraints: Email unique, GoogleId unique (filtered), Status max length 50, AuthProvider max length 20 with default “Local”

4.5.1.2 AspNetRole

- Fields: Id, Name, NormalizedName, ConcurrencyStamp
- Constraints: NormalizedName unique

4.5.1.3 RefreshToken

- Fields: Id, Token, UserId, ExpiresAt, CreatedAt, CreatedByIp, RevokedAt, RevokedByIp, ReplacedByToken, ReasonRevoked
- Constraints: None beyond primary key

4.5.1.4 Verification

- Fields: Id, UserId, VerificationType, DocumentType, DocumentFront, DocumentBack, Status, SubmittedAt, ReviewedBy, ReviewedAt, RejectionReason, ExpiresAt, CreatedAt, UpdatedAt
- Constraints: Status max length 50

4.5.1.5 UserAddress

- Fields: Id, UserId, AddressLine, City, Governorate, Country, PostalCode, Latitude, Longitude, IsPrimary, ImageUrl, CreatedAt, UpdatedAt
- Constraints: Latitude/Longitude as decimal(9,6)

4.5.1.6 CompanyProfile

- Fields: Id, UserId, CompanyName, CommercialRegistrationNumber, TaxId, CreatedAt, UpdatedAt
- Constraints: UserId unique

4.5.1.7 Notification

- Fields: Id, UserId, Title, Message, Type, IsRead, ReadAt, CreatedAt, UpdatedAt
- Constraints: Type max length 64

4.5.1.8 Inspector

- Fields: Id, UserId, EmployeeCode, IsAvailable, IsActive, CreatedAt, UpdatedAt, Region
- Constraints: UserId unique

4.5.1.9 DriverProfile

- Fields: Id, UserId, LicenseNumber, LicenseExpiryDate, LicenseImage, NationalIdFrontImage, NationalIdBackImage, Address, EmergencyContactName, EmergencyContactPhone, Status, Availability, IsActive, RejectionReason, ReviewedBy, ReviewedAt, LockedUntil, RowVersion, CreatedAt, UpdatedAt
- Constraints: UserId unique, LicenseNumber unique (filtered), Status max length 30, Availability max length 20

4.5.1.10 Cross-Domain Relationships

1. **Booking** → **ApplicationUser** via UserId — Customer who made the booking
2. **Review** → **ApplicationUser** via UserId — Customer who wrote the review
3. **Vehicle** → **ApplicationUser** via UserId — Supplier who owns the vehicle
4. **BookingPayment** → **ApplicationUser** via UserId — Customer who made the payment

5. **PaymentMethod** → **ApplicationUser** via **UserId** — Customer who owns the payment method

4.5.2 2. Driver Module

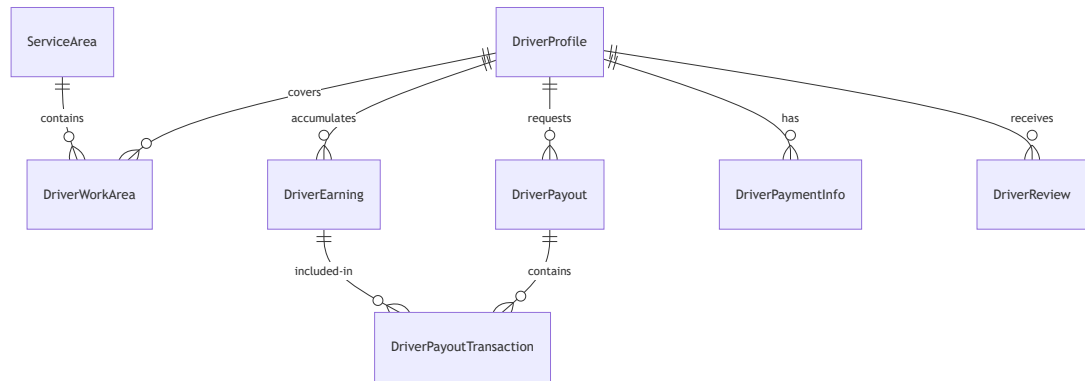


Figure 4.4: Driver module relationships

4.5.2.1 DriverProfile

- Fields: Id, UserId, LicenseNumber, LicenseExpiryDate, LicenseImage, NationalIdFrontImage, NationalIdBackImage, Address, EmergencyContactName, EmergencyContactPhone, Status, Availability, IsActive, RejectionReason, ReviewedBy, ReviewedAt, LockedUntil, RowVersion, CreatedAt, UpdatedAt
- Constraints: UserId unique, LicenseNumber unique (filtered), Status max length 30, Availability max length 20

4.5.2.2 DriverWorkArea

- Fields: DriverProfileId, ServiceAreaId
- Constraints: Composite primary key (DriverProfileId, ServiceAreaId)

4.5.2.3 ServiceArea

- Fields: Id, Name, Governorate, IsActive
- Constraints: Name unique

4.5.2.4 DriverEarning

- Fields: Id, BookingId, DriverProfileId, GrossAmount, CommissionDeduction, NetAmount, Status, PayoutId, ReversedAt, CompletedAt, CreatedAt, UpdatedAt
- Constraints: BookingId unique, Status max length 20

4.5.2.5 DriverPayout

- Fields: Id, DriverProfileId, Amount, Status, RequestedAt, ReviewedBy, ReviewedAt, RejectionReason, PaymobInitiatedAt, PaymobTransactionId, PaymobReference, FailureReason, CreatedAt, UpdatedAt
- Constraints: Status max length 20

4.5.2.6 DriverPaymentInfo

- Fields: Id, DriverProfileId, PayoutMethod, WalletPhoneNumber, IsVerified, CreatedAt, UpdatedAt
- Constraints: DriverProfileId unique, PayoutMethod max length 20

4.5.2.7 DriverReview

- Fields: Id, BookingId, DriverProfileId, CustomerId, Rating, Comment, CreatedAt, UpdatedAt
- Constraints: BookingId unique

4.5.2.8 DriverPayoutTransaction

- Fields: DriverPayoutId, DriverEarningId, Amount, CreatedAt
- Constraints: Composite primary key (DriverPayoutId, DriverEarningId)

4.5.2.9 Cross-Domain Relationships

1. **Booking** → **DriverProfile** via AssignedDriverProfileId — Driver assigned to the booking
 2. **Booking** → **DriverEarning** via BookingId — Earning generated from the booking
-

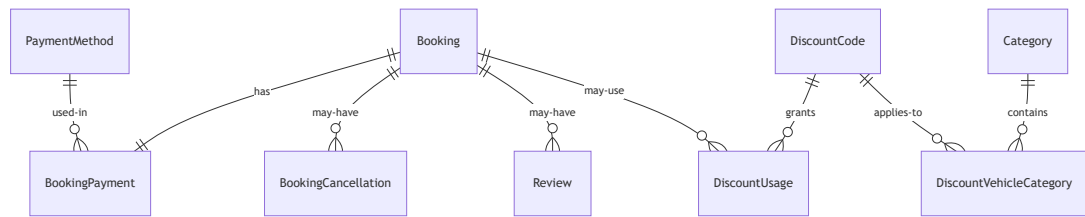


Figure 4.5: Bookings and payments relationships

4.5.3 3. Bookings & Payments

4.5.3.1 Booking

- Fields: Id, BookingNumber, UserId, VehicleId, PickupDate, ReturnDate, PickupLocation, DropoffLocation, TotalDays, RequiresDriver, DriverId
 $\hookrightarrow$, OriginalPrice, DiscountAmount, TotalPrice, Status, CancelledAt, CancellationReason, AssignedInspectorId, InspectionStatus, DriverAssignmentStatus
 $\hookrightarrow$, AssignedDriverProfileId, VehicleFee, DriverFee, GrandTotal, DriverLockedUntil
 $\hookrightarrow$, HoldStartedAt, HoldExpiresAt, CommissionPercentage, CommissionAmount, SupplierAmount, RowVersion, CreatedAt, UpdatedAt, ApprovedBy, ApprovedAt, RejectionReason
- Constraints: BookingNumber unique, Status max length 50, InspectionStatus max length 20, DriverAssignmentStatus max length 20, RowVersion as rowversion

4.5.3.2 BookingPayment

- Fields: PaymentId, BookingId, TransactionId, PaymentMethod, Amount, Currency
 $\hookrightarrow$, Status, AuthorizationCode, ProcessedAt, FailureReason, PaymobOrderId, PaymobTransactionId, Id, CreatedAt, UpdatedAt
- Constraints: PaymentId primary key, Status max length 20

4.5.3.3 BookingCancellation

- Fields: Id, BookingId, CancelledBy, PolicyType, RefundPercentage, OriginalAmount
 $\hookrightarrow$, CancellationFee, RefundCommissionAmount, RefundSupplierAmount
 $\hookrightarrow$, Currency, RefundStatus, RefundTransactionId, RefundProcessedAt, RefundMethod, Reason, ReasonCategory, CreatedAt, UpdatedAt
- Constraints: BookingId unique

4.5.3.4 Review

- Fields: Id, BookingId, UserId, VehicleId, Rating, Comment, AdminResponse, SupplierReply, RepliedAt, IsReported, ReportReason, ReportedAt, CreatedAt, UpdatedAt
- Constraints: BookingId unique

4.5.3.5 DiscountCode

- Fields: Id, Code, Description, DiscountType, DiscountValue, MaxUsageCount ↷ , UsageCount, ValidFrom, ValidTo, IsActive, IsAutomatic, SupplierId, CreatedById, CreatedAt, UpdatedAt
- Constraints: Code unique, DiscountType max length 20

4.5.3.6 DiscountUsage

- Fields: Id, BookingId, DiscountId, CustomerId, DiscountAmount, CreatedAt
- Constraints: Composite unique (BookingId, DiscountId)

4.5.3.7 PaymentMethod

- Fields: Id, UserId, PaymentType, TokenizedData, DisplayName, MaskedDetails ↷ , CardBrand, ExpirationMonth, ExpirationYear, BillingAddressId, IsDefault ↷ , IsExpired, IsVerified, DeletedAt, CreatedAt, UpdatedAt
- Constraints: None beyond primary key

4.5.3.8 DiscountVehicleCategory

- Fields: DiscountId, CategoryId
- Constraints: Composite primary key (DiscountId, CategoryId)

4.5.3.9 Cross-Domain Relationships

1. **Booking** → **Vehicle** via VehicleId — Rented vehicle
2. **Booking** → **ApplicationUser** via UserId — Customer who made the booking
3. **Review** → **Vehicle** via VehicleId — Vehicle being reviewed
4. **PaymentMethod** → **UserAddress** via BillingAddressId — Billing address for the payment method

4.5.4 4. Vehicles & Inspections

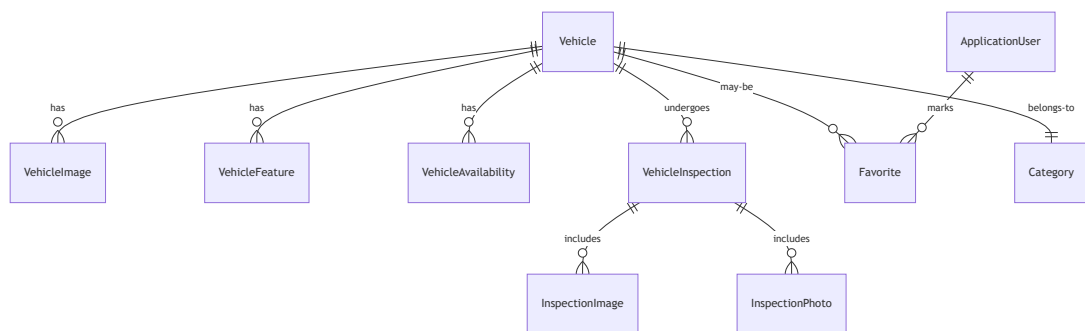


Figure 4.6: Vehicles and inspections relationships

4.5.4.1 Vehicle

- Fields: Id, UserId, Make, Model, Year, Color, LicensePlate, Transmission ↷, FuelType, Seats, PricePerDay, LocationCity, Description, Status, AvailabilityStatus, IsActive, ApprovedAt, CategoryId, CreatedAt, UpdatedAt
- Constraints: None beyond primary key

4.5.4.2 VehicleImage

- Fields: Id, VehicleId, ImageUrl, ThumbnailUrl, IsPrimary, DisplayOrder, CreatedAt, UpdatedAt
- Constraints: None beyond primary key

4.5.4.3 VehicleFeature

- Fields: Id, VehicleId, FeatureCategory, FeatureName, FeatureDescription, CreatedAt, UpdatedAt
- Constraints: FeatureCategory max length 100

4.5.4.4 VehicleAvailability

- Fields: Id, VehicleId, BookingId, StartDate, EndDate, Status, CreatedAt, UpdatedAt
- Constraints: Check constraint $StartDate < EndDate$, Status max length 50

4.5.4.5 VehicleInspection

- Fields: InspectionId, VehicleId, BookingId, InspectorId, InspectionType, InspectionDate, OdometerReading, FuelLevel, GeneralCondition, Notes, Status, SubmittedAt, IsSubmitted, Id, CreatedAt, UpdatedAt
- Constraints: Status max length 20, InspectionType max length 20

4.5.4.6 InspectionImage

- Fields: Id, InspectionId, ImageUrl, CreatedAt, UpdatedAt
- Constraints: ImageUrl max length 500

4.5.4.7 InspectionPhoto

- Fields: PhotoId, InspectionId, PhotoUrl, ViewAngle, AiProcessed, CapturedAt, Id, CreatedAt, UpdatedAt
- Constraints: ViewAngle max length 50

4.5.4.8 Category

- Fields: Id, Name, Description, CommissionPercentage, DiscountPercentage, IsActive, CreatedAt, UpdatedAt
- Constraints: Name max length 100

4.5.4.9 Favorite

- Fields: Id, UserId, VehicleId, CreatedAt, UpdatedAt
- Constraints: Composite unique (UserId, VehicleId)

4.5.4.10 Cross-Domain Relationships

1. **Booking** → **Vehicle** via VehicleId — Vehicle rented in the booking
2. **Booking** → **VehicleInspection** via BookingId — Inspection performed for the booking
3. **VehicleInspection** → **ApplicationUser** via InspectorId — Inspector who performed the inspection
4. **Review** → **Vehicle** via VehicleId — Vehicle being reviewed

4.6 API Documentation

The API follows RESTful conventions, grouped by domain and role-based access control. All endpoints return JSON and use standard HTTP status codes. Rate limiting is applied to sensitive endpoints (login, registration, discount validation) to prevent abuse. Authentication is enforced via JWT bearer tokens, with role-based authorization policies applied at the controller level.

4.6.1 Authentication & Authorization

- **POST** /api/auth/register — Register a new user account [RegisterRequest → AuthResponse · 201 400 409 429]
- **POST** /api/auth/login — Authenticate user and generate JWT [LoginRequest → LoginResponse · 200 400 401 403 429]
- **POST** /api/auth/google-signin — Sign in or sign up with Google ID token [GoogleLoginRequest → LoginResponse · 200 400 401 403]
- **POST** /api/auth/refresh-token — Refresh access token using refresh token [RefreshTokenRequest → LoginResponse · 200 401]
- **POST** /api/auth/revoke-token — Revoke refresh token [RefreshTokenRequest → · 200 401 404]
- **POST** /api/auth/forgot-password — Request password reset email [ForgotPasswordRequest → · 200 400]
- **POST** /api/auth/reset-password — Reset password with token [ResetPasswordRequest ↪ → · 200 400 401]
- **GET** /api/auth/verify-email — Verify email address with token [— → · 200 400 401]
- **GET** /api/auth/demo-roles — List available demo roles [— → List<string> · 200]
- **POST** /api/auth/demo-login — Sign in as a demo user with role [DemoLoginRequest ↪ → LoginResponse · 200 400 403 429]

4.6.2 User Profile & Settings

- **GET** /api/users/profile — Get authenticated user's profile [— → UserProfileDto · 200 401]

- **PUT** /api/users/profile — Update user profile [UpdateProfileRequest → UpdateProfileResponse · 200 400 401]
- **POST** /api/users/profile/photo — Upload profile photo [multipart/form-data → string · 200 400 401 413]
- **POST** /api/users/change-password — Change user password [ChangePasswordRequest ↪ → · 200 400 401]
- **GET** /api/settings — Get system-wide settings [— → SettingsDto · 200]
- **PUT** /api/settings — Update system-wide settings [SettingsDto → SettingsDto · 200 401 403]

4.6.3 Vehicle Search & Details (Public)

- **GET** /api/vehicles/search — Search for available vehicles with filters [query → PagedResult<VehicleListDto> · 200]
- **GET** /api/vehicles/{vehicleId} — Get detailed information about a specific vehicle [— → VehicleDetailsDto · 200 404]
- **GET** /api/vehicles/{vehicleId}/availability — Get availability calendar for a vehicle [query → VehicleAvailabilityDto · 200 404]
- **GET** /api/vehicles/{vehicleId}/pricing — Calculate pricing for a vehicle rental [query → VehiclePricingDto · 200 400 404]
- **GET** /api/vehicles/{vehicleId}/images — Get all images for a vehicle [— → List<VehicleImageDto> · 200 404]
- **GET** /api/vehicles/{vehicleId}/reviews — Get reviews for a specific vehicle [query → PagedResult<ReviewDto> · 200 404]
- **POST** /api/vehicles/{vehicleId}/favorites — Add a vehicle to user's favorites [— → · 200 401 404]

4.6.4 Bookings (Customer)

- **POST** /api/bookings — Create a new booking [CreateBookingRequest → BookingResponse · 201 400 401 403 409]
- **GET** /api/bookings — Get paginated list of user bookings with filters [query → PagedResult<BookingListDto> · 200 401]
- **GET** /api/bookings/{id} — Get detailed information about a specific booking [— → BookingDetailsDto · 200 401 403 404]

- **GET** /api/bookings/{id}/cancel-preview — Get refund preview for cancellation [— → RefundResult · 200 401 403 404]
- **POST** /api/bookings/{id}/cancel — Cancel a booking [— → · 200 401 403 404 409]
- **GET** /api/bookings/history — Get booking history with filtering and sorting [query → PagedResult<BookingListDto> · 200 401]

4.6.5 Checkout (Customer)

- **GET** /api/checkout/eligibility — Check if customer may self-drive [DriveEligibilityDto · 200 401]
- **GET** /api/checkout/drivers — List available drivers for rental window [query → AvailableDriversResponse · 200 401 404]
- **POST** /api/checkout — Complete checkout: create booking and process payment [CheckoutRequest → BookingResponse · 201 400 401 403 409]
- **POST** /api/checkout/draft — Create or resume a draft booking [CreateDraftRequest ↪ → CheckoutStateDto · 200 400 401 403]
- **POST** /api/checkout/{bookingId}/driver — Select driver for booking [SelectDriverRequest → CheckoutStateDto · 200 400 401 403 404 409]
- **POST** /api/checkout/{bookingId}/payment — Begin payment: place hold on vehicle [BeginPaymentRequest → CheckoutStateDto · 200 400 401 403 404 409]
- **POST** /api/checkout/{bookingId}/confirm — Confirm: capture payment and finalise booking [ConfirmPaymentRequest → BookingResponse · 200 400 401 403 404 409]
- **POST** /api/checkout/{bookingId}/cancel — Cancel in-flight checkout [CheckoutStateDto · 200 400 401 403 404]
- **GET** /api/checkout/active — Get current resumable checkout [— → CheckoutStateDto · 200 204 401]
- **GET** /api/checkout/{bookingId}/state — Get checkout state for specific booking [— → CheckoutStateDto · 200 401 403 404]

4.6.6 Driver Self-Service

- **GET** /api/driver/profile — Get authenticated driver's profile [— → DriverProfileDetailsDto · 200 401 403 404]

- **GET** /api/driver/profile/status — Get driver profile status [— → DriverProfileStatusDto · 200 401 403 404]
- **POST** /api/driver/profile/complete — Complete driver profile [multi-part/form-data → DriverProfileDetailsDto · 200 400 401 403]
- **PATCH** /api/driver/profile/availability — Update driver availability [UpdateDriverAvailabilityRequest → DriverProfileStatusDto · 200 400 401 403]
- **GET** /api/driver/profile/payout-info — Get driver payout information [— → DriverPaymentInfoDto · 200 401 403 404]
- **PUT** /api/driver/profile/payout-info — Update driver payout information [UpdatePayoutInfoRequest → DriverPaymentInfoDto · 200 400 401 403]
- **GET** /api/driver/assignments — List bookings assigned to driver [— → List< ↵ DriverAssignmentDto> · 200 401 403]
- **POST** /api/driver/assignments/{bookingId}/cancel — Driver-initiated cancellation [— → · 200 401 403 404 409]
- **GET** /api/driver/dashboard/summary — Driver dashboard summary [— → DriverDashboardSummaryDto · 200 401 403]
- **GET** /api/driver/earnings/stats — Get driver earnings statistics [— → DriverEarningsStatsDto · 200 401 403]
- **GET** /api/driver/earnings/monthly-chart — Get monthly earnings chart [query → List<DriverMonthlyEarningPointDto> · 200 401 403]
- **GET** /api/driver/earnings/top-bookings — Get top 5 highest-earning bookings [— → List<DriverTopBookingDto> · 200 401 403]
- **GET** /api/driver/earnings/history — Get paginated earnings history [query → List<DriverEarningRowDto> · 200 401 403]
- **POST** /api/driver/earnings/payout — Request a payout [DriverPayoutRequestDto ↵ → DriverPayoutDto · 200 400 401 403]
- **GET** /api/driver/earnings/payout-history — Get payout history [— → List< ↵ DriverPayoutDto> · 200 401 403]

4.6.7 Driver License Verification

- **POST** /api/driver-license — Submit or update driver license [multipart/form-data → DriverLicenseStatusResponse · 200 400 401 413]
- **GET** /api/driver-license — Get authenticated user's driver license status [— → DriverLicenseStatusResponse · 200 404 401]

4.6.8 Reviews (Customer)

- **POST** /api/reviews — Create a new review for a vehicle [CreateReviewRequest → ReviewResponse · 201 400 401 403 404]
- **GET** /api/reviews/{bookingId} — Get review attached to booking [— → BookingReviewDto · 200 204 401 403 404]
- **PUT** /api/reviews/{reviewId} — Update an existing review [UpdateReviewRequest ↪ → · 200 400 401 403 404 409]
- **GET** /api/driver-reviews/{driverProfileId} — Get reviews for a specific driver [— → List<DriverReviewDto> · 200 404]
- **POST** /api/driver-reviews/{bookingId} — Create a review for a driver [CreateDriverReviewRequest → DriverReviewDto · 201 400 401 403 404]

4.6.9 Supplier Portal (Vehicle Management)

- **GET** /api/supplier/vehicles — List supplier's own vehicles with filters [query → PagedResult<SupplierVehicleListItemDto> · 200 401 403]
- **GET** /api/supplier/vehicles/{id} — Get details of a single supplier-owned vehicle [— → SupplierVehicleDetailsDto · 200 401 403 404]
- **POST** /api/supplier/vehicles — Create a new vehicle [CreateSupplierVehicleRequest ↪ → VehicleResponse · 201 400 401 403]
- **PUT** /api/supplier/vehicles/{id} — Update an existing supplier-owned vehicle [UpdateSupplierVehicleRequest → VehicleResponse · 200 400 401 403 404 409]
- **DELETE** /api/supplier/vehicles/{id} — Soft-delete a vehicle [— → VehicleResponse · 200 401 403 404]
- **PATCH** /api/supplier/vehicles/{id}/availability — Toggle vehicle availability [SupplierAvailabilityToggleRequest → VehicleResponse · 200 400 401 403 404 409]

- **POST** /api/supplier/vehicles/{id}/images — Upload an image for a vehicle [multipart/form-data → VehicleImageDto · 200 400 401 403 404 413]

4.6.10 Supplier Portal (Bookings)

- **GET** /api/supplier/bookings — List supplier's own bookings with filters [query → PagedResult<SupplierBookingListItemDto> · 200 401 403]
- **GET** /api/supplier/bookings/{id} — Get details of a single supplier-owned booking [— → SupplierBookingDetailsDto · 200 401 403 404]

4.6.11 Supplier Portal (Reviews)

- **GET** /api/supplier/reviews — List supplier's own reviews with filters [query → PagedResult<SupplierReviewListItemDto> · 200 401 403]
- **GET** /api/supplier/reviews/statistics — Get aggregate review statistics [— → SupplierReviewStatisticsDto · 200 401 403]
- **PUT** /api/supplier/reviews/{reviewId}/reply — Create or update supplier reply [SupplierReplyRequest → SupplierReviewListItemDto · 200 400 401 403 404]
- **POST** /api/supplier/reviews/{reviewId}/report — Flag review as inappropriate [SupplierReportReviewRequest → SupplierReviewListItemDto · 200 400 401 403 404]

4.6.12 Supplier Portal (Earnings)

- **GET** /api/supplier/earnings/stats — Get supplier earnings statistics [— → SupplierEarningsStatsDto · 200 401 403]
- **GET** /api/supplier/earnings/chart — Get monthly earnings chart [query → List<MonthlyRevenuePointDto> · 200 401 403]
- **GET** /api/supplier/earnings/top-vehicles — Get top 5 highest-earning vehicles [query → List<SupplierTopVehicleDto> · 200 401 403]

4.6.13 Supplier Portal (Notifications)

- **GET** /api/supplier/notifications — List supplier's notifications with filter [query → PagedResult<SupplierNotificationDto> · 200 401 403]

- **GET** /api/supplier/notifications/unread-count — Get unread notification count [— → { unreadCount: number } · 200 401 403]
- **PATCH** /api/supplier/notifications/{id}/read — Mark a single notification as read [— → · 200 401 403]
- **PATCH** /api/supplier/notifications/read-all — Mark all notifications as read [— → · 200 401 403]
- **DELETE** /api/supplier/notifications/{id} — Delete a single notification [— → · 200 401 403 404]

4.6.14 Supplier Portal (Dashboard)

- **GET** /api/supplier/dashboard/stats — Get supplier dashboard statistics [— → SupplierDashboardStatsDto · 200 401 403]
- **GET** /api/supplier/dashboard/bookings-by-status — Get bookings grouped by status [— → BookingsByStatusDto · 200 401 403]
- **GET** /api/supplier/dashboard/vehicle-status-distribution — Get vehicle counts by status [— → Dictionary<string, int> · 200 401 403]

4.6.15 Admin (User Management)

- **GET** /api/admin/users — Get paginated list of users with filters [query → UserManagementListResponse · 200 401 403]
- **GET** /api/admin/users/{id} — Get user by ID [— → UserManagementDto · 200 401 403 404]
- **POST** /api/admin/users — Create a new user account [CreateUserRequest → UserManagementResponse · 201 400 401 403]
- **PUT** /api/admin/users/{id} — Update an existing user account [UpdateUserRequest ↪ → UserManagementResponse · 200 400 401 403 404]
- **PATCH** /api/admin/users/{id}/toggle-status — Toggle user status between Active and Blocked [— → · 200 401 403 404]
- **POST** /api/admin/users/{id}/photo — Upload profile photo for a user [multi-part/form-data → · 200 400 401 403 404 413]
- **DELETE** /api/admin/users/{id} — Permanently delete a user [— → DeleteUserResponse · 200 401 403 404 409]

4.6.16 Admin (Supplier Management)

- **GET** /api/admin/suppliers — Get paginated list of suppliers [query → PagedResult<SupplierManagementDto> · 200 401 403]
- **GET** /api/admin/suppliers/{id} — Get supplier by ID [— → SupplierManagementDto ↪ · 200 401 403 404]
- **POST** /api/admin/suppliers — Create a new supplier account [CreateSupplierRequest ↪ → SupplierManagementResponse · 201 400 401 403]
- **PUT** /api/admin/suppliers/{id} — Update an existing supplier account [UpdateSupplierRequest → SupplierManagementResponse · 200 400 401 403 404]
- **DELETE** /api/admin/suppliers/{id} — Delete a supplier account [— → SupplierManagementResponse · 200 401 403 404]

4.6.17 Admin (Vehicle Management)

- **GET** /api/vehicles/admin — Search vehicles for Admin/Supplier dashboard [query → PagedResult<VehicleListDto> · 200 401 403]
- **POST** /api/vehicles — Create a new vehicle [CreateVehicleRequest → VehicleResponse · 201 400 401 403]
- **PUT** /api/vehicles/{id} — Update an existing vehicle [UpdateVehicleRequest → VehicleResponse · 200 400 401 403 404]
- **DELETE** /api/vehicles/{id} — Delete a vehicle [— → VehicleResponse · 200 401 403 404]
- **GET** /api/vehicles/admin/stats — Get aggregate vehicle statistics [— → AdminVehicleStatsDto · 200 401 403]
- **GET** /api/vehicles/{id}/bookings-check — Check if vehicle has active bookings [— → bool · 200 401 403 404]

4.6.18 Admin (Booking Management)

- **GET** /api/admin/bookings/stats — Get operational booking stats [— → AdminBookingStatsDto · 200 401 403]
- **GET** /api/admin/bookings/analytics — Get complex analytics and queues [— → AdminBookingAnalyticsDto · 200 401 403]

- **GET** /api/admin/bookings — Search bookings for Admin/Supplier dashboard [query → PagedResult<BookingListDto> · 200 401 403]
- **GET** /api/admin/bookings/{id} — Get booking details for Admin/Supplier dashboard [— → BookingDetailsDto · 200 401 403 404]
- **PATCH** /api/admin/bookings/{id}/status — Update booking status [UpdateBookingStatusRequest → · 200 400 401 403 404]
- **PUT** /api/admin/bookings/{id} — Edit a booking [UpdateBookingRequest → BookingDetailsDto · 200 400 401 403 404]
- **POST** /api/admin/bookings/delete — Bulk delete selected bookings [DeleteBookingsRequest → · 200 400 401 403]
- **GET** /api/admin/bookings/customers — Searchable customer picker [query → List<CustomerPickerItemDto> · 200 401 403]
- **GET** /api/admin/bookings/vehicles — Searchable available-vehicles picker [query → List<VehiclePickerItemDto> · 200 401 403]
- **POST** /api/admin/bookings/{id}/approve — Approve a booking [— → BookingResponse · 200 401 403 404]
- **POST** /api/admin/bookings/{id}/reject — Reject a booking [RejectBookingRequest ↪ → BookingResponse · 200 400 401 403 404]

4.6.19 Admin (Driver Management)

- **GET** /api/admin/drivers — List all drivers with optional status filter [query → List<AdminDriverListItemDto> · 200 401 403]
- **GET** /api/admin/drivers/pending — List pending driver verification requests [— → List<AdminDriverListItemDto> · 200 401 403]
- **GET** /api/admin/drivers/{id} — Get driver details with workload stats [— → AdminDriverDetailsDto · 200 401 403 404]
- **POST** /api/admin/drivers/{id}/approve — Approve a driver [— → · 200 401 403 404]
- **POST** /api/admin/drivers/{id}/reject — Reject a driver [AdminRejectDriverRequest ↪ → · 200 400 401 403 404]
- **POST** /api/admin/drivers/{id}/enable — Enable a driver [— → · 200 401 403 404]

- **POST** /api/admin/drivers/{id}/disable — Disable a driver [— → · 200 401 403 404]

4.6.20 Admin (Driver License Reviews)

- **GET** /api/admin/driver-licenses — List driver license requests for admin review [query → PagedResult<AdminDriverLicenseDto> · 200 401 403]
- **POST** /api/admin/driver-licenses/{id}/approve — Approve a driver license request [— → AdminDriverLicenseDto · 200 401 403 404]
- **POST** /api/admin/driver-licenses/{id}/reject — Reject a driver license request [AdminRejectDriverLicenseRequest → AdminDriverLicenseDto · 200 400 401 403 404]

4.6.21 Admin (Inspector Management)

- **GET** /api/admin/inspectors — List all inspectors with optional active filter [query → List<InspectorDto> · 200 401 403]
- **GET** /api/admin/inspectors/{id} — Get inspector details with workload stats [— → InspectorDetailsDto · 200 401 403 404]
- **POST** /api/admin/inspectors — Provision a new inspector [CreateInspectorRequest ↪ → InspectorDto · 201 400 401 403]
- **PATCH** /api/admin/inspectors/{id}/status — Enable/disable an inspector [UpdateInspectorStatusRequest → InspectorDto · 200 400 401 403 404]

4.6.22 Admin (Identity Verification Reviews)

- **GET** /api/admin/verifications — List verification requests for admin review [query → PagedResult<AdminVerificationDto> · 200 401 403]
- **POST** /api/admin/verifications/{id}/approve — Approve a verification request [— → AdminVerificationDto · 200 401 403 404]
- **POST** /api/admin/verifications/{id}/reject — Reject a verification request [AdminRejectVerificationRequest → AdminVerificationDto · 200 400 401 403 404]

4.6.23 Admin (Booking Inspections)

- **POST** /api/admin/bookings/{bookingId}/inspections/assign — Assign an inspector to booking [AssignInspectorRequest → InspectionDetailsDto · 200 400

401 403 404 409]

4.6.24 Admin (Driver Earnings)

- **GET** /api/admin/driver-earnings/{driverProfileId}/overview — Get driver earnings overview [— → AdminDriverEarningsOverviewDto · 200 401 403 404]
- **GET** /api/admin/driver-earnings/pending-payouts — List pending payouts [— → List<AdminDriverPayoutListItemDto> · 200 401 403]
- **GET** /api/admin/driver-earnings/pending-verification — List pending wallet verifications [— → List<AdminDriverPayoutListItemDto> · 200 401 403]
- **POST** /api/admin/driver-earnings/payouts/{payoutId}/approve — Approve a payout [— → · 200 401 403 404]
- **POST** /api/admin/driver-earnings/payouts/{payoutId}/reject — Reject a payout [AdminRejectPayoutRequest → · 200 400 401 403 404]
- **POST** /api/admin/driver-earnings/payouts/{payoutId}/retry — Retry a failed payout [— → · 200 401 403 404]
- **POST** /api/admin/driver-earnings/{driverProfileId}/verify-wallet — Verify wallet information [— → · 200 401 403 404]
- **GET** /api/admin/driver-earnings/{driverProfileId}/history — Get driver earnings history [query → List<DriverEarningRowDto> · 200 401 403 404]
- **GET** /api/admin/driver-earnings/platform-summary — Get platform-wide driver earnings summary [— → PlatformDriverEarningsSummaryDto · 200 401 403]

4.6.25 Admin (Commission Settings)

- **GET** /api/admin/commission/global — Get global commission percentage [— → { percentage: number } · 200 401 403]
- **PUT** /api/admin/commission/global — Update global commission percentage [UpdateGlobalCommissionRequest → { percentage: number } · 200 400 401 403]
- **GET** /api/admin/commission/driver/global — Get global driver commission percentage [— → { percentage: number } · 200 401 403]
- **PUT** /api/admin/commission/driver/global — Update global driver commission percentage [UpdateDriverCommissionRequest → { percentage: number } · 200 400 401 403]

4.6.26 Admin (Service Areas)

- **GET** /api/service-areas — List all service areas [— → List<ServiceAreaDto> · 200]
- **POST** /api/service-areas — Create a new service area [ServiceAreaDto → ServiceAreaDto · 201 400 401 403]
- **PUT** /api/service-areas/{id} — Update an existing service area [ServiceAreaDto ↪ → ServiceAreaDto · 200 400 401 403 404]
- **DELETE** /api/service-areas/{id} — Delete a service area [— → · 200 401 403 404]

4.6.27 Admin (Locations)

- **GET** /api/locations/autocomplete — Get location autocomplete suggestions [query → List<LocationSuggestionDto> · 200]
- **GET** /api/locations — Get paginated locations [query → PagedResult< ↪ LocationDto> · 200]
- **POST** /api/locations — Create a new location [CreateLocationRequest → LocationDto · 201 400 401 403]
- **PUT** /api/locations/{id} — Update an existing location [UpdateLocationRequest ↪ → LocationDto · 200 400 401 403 404]
- **GET** /api/locations/{id} — Get location by ID [— → LocationDto · 200 401 403 404]
- **DELETE** /api/locations/{id} — Delete a location [— → · 200 401 403 404]
- **POST** /api/locations/{id}/image — Upload an image for a location [multi-part/form-data → · 200 400 401 403 404 413]

4.6.28 Admin (Categories)

- **GET** /api/categories — List all categories [— → List<CategoryResponseDto> · 200]
- **GET** /api/categories/summary — Get category statistics [— → AdminCategoryStatsDto ↪ · 200 401 403]
- **GET** /api/categories/search — Search categories with filters [query → PagedResult<AdminCategoryListDto> · 200 401 403]

- **POST** /api/categories — Create a new category [CategoryDto → CategoryResponseDto
↪ · 201 400 401 403]
- **PUT** /api/categories/{id} — Update an existing category [CategoryDto → CategoryResponseDto · 200 400 401 403 404]
- **DELETE** /api/categories/{id} — Delete a category [— → · 200 401 403 404 409]
- **GET** /api/categories/{id}/details — Get category details with vehicle and booking stats [— → object · 200 401 403 404]
- **POST** /api/categories/bulk-assign — Bulk assign vehicles to category [BulkAssignDto → · 200 400 401 403]
- **POST** /api/categories/{id}/image — Upload an image for a category [multi-part/form-data → · 200 400 401 403 404 413]

4.6.29 Admin (Promotions)

- **GET** /api/admin/promotions — List discount codes with filters [query → PagedResult<DiscountCodeResponse> · 200 401 403]
- **POST** /api/admin/promotions — Create a new discount code [DiscountCodeCreateRequest
↪ → DiscountCodeResponse · 201 400 401 403]
- **PUT** /api/admin/promotions/{id} — Update an existing discount code [DiscountCodeUpdateRequest → DiscountCodeResponse · 200 400 401 403 404]
- **DELETE** /api/admin/promotions/{id} — Delete a discount code [— → · 200 401 403 404]
- **GET** /api/admin/promotions/{id}/analytics — Get discount code analytics [query → DiscountAnalyticsDto · 200 401 403 404]

4.6.30 Public (Promotions)

- **GET** /api/public/promotions/offers — List public offers [query → PagedResult<PublicOfferDto> · 200]
- **POST** /api/public/promotions/validate — Validate discount code [DiscountValidationRequest → DiscountValidationResponse · 200 400 401 429]
- **POST** /api/v1/promotions/apply — Apply discount to booking [DiscountApplyRequest
↪ → BookingDiscountResult · 200 400 401 429]

- **GET** /api/v1/promotions/automatic — Get automatic discounts [query → List
↪ <AutomaticDiscountDto> · 200 401]

4.6.31 Public (Landing Page)

- **GET** /api/public/landing — Get landing page content [— → LandingPageContentDto
↪ · 200]
- **GET** /api/public/suppliers — Get paginated list of public suppliers [query →
PagedResult<PublicSupplierDto> · 200]
- **GET** /api/public/suppliers/{id} — Get public supplier details [— →
PublicSupplierDto · 200 404]
- **GET** /api/public/destinations — Get popular destinations [query → List<
↪ PublicDestinationDto> · 200]

4.6.32 Inspector (Dashboard)

- **GET** /api/inspector/dashboard/today-stats — Get today's KPI stats [— →
InspectorTodayStatsDto · 200 401 403]
- **GET** /api/inspector/dashboard/today-tasks — Get today's enriched task list
[— → List<InspectorTaskDto> · 200 401 403]
- **GET** /api/inspector/dashboard/{inspectionId} — Get full details for one in-
spection [— → InspectionDetailsDto · 200 401 403 404]
- **PUT** /api/inspector/dashboard/{inspectionId}/draft — Update notes/odome-
ter/fuel/condition on draft [UpdateInspectionDraftRequest → InspectionDetailsDto
↪ · 200 400 401 403 404 409]
- **POST** /api/inspector/dashboard/{inspectionId}/images — Attach hosted im-
age URL to inspection [AddInspectionImageRequest → InspectionImageDto · 200
400 401 403 404]
- **POST** /api/inspector/dashboard/{inspectionId}/upload-images — Upload
raw image files [multipart/form-data → List<InspectionImageDto> · 200 400 401
403 404 413]
- **POST** /api/inspector/dashboard/{inspectionId}/submit — Submit inspection
(approve/reject) [SubmitInspectionRequest → InspectionDetailsDto · 200 400
401 403 404 409]

- **GET** /api/inspector/dashboard/history — Get submitted inspection history [— → List<InspectionDto> · 200 401 403]

4.6.33 Notifications (User)

- **GET** /api/notifications — Get all notifications for authenticated user [— → List<NotificationDto> · 200 401]
- **PATCH** /api/notifications/{id}/read — Mark a notification as read [— → · 200 401]
- **PUT** /api/notifications/{id}/read — Mark a notification as read (backwards compatible) [— → · 200 401]
- **PATCH** /api/notifications/read-all — Mark all notifications as read [— → · 200 401]
- **GET** /api/notifications/counter — Get count of unread notifications [— → { count: number } · 200 401 403]
- **POST** /api/notifications/seed — Seed dummy notifications for testing [— → · 200 401 403]
- **DELETE** /api/notifications/{id} — Delete a notification [— → · 200 401 404]

4.6.34 Payments

- **GET** /api/payments/history — Get payment history for authenticated user [query → PagedResult<PaymentDto> · 200 401]
- **GET** /api/payments/{transactionId} — Get details of specific payment [— → PaymentDto · 200 401 404]
- **GET** /api/payments/{transactionId}/receipt — Generate and download receipt [query → file · 200 401 404]
- **GET** /api/payments/pending — Get pending payment transactions [— → List< ↪ PaymentDto> · 200 401]
- **GET** /api/payments/failed — Get recent failed payment attempts [query → List<PaymentDto> · 200 401]
- **POST** /api/payments — Create a new payment for a booking [PaymentRequest → PaymentResponse · 200 400 401 403]

- **POST** /api/payments/initiate — Initiate payment with Paymob [InitiatePaymentRequest ↪ → PaymobInitiateResponse · 200 400 401 403]
- **POST** /api/payments/paymob-callback — Paymob callback endpoint [form → · 200]
- **POST** /api/payments/paymob-webhook — Paymob webhook endpoint [json → · 200]
- **POST** /api/payments/{bookingId}/refund — Admin refund [AdminRefundRequest ↪ → RefundResponse · 200 400 401 403 404]

4.6.35 Identity Verification

- **POST** /api/verifications — Submit identity verification documents [multi-part/form-data → UserVerificationDto · 200 400 401 413]
- **GET** /api/verifications — Get current user verification status [— → UserVerificationDto · 200 401 404]

4.6.36 Dashboard (Admin)

- **GET** /api/dashboard/summary — Get dashboard summary statistics [— → DashboardSummaryDto · 200 401 403]
- **GET** /api/dashboard/recent-summary — Get recent activity summary [— → List<RecentActivityItemDto> · 200 401 403]
- **GET** /api/dashboard/recent-bookings — Get recent bookings [query → List<↪ RecentBookingDto> · 200 401 403]
- **GET** /api/dashboard/upcoming-bookings — Get upcoming bookings [query → List<UpcomingBookingDto> · 200 401 403]
- **GET** /api/dashboard/revenue-week — Get revenue data for the week [— → List<RevenueDataPointDto> · 200 401 403]
- **GET** /api/dashboard/live-tracking — Get live tracking data [— → LiveTrackingDto · 200 401 403]
- **GET** /api/dashboard/top-vehicles — Get top performing vehicles [query → List<TopVehicleDto> · 200 401 403]
- **GET** /api/dashboard/system-status — Get system status [— → SystemStatusDto ↪ · 200]

- **GET** /api/dashboard/revenue-overview — Get revenue overview [query → RevenueOverviewDto · 200 401 403]

4.6.37 Content Management

- **GET** /api/about — Get all about sections [query → List<AboutSectionDto> · 200]
- **GET** /api/about/{id} — Get about section by ID [query → AboutSectionDto · 200 404]
- **POST** /api/about — Create a new about section [CreateAboutSectionRequest → AboutSectionDto · 201 400 401 403]
- **PUT** /api/about/{id} — Update an about section [UpdateAboutSectionRequest → AboutSectionDto · 200 400 401 403 404]
- **DELETE** /api/about/{id} — Delete an about section [— → · 200 401 403 404]
- **GET** /api/privacy — Get all privacy sections [query → List<PrivacySectionDto> ↪ > · 200]
- **GET** /api/privacy/{id} — Get privacy section by ID [query → PrivacySectionDto ↪ · 200 404]
- **POST** /api/privacy — Create a new privacy section [CreatePrivacySectionRequest ↪ → PrivacySectionDto · 201 400 401 403]
- **PUT** /api/privacy/{id} — Update a privacy section [UpdatePrivacySectionRequest ↪ → PrivacySectionDto · 200 400 401 403 404]
- **DELETE** /api/privacy/{id} — Delete a privacy section [— → · 200 401 403 404]
- **GET** /api/terms — Get all terms sections [query → List<TermsSectionDto> · 200]
- **GET** /api/terms/{id} — Get terms section by ID [query → TermsSectionDto · 200 404]
- **POST** /api/terms — Create a new terms section [CreateTermsSectionRequest → TermsSectionDto · 201 400 401 403]
- **PUT** /api/terms/{id} — Update a terms section [UpdateTermsSectionRequest → TermsSectionDto · 200 400 401 403 404]
- **DELETE** /api/terms/{id} — Delete a terms section [— → · 200 401 403 404]

4.6.38 Health Checks

- **GET** /api/health — Basic health check [— → HealthResponse · 200]
- **GET** /api/health/detailed — Detailed health check [— → DetailedHealthResponse
↪ · 200 503]



Figure 4.7: API controller domains and their relationships

4.7 Architecture Decision Records (ADRs)

4.7.1 1. Clean Architecture with Layered Separation

Decision: Adopt Clean Architecture with clear separation of concerns across Presentation (API), Application, Domain, and Infrastructure layers. **Rationale:** - The repository structure (backend/Api, backend/Application, backend/Domain, backend/↪ Infrastructure) evidences a deliberate separation of concerns. - Controllers in Api/Controllers depend on interfaces defined in Application/Interfaces, not on concrete implementations. - Domain entities (Domain/Entities) contain no infrastructure concerns (e.g., no DbContext references). - Infrastructure layer (Infrastructure/↪ Repositories) implements interfaces and handles persistence, email, and external services. - This separation enables independent testing, easier maintenance, and clearer boundaries between business logic and technical details.

4.7.2 2. MediatR for CQRS and Vertical Slice Architecture

Decision: Use MediatR to implement Command Query Responsibility Segregation (CQRS) and vertical slice architecture. **Rationale:** - MediatR is registered in Program.cs (builder.Services.AddMediatR). - Inspection-related endpoints in InspectorDashboardController dispatch MediatR commands and queries (e.g., GetInspectorTodayStatsQuery, GetInspectorTasksQuery). - This pattern decouples request handling from controllers, enabling independent scaling of reads and writes. - Vertical slices (e.g., GetInspectorTodayStatsQueryHandler) colocate all logic for a

single feature, improving maintainability and reducing cross-cutting changes.

4.7.3 3. JWT Authentication with Role-Based Authorization

Decision: Implement JWT-based authentication with role-based authorization policies. **Rationale:** - JWT configuration is visible in `Program.cs` (`AddAuthentication`, `AddJwtBearer`). - Controllers use `[Authorize]` and `[Authorize(Roles = "Admin")]` attributes to enforce role-based access. - `AuthController` issues JWT tokens with role claims (`LoginResponse` includes `user.roles`). - Role-based policies are enforced at the controller level, ensuring security is applied consistently across all endpoints. - This approach is scalable, stateless, and integrates seamlessly with frontend authentication libraries.

4.7.4 4. FluentValidation for Input Validation

Decision: Use `FluentValidation` for all input validation at the API boundary. **Rationale:** - `FluentValidation` is registered in `Program.cs` (`AddValidatorsFromAssembly`). - Every request DTO has a corresponding validator (e.g., `RegisterRequestValidator`, `CreateBookingRequestValidator`). - Validators are injected into controllers and executed before business logic, ensuring invalid data never reaches the service layer. - Validation rules are expressive, reusable, and maintainable, reducing boilerplate in controllers. - Error responses follow a consistent schema (`ValidationError`), improving frontend error handling.

4.7.5 5. AutoMapper for DTO Mapping

Decision: Use `AutoMapper` to map between domain entities and DTOs. **Rationale:** - `AutoMapper` is registered in `Program.cs` (`AddAutoMapper`). - Mapping profiles (e.g., `AuthMappingProfile`) define transformations between entities and DTOs. - This reduces repetitive mapping code in services and controllers, improving maintainability. - `AutoMapper` supports complex nested mappings (e.g., `BookingDetailsDto` includes customer, vehicle, payment, and inspection data).

4.7.6 6. Rate Limiting Middleware

Decision: Implement custom rate limiting middleware to prevent abuse of sensitive endpoints. **Rationale:** - `RateLimitingMiddleware` is registered in `Program.cs` (app.
↳ `UseMiddleware<RateLimitingMiddleware>()`). - Rate limits are applied to `/api/auth`
↳ `/login` (5 attempts/15 minutes), `/api/auth/register` (5 attempts/hour), and discount validation endpoints. - The middleware uses an in-memory store (`ConcurrentDictionary`) to track request counts per IP and endpoint. - This protects against brute-force attacks and ensures fair usage of public endpoints.

4.7.7 7. Global Exception Handling Middleware

Decision: Implement global exception handling middleware to standardize error responses. **Rationale:** - `GlobalExceptionHandlerMiddleware` is registered in `Program.cs`
↳ (app.`UseMiddleware<GlobalExceptionHandlerMiddleware>()`). - The middleware catches exceptions and returns consistent error responses (`ErrorResponse`). - Different exception types map to appropriate HTTP status codes (e.g., `NotFoundException` → 404, `ConflictException` → 409). - This removes repetitive try-catch blocks from controllers and ensures consistent error responses across the API.

4.7.8 8. Repository Pattern with Entity Framework Core

Decision: Use the Repository pattern with Entity Framework Core for data access. **Rationale:** - Generic `IRepository<T>` and `IPaginatedRepository<T>` interfaces are defined in `Application/Interfaces`. - Concrete implementations (`Repository<T>`, `PaginatedRepository<T>`) are in `Infrastructure/Repositories`. - Specialized repositories (e.g., `IBookingRepository`, `IVehicleRepository`) extend the generic interfaces with domain-specific queries. - This abstraction decouples business logic from EF Core, enabling easier testing and potential future database changes.

4.7.9 9. Soft Delete Pattern

Decision: Implement soft delete for entities (e.g., vehicles, users) using an `IsActive` flag. **Rationale:** - `Vehicle` entity includes an `IsActive` boolean field. - Soft-deleted vehicles are excluded from public listings and admin searches (`query = query.Where(`
↳ `v => v.IsActive)`). - This preserves historical data (bookings, payments, reviews)

while removing entities from active use. - Hard deletes are reserved for admin operations (IUserDeletionService) with strict validation to prevent data loss.

4.7.10 10. Role-Based Access Control (RBAC)

Decision: Implement role-based access control with distinct roles: Admin, Supplier, Customer, Driver, Inspector. **Rationale:** - Roles are defined in the database and assigned to users (AspNetUserRoles). - Controllers use [Authorize(Roles = "Admin")] to restrict access. - Role-based policies are enforced at the service level (e.g., SupplierBookingsController filters by Vehicle.UserId). - This enables fine-grained access control and supports the multi-tenant nature of the platform.

4.7.11 11. Rate Limiting for Authentication Endpoints

Decision: Apply rate limiting to authentication endpoints to prevent brute-force attacks. **Rationale:** - RateLimitingMiddleware enforces 5 login attempts per 15 minutes per IP. - Registration is limited to 5 attempts per hour per IP. - This protects against credential stuffing and automated registration bots. - Rate limits are configurable and documented in the code, allowing for adjustments based on security requirements.

4.7.12 12. Token Rotation for Refresh Tokens

Decision: Implement token rotation for refresh tokens to enhance security. **Rationale:** - AuthService.RefreshTokenAsync revokes the old refresh token and issues a new one. - This prevents token replay attacks and ensures that a compromised refresh token cannot be used indefinitely. - The frontend must handle token rotation by updating its stored refresh token after each refresh.

4.7.13 13. Google OAuth Integration

Decision: Integrate Google OAuth for social sign-in/sign-up. **Rationale:** - GoogleAuthService validates Google ID tokens and handles account creation/linking. - AuthController.GoogleSignIn endpoint accepts Google ID tokens and returns JWT tokens. - This reduces friction for users and leverages Google's authentication infrastructure for security.

4.7.14 14. Paymob Payment Gateway Integration

Decision: Use Paymob as the payment gateway for processing payments. **Rationale:** - PaymobClient implements IPaymobClient with methods for payment initiation, refunds, and disbursements. - Payment endpoints (PaymentsController.InitiatePayment, PaymobCallback, PaymobWebhook) handle Paymob integration. - This enables secure payment processing and supports features like refunds and payouts to drivers.

4.7.15 15. Driver Module with Professional Profile

Decision: Implement a dedicated Driver module with professional profile requirements. **Rationale:** - DriverProfile entity includes fields for license number, expiry date, service areas, and payout information. - DriverProfileController provides endpoints for completing and updating driver profiles. - Driver-specific endpoints (e.g., DriverAssignmentsController, DriverDashboardController) support driver workflows. - This enables the platform to support professional drivers as a distinct user type with unique requirements.

4.7.16 16. Supplier Portal with Scoped Access

Decision: Implement a Supplier portal with scoped access to supplier-owned data. **Rationale:** - Supplier endpoints (e.g., SupplierBookingsController, SupplierVehiclesController) filter data by the authenticated supplier's ID. - Suppliers cannot access data owned by other suppliers, ensuring data isolation. - This supports the multi-tenant nature of the platform and enables suppliers to manage their own fleet and bookings.

4.7.17 17. Identity Verification Workflow

Decision: Implement an identity verification workflow for customers. **Rationale:** - Verification entity tracks document submissions and review status. - VerificationsController allows users to submit documents and check status. - Admin endpoints (AdminVerificationsController) enable review and approval/rejection. - This ensures compliance with regulatory requirements and reduces fraud.

4.7.18 18. Driver License Verification Workflow

Decision: Implement a driver license verification workflow for drivers. **Rationale:** - DriverLicense entity tracks license submissions and review status. - DriverLicenseController allows drivers to submit and update licenses. - Admin endpoints (AdminDriverLicensesController) enable review and approval/rejection. - This ensures that only verified drivers can be assigned to bookings, improving safety and compliance.

4.7.19 19. Inspection Workflow for Bookings

Decision: Implement a vehicle inspection workflow for bookings. **Rationale:** - VehicleInspection entity tracks pre-pickup and post-return inspections. - InspectorDashboardController provides endpoints for inspectors to view, update, and submit inspections. - Admin endpoints (AdminBookingInspectionsController) enable inspector assignment. - This ensures vehicle condition is documented before and after each rental, reducing disputes and improving accountability.

4.7.20 20. Driver Earnings and Payouts

Decision: Implement a driver earnings and payout system. **Rationale:** - DriverEarning entity tracks earnings per booking. - DriverPayout entity tracks payout requests and status. - DriverEarningsController provides endpoints for drivers to view earnings and request payouts. - Admin endpoints (AdminDriverEarningsController) enable review and approval of payouts. - This enables drivers to earn income from the platform and supports financial transparency.

4.7.21 21. Supplier Earnings and Analytics

Decision: Implement supplier earnings and analytics dashboards. **Rationale:** - SupplierEarningsController provides endpoints for earnings statistics, monthly charts, and top vehicles. - SupplierDashboardController provides endpoints for dashboard statistics and vehicle status distribution. - All data is scoped to the authenticated supplier, ensuring data isolation. - This enables suppliers to track their performance and make informed business decisions.

4.7.22 22. Review System with Supplier Replies

Decision: Implement a review system with support for supplier replies. **Rationale:**

- Review entity tracks customer reviews for vehicles.
- SupplierReview entity tracks supplier replies and reports.
- ReviewsController provides endpoints for customers to create and update reviews.
- SupplierReviewsController provides endpoints for suppliers to list reviews, reply, and report inappropriate content.
- This enables feedback-driven improvement and supports supplier engagement with customers.

4.7.23 23. Content Management System (CMS)

Decision: Implement a lightweight CMS for static content (About, Privacy, Terms). **Rationale:**

- AboutController, PrivacyController, TermsController provide endpoints for managing content sections.
- Content is stored in the database and can be localized.
- This enables non-technical users to update static content without code changes.

4.7.24 24. Health Check Endpoints

Decision: Implement health check endpoints for monitoring. **Rationale:**

- HealthController provides /api/health and /api/health/detailed endpoints.
- These endpoints return application status, version, and component health.
- This enables monitoring systems to track application health and uptime.

4.7.25 25. Swagger/OpenAPI Documentation

Decision: Use Swagger/OpenAPI for API documentation. **Rationale:**

- Swagger is configured in Program.cs (AddSwaggerGen).
- XML comments from controllers and DTOs are included in the documentation.
- Custom operation filters (AuthenticationRequirementOperationFilter, RateLimitOperationFilter) enhance documentation with authentication and rate limiting information.
- This provides interactive API documentation for developers and supports API-first development.

4.7.26 26. Environment-Based Configuration

Decision: Use environment-based configuration with appsettings.json and appsettings.{Environment}.json. **Rationale:**

- Configuration is loaded based on the ASPNETCORE_ENVIRONMENT variable.
- Sensitive settings (e.g., JWT secret, Paymob

API keys) are configured per environment. - This enables environment-specific configurations and supports secure deployment practices.

4.7.27 27. Serilog for Structured Logging

Decision: Use Serilog for structured logging. **Rationale:** - Serilog is configured in Program.cs (UseSerilog). - Logs include contextual information (e.g., request host, IP address, user ID). - This enables centralized logging, easier debugging, and better observability.

4.7.28 28. Database Migrations with Entity Framework Core

Decision: Use Entity Framework Core migrations for database schema management. **Rationale:** - Migrations are applied on startup (db.Database.MigrateAsync()). - This ensures the database schema is always in sync with the application model. - Migrations support versioned schema changes and enable rollback if needed.

4.7.29 29. Seeding Demo Data

Decision: Seed demo data on startup when enabled. **Rationale:** - DbInitializer. InitializeAsync seeds roles, demo users, and sample data. - Demo data is enabled via SEED_DEMO_DATA environment variable or --seed-only command-line argument. - This enables rapid development and testing without manual data setup.

4.7.30 30. CORS Configuration

Decision: Configure CORS to restrict API access to allowed origins. **Rationale:** - CORS policy (AllowFrontend) is configured in Program.cs. - Allowed origins are specified in appsettings.json. - This prevents unauthorized frontend applications from accessing the API and enhances security.

4.8 Key Algorithms and Workflows (Mermaid)

This section presents the core workflows implemented in the Ares platform, documented as narrow Mermaid flowcharts that compile via mmdc to fit the A4 column. Each diagram reflects the actual service logic evidenced in the repository context.

4.8.1 Booking Checkout Workflow

The checkout workflow spans CheckoutService and BookingService, implementing a staged reservation pattern with optimistic pre-checks and a final pessimistic lock to prevent double-booking.

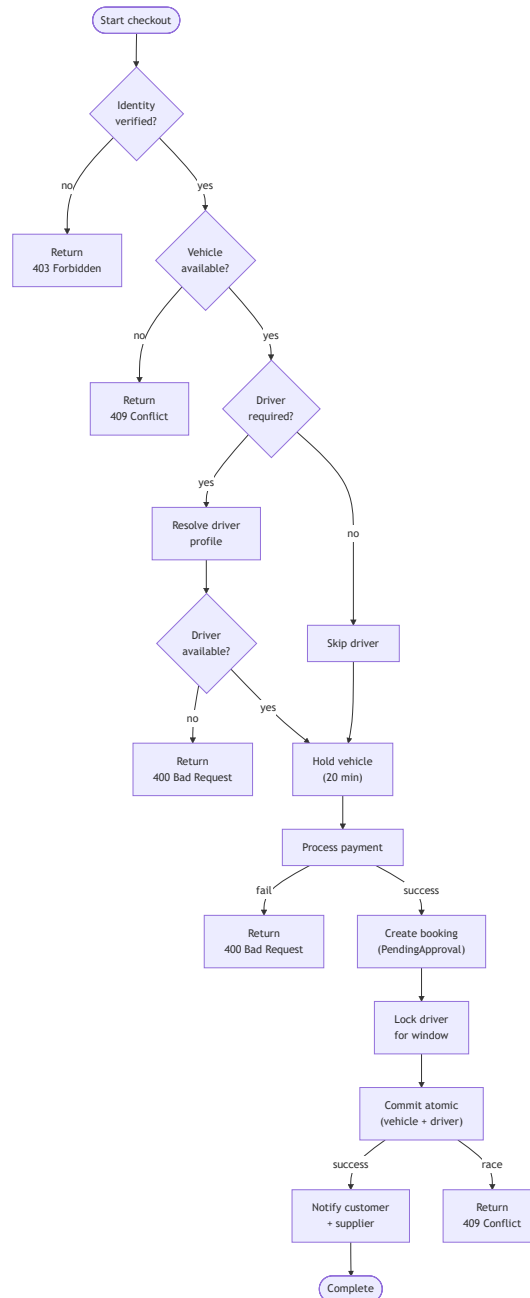


Figure 4.8: Booking checkout workflow

4.8.1.1 Key Steps

1. **Identity Gate** – Only customers with an approved identity verification (`Verification.Status == "Approved"`) may proceed.
 2. **Vehicle Availability** – Optimistic pre-check via `IVehicleRepository`.
↳ `IsAvailableAsync`.
 3. **Driver Resolution** – When the customer lacks an approved license (`Driver` entity), a driver is mandatory (`Booking.NeedDriver = true`). The selected driver is validated for overlapping assignments via `IDriverProfileRepository`.
↳ `HasOverlappingAssignmentAsync`.
 4. **Vehicle Hold** – A 20-minute hold (`Booking.HoldExpiresAt`) is placed, reserving the vehicle for the customer.
 5. **Payment Simulation** – `CheckoutService.SimulatePaymentAsync` mirrors `PaymentService` logic, always succeeding in this environment.
 6. **Atomic Commit** – `IBookingRepository.ReserveVehicleAtomicAsync` opens a serializable transaction, re-checks availability under a `UPDLOCK, HOLDLOCK`, and commits the booking, payment, and driver lock atomically. Concurrent checkouts for the same vehicle/window result in a 409 Conflict for the loser.
 7. **Notifications** – Best-effort notifications (`BookingCompletedEvent`) are fired via `MediatR`.
-

4.8.2 Payment Processing Workflow

The payment workflow (`PaymentService`) integrates with `Paymob` for real transactions, enforcing business rules such as mandatory driver assignment before payment completion.

4.8.2.1 Key Steps

1. **Mandatory Driver Gate** – A booking requiring a driver (`Booking.NeedDriver = true`) must have `Booking.AssignedDriverProfileId` populated before payment can complete.
2. **Paymob Integration** –

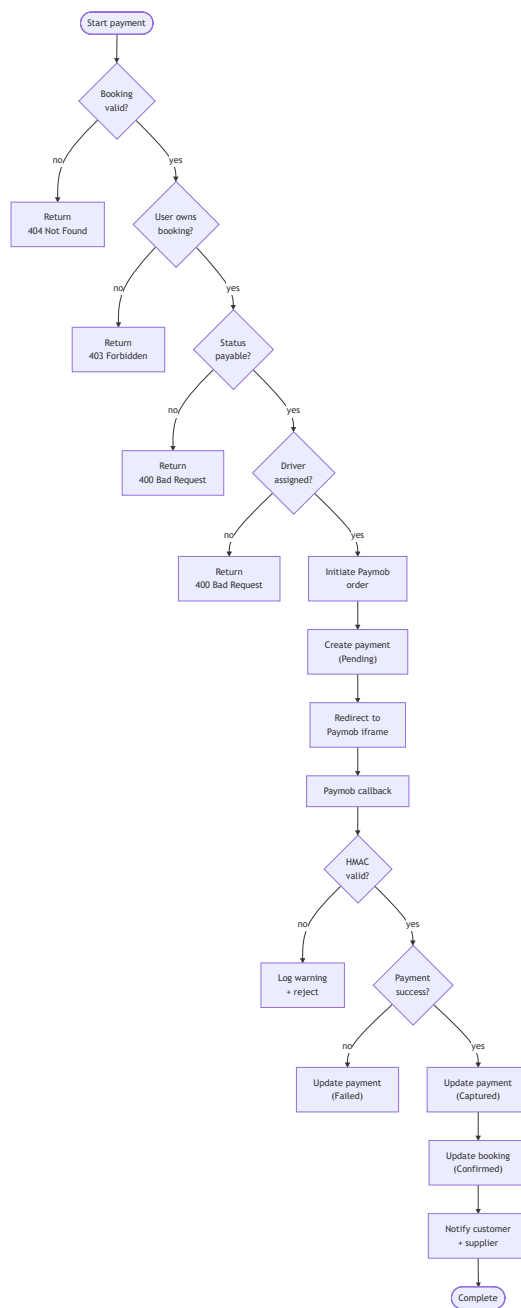


Figure 4.9: Payment processing workflow

- `InitiatePaymentAsync` fetches an auth token, creates an order, and requests a payment key.
- The customer is redirected to Paymob's iframe (`iframeUrl`).
- `ProcessPaymobCallbackAsync` verifies the HMAC signature using `PaymobSettings.HmacSecret` and updates the payment status.

3. **Atomic Updates** – The booking status transitions from `PendingApproval` to `Confirmed` only after the payment is captured.
4. **Notifications** – Best-effort notifications (`BookingCompletedEvent`) are published via `MediatR`.

4.8.3 Refund Calculation Workflow

The refund workflow (`RefundCalculator`) implements a sliding-scale cancellation policy based on the time remaining until pickup.

4.8.3.1 Key Steps

1. **Policy Windows** –
 - **Free:** >7 days until pickup → 100% refund.
 - **Standard:** 3–7 days → 70% refund.
 - **Strict:** <3 days → 30% refund.
2. **Cancellation Fee** – The fee is the difference between the total amount and the refund amount.
3. **Refund Initiation** – `PaymentService.RefundAsync` calls Paymob's refund API (`/api/acceptance/void_refund/refund`) with the calculated amount.

4.8.4 Vehicle Inspection and Assignment Workflow

The inspection workflow (`InspectionService`) and auto-assignment background service (`VehicleInspectionAutoAssignmentBackgroundService`) coordinate pickup and return inspections.

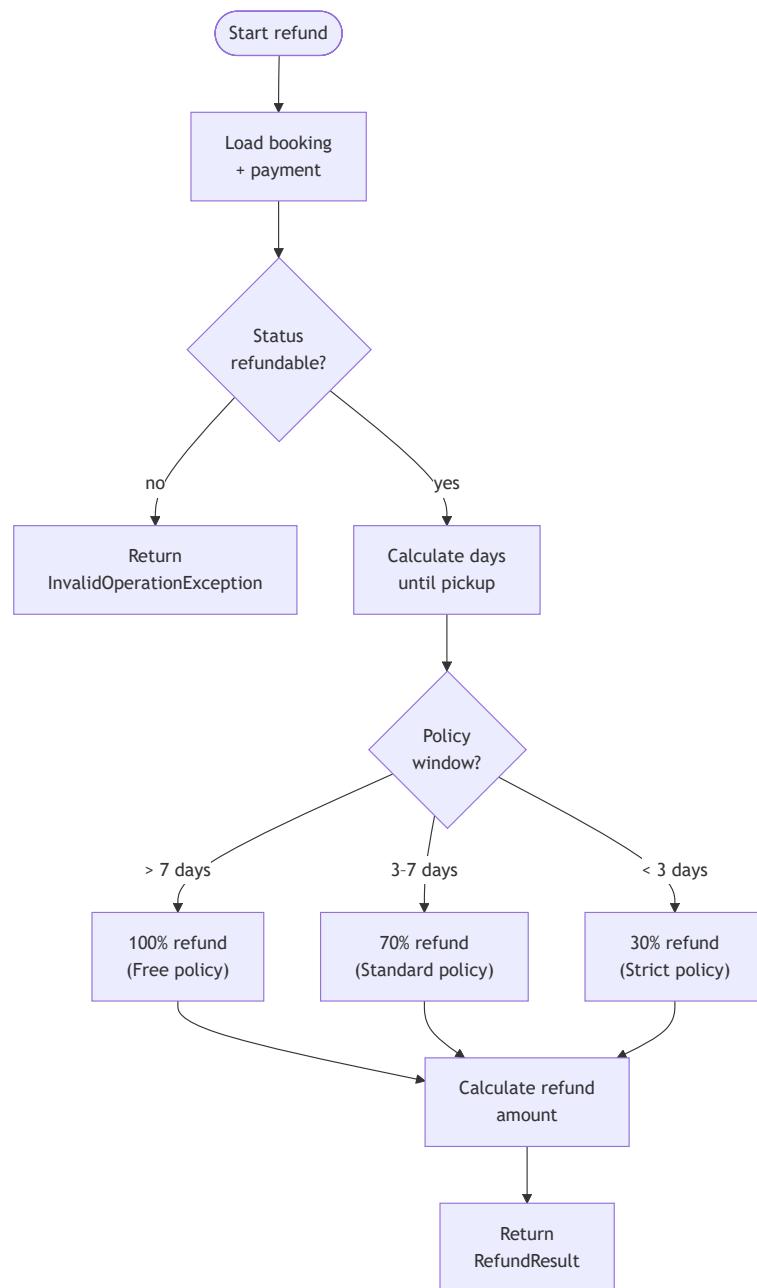


Figure 4.10: Refund calculation workflow

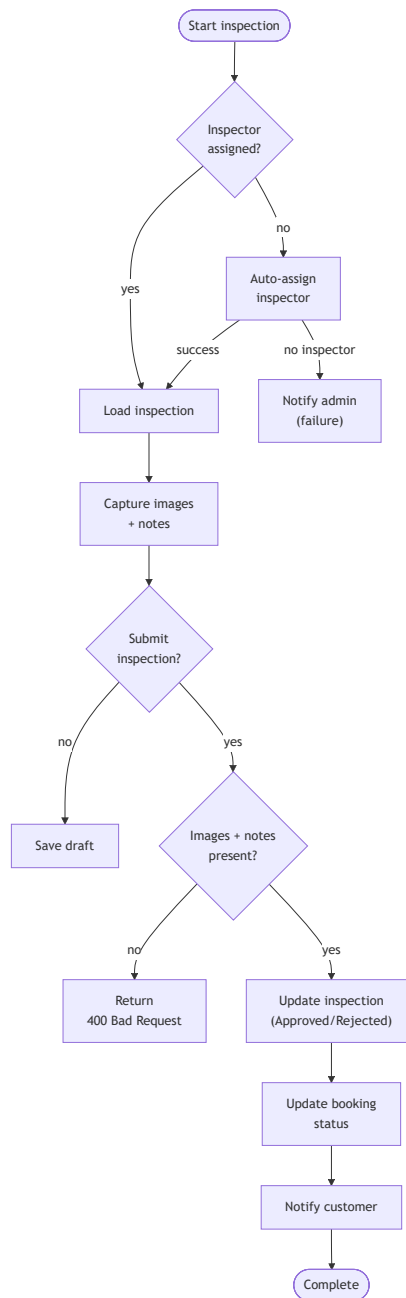


Figure 4.11: Vehicle inspection and assignment workflow

4.8.4.1 Key Steps

1. **Auto-Assignment** – The background service (`VehicleInspectionAutoAssignmentBackgroundService`) assigns inspectors to bookings 24 hours before pickup/return via `AssignInspectorCommand`.
 2. **Inspection Submission** – Inspectors must attach at least one image and provide notes before submission.
 3. **Booking State Transitions** –
 - Pickup inspection approved → `Booking.Status = Confirmed`.
 - Pickup inspection rejected → `Booking.Status = Cancelled`.
 - Return inspection approved → `Booking.Status = Completed`.
 4. **Notifications** – Best-effort notifications are sent to customers and admins via `INotificationService`.
-

4.8.5 Discount Validation and Application Workflow

The discount workflow (`DiscountValidationService` and `DiscountApplicationService`) enforces eligibility rules and stacking constraints.

4.8.5.1 Key Steps

1. **Eligibility Rules** –
 - Customer segment (e.g., "new" or "returning").
 - Vehicle category (discounts may be restricted to specific categories).
 - Stacking constraints (`Discount.AllowStacking`).
2. **Atomic Application** – `DiscountApplicationService.ApplyDiscountToBookingAsync` uses a transaction to update `Booking.AppliedDiscountCodes` and increment `Discount.UsageCount`.
3. **Validation Logging** – Each validation attempt is logged via `DiscountValidationLog`.

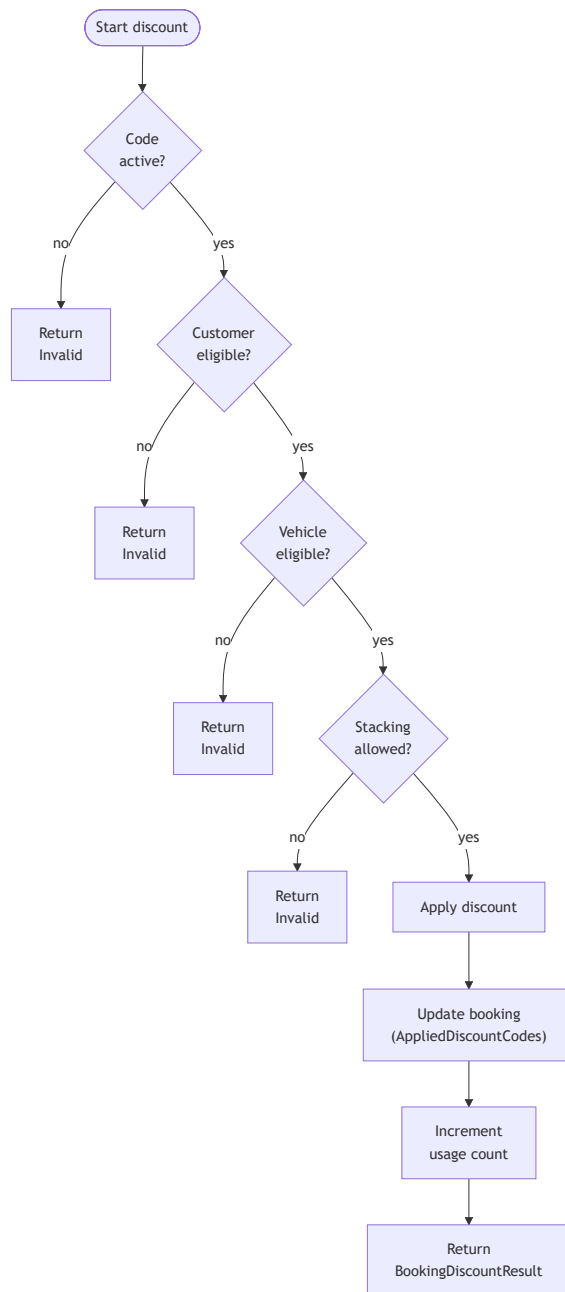


Figure 4.12: Discount validation and application workflow

4.9 Supporting Service Workflows

This section documents the business logic and workflow patterns implemented by the platform's secondary services, organized by domain. Each workflow is derived from the actual service implementations evidenced in the codebase.

4.9.1 Authentication & Authorization Workflows

4.9.1.1 User Registration & Profile Completion

The registration workflow (@fig-3-4-registration) begins when a user submits credentials via `AuthService.RegisterAsync`. The service: 1. Validates that the email is not already registered 2. Creates a new `ApplicationUser` with status "Pending" and assigns the requested role (Customer, Supplier, or Driver) 3. Generates an email verification token and sends a confirmation link 4. For Driver registrations, creates an empty `DriverProfile` with status "Incomplete" 5. Notifies admins of the new registration (best-effort)

After registration, users must complete their profile via `UserProfileService`. ↪ `UpdateProfileAsync` before accessing role-specific features. Profile completion includes: - Personal details (name, phone, address) - Emergency contact information - Language and currency preferences - For Drivers: license upload, work areas, and payment information

4.9.1.2 Identity Verification

Users must verify their identity via `VerificationService.SubmitVerificationAsync` before booking vehicles or listing vehicles for rent. The workflow: 1. Validates that the user has no approved verification 2. Validates that no pending verification exists 3. Validates document files (type, size) 4. Saves files to `wwwroot/uploads/verifications` 5. Creates a `Verification` record with status "Pending" 6. Notifies admins of the new submission (best-effort)

Admins review submissions via `VerificationService.ApproveVerificationAsync` or `RejectVerificationAsync`, which: - Updates the verification status - Notifies the user (best-effort) - For approvals, enables booking capabilities

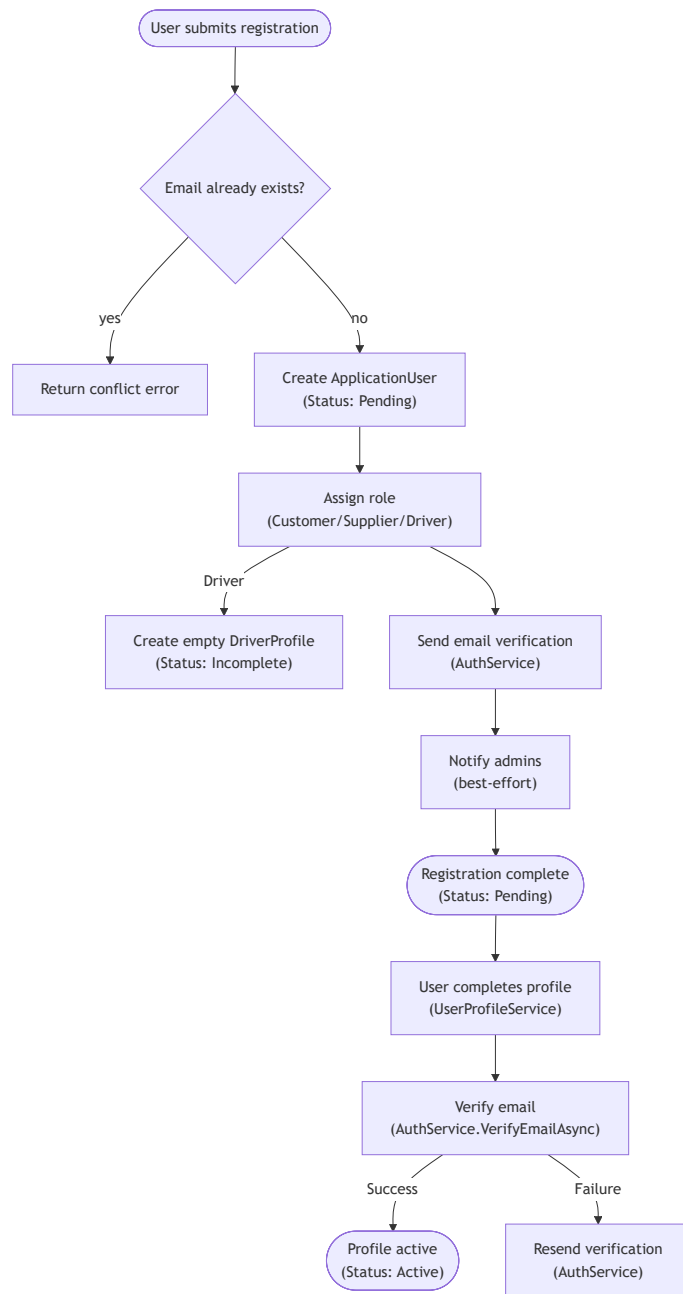


Figure 4.13: User registration and profile completion workflow

4.9.1.3 Driver License Verification

Drivers must submit their license via `DriverLicenseService.SubmitOrUpdateAsync`, which: 1. Validates license number and expiry date 2. Validates the license image file 3. Saves the image to `wwwroot/uploads/driver-licenses` 4. Creates or updates a Driver record with status “Pending” 5. Notifies admins of the new submission (best-effort)

Admins review licenses via `DriverLicenseService.ApproveDriverLicenseAsync` or `RejectDriverLicenseAsync`, which update the verification status and notify the driver.

4.9.1.4 Google OAuth Integration

The `GoogleAuthService.SignInWithGoogleAsync` workflow: 1. Validates the Google ID token 2. Attempts to find an existing user by GoogleId or email 3. Creates a new user if none exists, linking the Google account 4. Issues JWT and refresh tokens via `JwtTokenService` 5. For Drivers, ensures a `DriverProfile` exists (self-healing)

4.9.1.5 Token Management

The `JwtTokenService` handles: - Token generation with role claims - Refresh token rotation via `AuthService.RefreshTokenAsync` - Token revocation via `AuthService`.

↪ `RevokeTokenAsync`

Refresh tokens are stored in the database and rotated on each use, with a 90-day TTL for audit trail retention.

4.9.2 Driver Management Workflows

4.9.2.1 Driver Profile Submission & Approval

Drivers complete their profile via `DriverProfileService.CompleteProfileAsync`, which: 1. Validates that the profile is incomplete or rejected 2. Validates required documents (license, ID, address proof) 3. Saves documents to `wwwroot/uploads/driver` ↪ `-documents` 4. Sets profile status to “PendingVerification” 5. Notifies admins of the new submission (best-effort)

Admins review profiles via `AdminDriverService.ApproveDriverAsync` or `RejectDriverAsync` ↪ , which: - Updates the profile status - For approvals, sets availability to “Available” - Notifies the driver (best-effort)

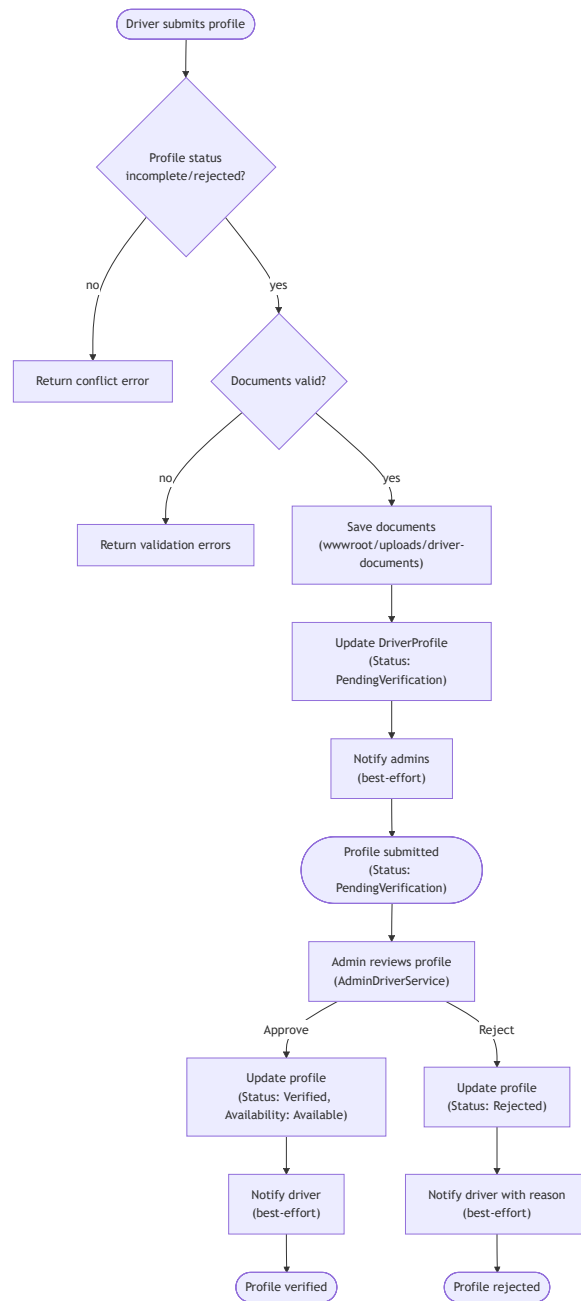


Figure 4.14: Driver profile submission and approval workflow

4.9.2.2 Driver Assignment

The `DriverAssignmentService` handles:

- Listing available drivers for a booking (`GetAvailableDriversForBookingAsync`)
- Selecting a driver (`SelectDriverAsync` ↪)
- Changing or cancelling a driver (`ChangeDriverAsync`, `CancelDriverAsync`)
- Driver-initiated cancellation (`DriverCancelAssignmentAsync`)

Key business rules:

- Drivers must be verified and available for the booking window
- Overlapping assignments are prevented via `IDriverProfileRepository`.
↪ `HasOverlappingAssignmentAsync`
- The 24-hour rule prevents changes within 24 hours of pickup
- Driver fees are calculated via `DriverPricingService`

4.9.2.3 Driver Earnings & Payouts

The `DriverEarningsService` manages:

- Earnings calculation for completed bookings (`CreateEarningForBookingAsync`)
- Earnings reversal for cancelled bookings (`ReverseEarningForBookingAsync`)
- Payout requests (`RequestPayoutAsync`)
- Payout history and statistics

Earnings are calculated as:

$$\text{DailyRate} \times \text{NumberOfDays} \times (1 - \text{PlatformCommission})$$

Platform commission is configurable via `driver.commission_percentage` system setting (default: 20%).

Payouts are processed via `DriverEarningsAdminService.ApprovePayoutAsync`, which:

1. Validates the driver's wallet information
2. Initiates a Paymob disbursement
3. Updates payout status to "Completed"
4. Updates associated earnings to "Paid"
5. Notifies the driver (best-effort)

4.9.3 Supplier Management Workflows

4.9.3.1 Vehicle Listing & Approval

Suppliers list vehicles via `SupplierVehicleService.CreateVehicleAsync`, which:

1. Validates the license plate is unique
2. Creates a `Vehicle` with status "Pending"
3. Saves images to `wwwroot/uploads/vehicles`
4. Notifies admins of the new submission (best-effort)

Admins review vehicles via `VehicleService.UpdateVehicleAsync`, which: - Updates the vehicle status (“Approved” or “Rejected”) - For approvals, sets availability to “Available” - Notifies the supplier (best-effort)

4.9.3.2 Supplier Restrictions

The `SupplierRestrictionService` handles: - Applying restrictions (`ApplyRestrictionAsync` ↪) - Removing restrictions (`RemoveRestrictionAsync`) - Converting restricted suppliers to blocked (`CheckAndConvertToBlockedAsync`)

When a restriction is applied: 1. All supplier vehicles are disabled 2. Future bookings are cancelled and refunded 3. Customers are notified 4. The supplier’s status is set to “Restricted” or “Blocked” based on active bookings

4.9.3.3 Supplier Notifications

The `SupplierNotificationService` provides a dedicated notification feed for suppliers, with deep links to relevant entities (vehicles, bookings). Notifications are generated by: - Vehicle status changes (`SupplierVehicleService`) - New bookings (`BookingService`) - Booking status changes (`BookingService`) - Review replies (`SupplierReviewService`)

Notification types are encoded in the `Notification.Type` field using `SupplierNotificationTypes` ↪ `.Format`, e.g., `"VehicleApproved:8a3b..."`.

4.9.4 Notification Workflows

4.9.4.1 Notification Delivery

The `NotificationService` handles: - Creating notifications (`CreateNotificationAsync` ↪) - Marking notifications as read - Retrieving user notifications - Fan-out to admins (`NotifyAdminsAsync`)

Notifications are stored in the `Notifications` table with fields: - `UserId`: Recipient - `Title/Message`: Display text - `Type`: Structured tag (e.g., `"BookingApproved"`, `"VehiclePendingReview:8a3b..."`) ↪ - `RedirectUrl`: Deep link to the relevant entity - `IsRead`: Read status

4.9.4.2 Driver Notifications

The `DriverNotificationService` provides a thin wrapper over `NotificationService` for driver-specific notifications, including: - Approval/rejection notifications - Assignment notifications - Earnings notifications - Payout notifications

All driver notifications are best-effort and wrapped in try-catch blocks to avoid breaking business operations.

4.9.5 Review Workflows

4.9.5.1 Customer Reviews

Customers submit reviews via `ReviewService.CreateReviewAsync`, which: 1. Validates the booking is completed and belongs to the customer 2. Validates the review is submitted within 14 days of completion 3. Creates a Review record 4. Notifies admins (best-effort)

4.9.5.2 Supplier Review Management

Suppliers manage reviews via `SupplierReviewService`: - Listing reviews with filters (rating, reply status, date range) - Replying to reviews (`SaveReplyAsync`) - Reporting reviews (`ReportReviewAsync`)

Replying to a review: 1. Updates the `Review.SupplierReply` field 2. Notifies the customer (best-effort)

4.9.5.3 Driver Reviews

Customers review drivers via `DriverReviewService.CreateReviewAsync`, which: 1. Validates the booking is completed and has an assigned driver 2. Creates a `DriverReview` record 3. Updates driver rating statistics

4.9.6 Commission & Pricing Workflows

4.9.6.1 Commission Calculation

The `CommissionService` calculates commission for each booking: 1. Retrieves the effective commission percentage for the vehicle - Checks for category-specific

commission - Falls back to global commission setting (`global.commission`) - Defaults to 15% if no settings are configured 2. Calculates commission amount and supplier amount:
 $\text{CommissionAmount} = \text{TotalPrice} \times (\text{CommissionPercentage} / 100)$ $\text{SupplierAmount} \hookrightarrow = \text{TotalPrice} - \text{CommissionAmount}$

4.9.6.2 Discount Application

The `DiscountApplicationService` applies discounts to bookings: 1. Validates the discount code via `DiscountValidationService` 2. Checks for existing discounts and stacking rules 3. Updates the booking's `AppliedDiscountCodes` field 4. Creates a `DiscountUsage` record 5. Increments the discount's `CurrentUsageCount`

The `DiscountValidationService` validates discounts based on: - Active status - Validity dates - Customer segment - Vehicle category - Usage limits

4.9.6.3 Automatic Discounts

The `AutomaticDiscountService` applies automatic discounts during vehicle search: 1. Determines the customer's segment 2. Retrieves active automatic discounts 3. Filters discounts by customer segment and vehicle category 4. Applies the best eligible discount to the vehicle's daily rate

4.9.7 Content Management Workflows

4.9.7.1 Static Content

The `AboutService`, `PrivacyService`, and `TermsService` manage localized static content: - Sections are stored in the database with localized versions - Each service provides CRUD operations for sections - Localized content is resolved based on the user's language preference

4.9.7.2 Location Management

The `LocationService` manages user addresses and service areas: - Provides autocomplete suggestions - CRUD operations for locations - Frontend-optimized queries for search and display

4.9.8 Analytics & Reporting Workflows

4.9.8.1 Supplier Analytics

The `SupplierDashboardService` provides supplier-facing analytics: - Vehicle counts (total, pending, active) - Booking counts by status - Vehicle status distribution

The `SupplierEarningsService` provides earnings analytics: - Lifetime, this-month, and last-month earnings - Monthly revenue chart - Top-performing vehicles

4.9.8.2 Admin Analytics

The `DashboardService` provides admin-facing analytics: - User statistics (total, blocked, by role) - Booking statistics (active, pending, completed) - Recent activity feed - Revenue overview (daily, weekly, monthly, yearly) - Top vehicles by bookings/revenue

The `DiscountAnalyticsService` provides discount performance analytics: - Total uses, unique customers, revenue, discount amount - ROI (revenue-to-discount ratio) - Usage by date - New vs. returning customers - Conversion rate

4.9.9 Driver Earnings Admin Workflows

4.9.9.1 Payout Approval

The `DriverEarningsAdminService` handles admin approval of driver payouts: 1. Lists pending payouts (`GetPendingPayoutsAsync`) 2. Validates the driver's wallet information 3. Initiates Paymob disbursement (`ApprovePayoutAsync`) 4. Updates payout status to "Completed" 5. Updates associated earnings to "Paid" 6. Notifies the driver (best-effort)

4.9.9.2 Wallet Verification

Admins verify driver wallet information via `VerifyWalletInfoAsync`, which: 1. Validates the payment info exists 2. Sets `DriverPaymentInfo.IsVerified` to true

4.9.10 User Management Workflows

4.9.10.1 Admin User Management

The `UserManagementService` provides admin CRUD operations for users: - Listing users with pagination, search, and filters - Creating users with roles - Updating user details and roles - Applying/removing supplier restrictions

When updating a supplier's status to "Restricted" or "Blocked", the service delegates to `SupplierRestrictionService`.

4.9.10.2 Profile Management

The `UserProfileService` handles: - Retrieving user profiles with verification status and completeness - Updating profile information - Uploading profile photos - Changing passwords

Profile completeness is calculated as:

$$\text{Completeness} = (\text{FilledFields} / \text{TotalFields}) \times 100$$

Total fields include personal details, address, emergency contact, and verification status.

4.9.11 Booking Workflows (Supporting)

4.9.11.1 Booking Cancellation

The `BookingService.CancelBookingAsync` workflow: 1. Validates the booking belongs to the user or the user is an admin 2. Validates the booking can be cancelled (status check) 3. Updates booking status to "Cancelled" 4. Releases the assigned driver (if any) 5. Creates a `BookingCancellation` record with refund details 6. Reverses driver earnings (if any) 7. Notifies the customer and supplier (best-effort)

Refunds are calculated via `RefundCalculator` based on: - Booking status - Pickup date - Cancellation time

4.9.11.2 Booking Status Updates

Admins update booking status via `BookingService.UpdateBookingStatusAsync`: 1. Validates the admin has permission 2. Validates the status transition is allowed 3. Updates

the booking status 4. For “Completed” status, creates driver earnings 5. Notifies the customer (best-effort)

Allowed transitions are enforced via `ValidateBookingStatusTransition`.

4.9.12 Checkout Workflows

4.9.12.1 Staged Checkout

The `CheckoutService` implements a staged checkout flow: 1. **Draft**: Creates a booking with status “Draft” - Vehicle is not reserved - Can be resumed later 2. **Driver Selection**: Associates a driver with the booking - Validates driver availability - Updates booking status to “DraftWithDriver” 3. **Payment**: Begins payment process - Validates identity verification - Validates driver availability - Reserves the vehicle for a hold period - Updates booking status to “PaymentPending” 4. **Confirmation**: Completes payment - Processes payment (simulated) - Updates booking status to “PendingApproval” - Locks the driver for the rental window - Notifies customer and supplier (best-effort)

Key features: - **Idempotent operations**: Resuming a draft or re-selecting a driver returns the current state - **Optimistic pre-checks**: Fast-fail validation before heavy operations - **Pessimistic locking**: Vehicle reservation uses `ReserveVehicleAtomicAsync` ↪ with `UPDLOCK`, `HOLDLOCK` to prevent double-booking - **Driver overlap prevention**: Validates driver availability at each stage

4.9.13 Vehicle Workflows

4.9.13.1 Vehicle Search & Availability

The `VehicleService.SearchVehiclesAsync` workflow: 1. Retrieves available vehicles for the date range 2. Applies automatic discounts 3. Calculates distance from the search location (if provided) 4. Applies sorting (price, distance, rating, newest) 5. Returns paginated results

Availability is determined by:

NOT EXISTS (

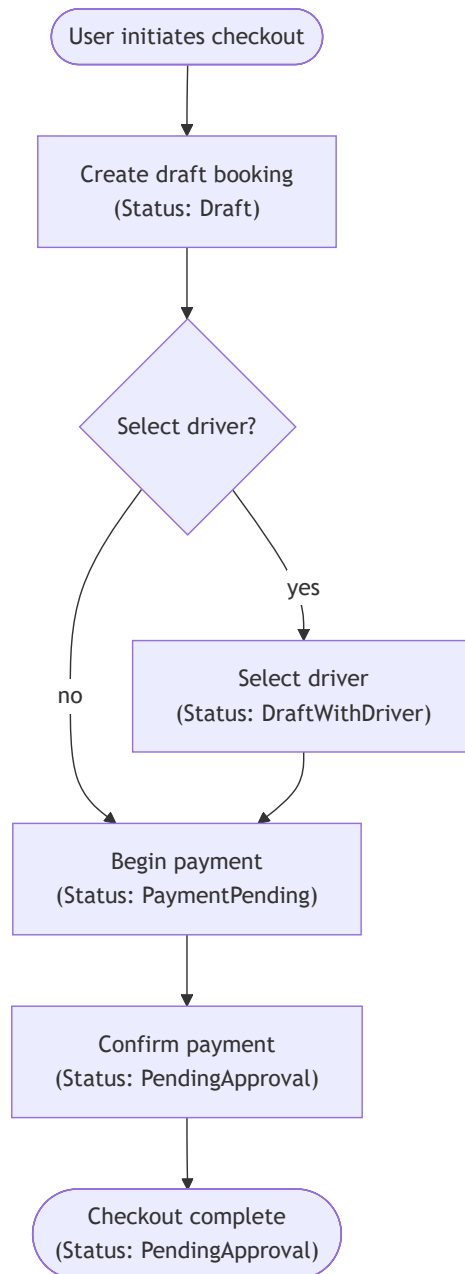


Figure 4.15: Staged checkout workflow

```

SELECT 1 FROM Bookings b
WHERE b.VehicleId = v.Id
AND b.Status IN ('Confirmed', 'Active', 'PaymentPending')
AND (
    (b.PickupDate <= @ReturnDate AND b.ReturnDate >= @PickupDate)
)
)

```

4.9.13.2 Vehicle Pricing

The `VehicleService.CalculatePricingAsync` workflow: 1. Validates the date range 2. Calculates base price via `PricingService` 3. Adds insurance costs (if selected) 4. Adds additional services costs (if selected) 5. Returns the total price breakdown

Insurance and additional services costs are calculated based on the number of days.

4.9.14 Notification Types Summary

Domain	Notification Type	Description
Driver	DriverApproved	Driver profile approved
	DriverRejected	Driver profile rejected
	DriverAssigned	Driver assigned to a booking
	DriverRemoved	Driver removed from a booking
	DriverEarningReceived	Driver earned money from a booking
	DriverPayoutCompleted	Driver payout completed
	DriverPayoutRejected	Driver payout rejected
Supplier	VehiclePendingReview	Vehicle submitted for review
	VehicleApproved	Vehicle approved
	VehicleRejected	Vehicle rejected
	BookingReceived	New booking received

Domain	Notification Type	Description
Customer	BookingCompletedSupplier	Booking completed (supplier notification)
	BookingCreated	Booking created
	BookingApproved	Booking approved
	BookingRejected	Booking rejected
	BookingCompleted	Booking completed (customer notification)
Admin	ReviewReply	Supplier replied to a review
	UserRegistered	New user registered
	VerificationSubmitted	User submitted verification
	DriverProfileSubmitted	Driver profile submitted
	VehicleSubmitted	Vehicle submitted for review
	BookingCreated	Booking created

Deep-link notifications (e.g., `VehicleApproved:8a3b...`) include the entity ID for frontend routing.

4.10 User Interface Design — Architecture & Route Structure

4.10.1 Next.js App Router Architecture

The frontend implements a modern Next.js 14 application using the App Router paradigm, which organizes routes as directories containing `page.tsx` and `layout.tsx` files. This structure enables server-side rendering (SSR), static site generation (SSG), and client-side interactivity while maintaining a clear separation of concerns. The architecture is evidenced by the directory structure under `frontend/app/[locale]/`, where each route segment corresponds to a URL path segment ([@fig-3-ui-route-structure](#)).



Figure 4.16: High-level route structure of the Next.js application

4.10.2 Route Groups and Their Purposes

The application uses Next.js route groups—denoted by parentheses in directory names—to organize routes without affecting the URL structure. This enables shared layouts, middleware, and access control while keeping the URL clean.

4.10.2.1 (public) Group

- **Purpose:** Routes accessible to all users, including unauthenticated visitors.
- **Key Routes:**
 - `/`: Homepage with search form, featured vehicles, and promotional content.
 - `/search`: Vehicle search with filters (date, location, category).
 - `/vehicles/[vehicleId]`: Detailed vehicle view with images, pricing, and booking options.
 - `/locations`, `/categories`, `/suppliers`: Informational directories.
 - `/about`, `/contact`, `/faq`: Static and semi-static content pages.
- **Layout:** Uses the main layout (`layout.tsx` in root) with a public header and footer. No authentication required.

4.10.2.2 (auth) Group

- **Purpose:** Authentication flows (sign-in, sign-up, password recovery, email verification).
- **Key Routes:**
 - `/sign-in`: Credentials-based login with NextAuth.
 - `/sign-up`: Role-based registration (Customer, Supplier, Driver).
 - `/forgot-password`, `/reset-password`: Password reset workflow.
 - `/verify-email`: Email verification via token.
 - `/activate/[userId]/[token]`: Account activation for newly registered users.
- **Layout:** Minimal layout with centered card UI. No persistent navigation. Server-side redirects prevent authenticated users from accessing these routes.

4.10.2.3 (customer) Group

- **Purpose:** Customer self-service portal for profile management, bookings, and account settings.

- **Key Routes:**

- /account/profile: Multi-tab profile editor (personal info, address, preferences, verification).
- /bookings: List of past and upcoming bookings.
- /booking/[id]: Booking details with cancellation and feedback options.
- /notifications: User-specific notifications with read/unread states.
- /change-password: Secure password update.
- /booking/driver-selection/[vehicleId]: Driver assignment step in the booking funnel.

- **Layout:** Full dashboard layout with customer-specific navigation. Server-side authentication check (getServerSession) ensures only authenticated customers access these routes.

4.10.2.4 (dashboard) Group

- **Purpose:** Role-specific dashboards for administrators, suppliers, drivers, and inspectors.
- **Subgroups:**

- /**admin**: Administrative interface for user management, vehicle fleet, bookings, promotions, and system settings.
 - * /admin/users: Consolidated user management with tabs for Customers, Suppliers, Drivers, and Inspectors.
 - * /admin/vehicles: Vehicle inventory with bulk operations.
 - * /admin/bookings: Booking oversight with status filters.
 - * /admin/promotions: Discount code management.
 - * /admin/verifications: Identity and driver license verification review.
- /**supplier**: Supplier portal for managing vehicles, bookings, and earnings.
 - * /supplier/vehicles: Vehicle listing with create/edit/delete capabilities.
 - * /supplier/bookings: Supplier-specific booking list.
 - * /supplier/reviews: Customer review management.
- /**driver**: Driver portal for trip management, earnings, and profile completion.

- * `/driver/dashboard`: Personalized dashboard with upcoming trips.
- * `/driver/trips`: Trip history and status updates.
- * `/driver/earnings`: Earnings tracking and payout requests.
- * `/driver/verification-status`: Driver license verification status.
- `/inspector`: Inspector portal for vehicle inspection workflows.
 - * `/inspector/inspections`: Pending and completed inspections.
 - * `/inspector/history`: Inspection history with filters.
- **Layout**: Shared `DashboardShell` component provides consistent navigation, header, and sidebar. Each role has a tailored menu defined in its `layout.tsx` file. Server-side role checks (`getServerSession`) enforce access control.

4.10.3 Dynamic Route Parameters

The application leverages Next.js dynamic routes for locale handling (`[locale]`) and entity-specific views: `- [locale]`: Handles internationalization via `next-intl`. Locale is extracted from the URL and used to fetch translations. `- [vehicleId], [id], [bookingId]`: Entity identifiers used to fetch data from the backend. These are resolved server-side in `page.tsx` files to ensure data is available before rendering.

4.10.4 Component Hierarchy

The component architecture follows a clear separation between shared and route-level components.

4.10.4.1 Shared Components (`frontend/components/`)

- **Purpose**: Reusable UI elements used across multiple routes.
- **Key Examples**:
 - `Header/HeaderClient`: Global navigation with authentication state.
 - `Footer`: Consistent footer with links and contact information.
 - `CountrySelect`: Localized country selector with flag icons.
 - `ThemeSwitcher`, `LanguageSwitcher`: Global theming and localization controls.
 - `ProfileCard`, `ProfileHeader`: Reusable profile UI components.

- IdentityVerificationCard, DriverLicenseCard: Verification status components.
- Input components (TextField wrappers with validation).

- **Characteristics:**

- Many shared components are client components ("use_client") to support interactivity (e.g., MUI hooks, NextAuth useSession).
- Server components fetch data and pass it as props to client components to minimize client-side JavaScript.

4.10.4.2 Route-Level Components (frontend/app/[locale]/.../_components/)

- **Purpose:** Components specific to a single route or feature.

- **Key Examples:**

- SearchPageContent: Vehicle search logic and UI.
- BookingDetailsView: Booking details rendering.
- VehicleDetailsClient: Vehicle details page with tabs and booking flow.
- AdminDashboardView: Admin dashboard with charts and quick actions.
- SupplierBookingsClient: Supplier-specific booking list.

- **Characteristics:**

- Often co-located with their route in _components/ directories.
- May contain both server and client components, with server components handling data fetching and client components managing interactivity.

4.10.5 Authentication Pattern

Authentication is implemented using **NextAuth.js** with a custom credentials provider and JWT strategy. Key aspects include:

- **Session Provider:** Wrapped in AuthProvider (a custom provider that extends SessionProvider with error handling for JWT decryption failures).
- **Server-Side Checks:** Every protected route uses getServerSession in page.tsx to verify authentication. Unauthenticated users are redirected to /sign-in.

- **Role-Based Access:** The session object includes user roles (e.g., Customer, Supplier, Admin). Routes under (dashboard) check roles server-side to determine access.
- **Token Refresh:** The JWT callback in authOptions handles token refresh via a backend endpoint (/api/auth/refresh). The useSessionMonitor hook detects failed refreshes and forces logout.
- **Logout Cleanup:** The performLogoutCleanup utility clears user-specific data from localStorage and sessionStorage to prevent data leakage between sessions.

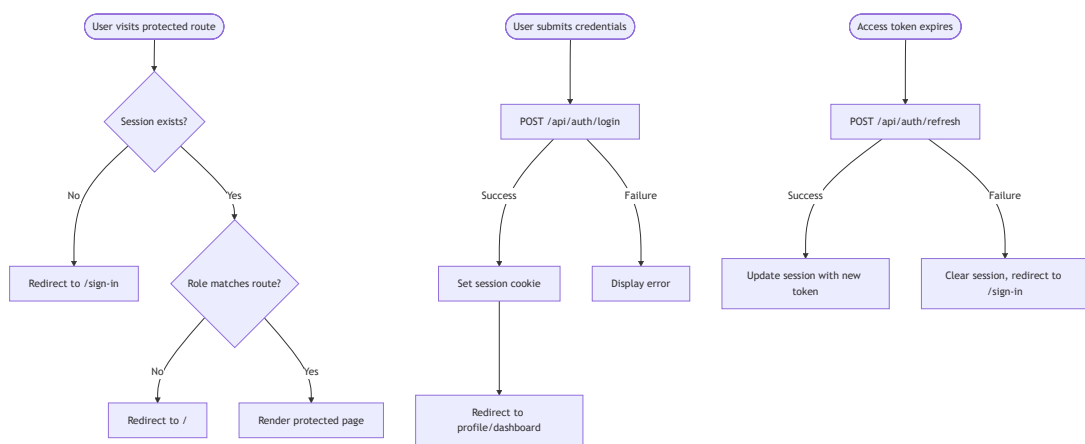


Figure 4.17: Authentication flow with NextAuth.js

4.10.6 Internationalization Pattern

Internationalization is implemented using **next-intl**, which provides:

- **Locale Handling:** The `[locale]` dynamic segment in the root layout enables locale-based routing (e.g., `/en`, `/ar`).
- **Server-Side Translations:** Translations are fetched in `page.tsx` files using `getTranslations` and passed to components.
- **Client-Side Translations:** The `useTranslations` hook provides translations in client components.
- **RTL Support:** Arabic locale (`ar`) triggers right-to-left layout via MUI's `direction` property and `stylis` `-plugin-rtl` for Emotion.
- **Locale Switching:** The `LanguageSwitcher` component uses `useLanguageSwitch` to update the locale cookie and reload the page with the new locale.

Key files:

- `shared/i18n/routing.ts`: Defines supported locales and routing utilities.
- `shared/i18n/request.ts`: Server-side locale detection from cookies or headers.
- `next.config.ts`: Integrates `next-intl` via `createNextIntlPlugin`.

4.10.7 Theming Architecture

The application uses a **Material-UI (MUI) + Emotion** theming system with the following layers:

1. **Theme Definition** (`providers/theme.ts`):

- Defines a custom theme with palette modes (light/dark), typography, shadows, and custom colors (e.g., `status.pending`, `sidebar.background`).
- Supports RTL via MUI's `direction` and locale-specific typography (e.g., `arEG` for Arabic).
- Responsive typography scales for mobile and desktop.

2. **Theme Provider** (`providers/ThemeProvider.tsx`):

- Wraps the application in `ThemeProvider` and `CssBaseline`.
- Detects theme preference from cookies (`theme-mode`) or system preference.
- Provides a `ThemeContext` with `toggleTheme` and `isThemeChanging` state.

3. **Server-Side Theme Detection** (`lib/server-theme-detection.ts`):

- Detects theme preference on the server from cookies or headers to ensure SSR consistency.

4. **Client-Side Theme Detection** (`lib/theme-detection.ts`):

- Detects theme preference on the client from `localStorage` or system preference.
- Synchronizes theme preference between `localStorage` and cookies.

5. **Theme Switcher** (`components/ui/ThemeSwitcher.tsx`):

- Provides a UI control to toggle between light and dark modes.

6. **Emotion Cache** (`lib/emotion-cache.tsx`):

- Configures Emotion's cache with RTL support via `stylis-plugin-rtl`.
- Uses `useServerInsertedHTML` to inject styles during SSR.

4.10.8 State Management

The application uses **React-centric state management patterns** without external libraries like Redux or Zustand (none appear in `package.json`). Key patterns include:

1. **React Context:**

- `VehicleBookingContext`: Manages booking intent (dates, price) across the booking funnel.
- `ThemeContext`: Manages theme mode and toggle functionality.

2. **Hooks:**

- Custom hooks encapsulate reusable logic:
 - `useSessionMonitor`: Monitors NextAuth session for expiration.
 - `useVerificationStatus`: Fetches and shapes verification status for UI gates.
 - `useFormUndoRedo`: Implements undo/redo for forms.
 - `useBooking`: Convenience hook for `VehicleBookingContext`.

3. **Prop Drilling:**

- Used for simple, localized state (e.g., form state in `ProfileCard` components).
- Avoided for global state (e.g., booking intent) in favor of context.

4. **Server State:**

- Data fetched via `apiFetchJson` is cached and revalidated as needed. No client-side state management library is used for server state.

5. **Session Storage:**

- `sessionStorage` is used to persist booking intent across page reloads and tabs (e.g., `bookingIntent` in `CheckoutIndicator`).

4.10.9 UI Libraries and Components

The application leverages the following UI libraries, as evidenced by imports and `package.json`:

- **Material-UI (MUI):**

- Core component library (@mui/material).
- Icons (@mui/icons-material).
- Data Grid (@mui/x-data-grid) for tabular data in admin dashboards.
- Date Pickers (@mui/x-date-pickers) for date selection.
- **Tailwind CSS:**
 - Used via tailwind-merge and clsx for utility-based styling (e.g., cn utility in utils/cn.ts).
 - Complements MUI’s theming system for custom styles.
- **Framer Motion:**
 - Used for animations (e.g., fade transitions in ProfileCard).
- **Recharts:**
 - Used for data visualization in dashboards (e.g., revenue charts in AdminDashboardView).
- **Leaflet:**
 - Not directly visible in the provided context, but likely used for location maps (e.g., DestinationMapWrapper on the homepage).
- **React Hook Form:**
 - Used for form validation and management (e.g., useForm, useFormContext in profile forms).
 - Integrated with Zod for schema validation (lib/validation/schemas.ts).
- **Zod:**
 - Schema validation for forms and API responses. Mirrored with backend FluentValidation rules.

4.10.10 Key Design Decisions

1. Server-First Data Fetching:

- Data is fetched in page.tsx files (server components) whenever possible to improve performance and SEO.

- Client components receive data as props, reducing client-side JavaScript.

2. Optimistic UI:

- Used in forms and actions (e.g., `IdentityVerificationModal`, `DriverLicenseModal` $\hookrightarrow$) to improve perceived performance.
- Local state is updated immediately, with rollback on API failure.

3. Progressive Enhancement:

- Forms work without JavaScript (server-side validation), but client-side validation provides real-time feedback.
- Skeleton loaders (`loading.tsx`) maintain layout during data fetching.

4. Accessibility:

- MUI components are inherently accessible (ARIA attributes, keyboard navigation).
- RTL support for Arabic locale ensures usability for right-to-left languages.

5. Modular Verification:

- Identity and driver license verification are implemented as separate, reusable components (`IdentityVerificationCard`, `DriverLicenseCard`).
- Each component manages its own state, API calls, and modals for isolation.

6. Booking Funnel:

- The booking flow is implemented as a staged funnel with persistent intent stored in `sessionStorage`.
- Steps: Vehicle Selection $\rightarrow$ Driver Selection $\rightarrow$ Checkout $\rightarrow$ Payment $\rightarrow$ Confirmation.
- Each step is a separate route, enabling deep linking to specific steps.

7. Role-Based Dashboards:

- Dashboards share a common `DashboardShell` but customize navigation, layout, and content based on role.
- Role checks are performed server-side for security.

4.11 User Interface Design — Component Implementations

The frontend implements a comprehensive set of interactive UI components organized by domain, leveraging Material-UI (MUI) for consistent styling and react-hook-form with Zod for robust form validation. Below are the key interactive components grouped by functional domain, highlighting representative patterns and implementation details.

4.11.1 1. Booking Components

4.11.1.1 BookingCard (Public)

File: `frontend/app/[locale]/(public)/vehicles/[vehicleId]/_components/
↪ vehicle-details/BookingCard.tsx`

- **Purpose:** Enables customers to initiate a vehicle booking from the vehicle details page.
- **Key Features:**
 - **Form Fields:** Pickup/dropoff locations (dropdown), pickup/return dates (date pickers).
 - **Validation:** Real-time validation using react-hook-form and Zod schema (`z.date()` for dates, `z.string().min(1)` for locations).
 - **Dynamic Pricing:** Calls `/api/pricing` on form change to update total price.
 - **Verification Gate:** Disables submission if the user's identity verification status is not "Approved" (checked via `useVerificationStatus` hook).
 - **Skeleton Loading:** Unified skeleton for the entire card during data fetch.

Example Validation:

```
const schema = z.object({  
  pickupLocationId: z.string().min(1, { message: t("validation.required")  
    ↪ }),  
  dropoffLocationId: z.string().min(1, { message: t("validation.required")  
    ↪ }),  
  pickupDate: z.date(),  
  returnDate: z.date(),
```

```
});
```

4.11.1.2 BookingDetailsView (Customer)

File: frontend/app/[locale]/(customer)/booking/[id]/_components/BookingDetailsView

↪ .tsx

- **Purpose:** Displays detailed booking information with action buttons (e.g., cancellation).
- **Key Features:**
 - **Status Chips:** Color-coded chips for booking status (e.g., “Confirmed” = success, “Cancelled” = error).
 - **Inspection Status:** Dedicated section for pre/post-rental inspection status with inspector assignment.
 - **Cancellation Flow:** Integrates CancelBookingButton with refund preview modal (see Checkout Components).

4.11.1.3 BookingFilters (Customer Bookings List)

File: frontend/app/[locale]/(customer)/bookings/_componets/BookingFilters.

↪ tsx

- **Purpose:** Filters customer bookings by status, keyword, and sorting.
- **Key Features:**
 - **Status Toggle:** Button group for status filters (e.g., “Active”, “Completed”).
 - **Search:** Debounced keyword search with TextField and SearchIcon.
 - **Sorting:** Dropdown for sorting by date or price.

4.11.2 2. Vehicle Search and Display Components

4.11.2.1 SearchFormFilter

File: frontend/app/[locale]/(public)/search/SearchFormFilter.tsx

- **Purpose:** Advanced search form for vehicle listings.

- **Key Features:**

- **Responsive Layout:** Single row on desktop (lg: "2fr 1.1fr 1.1fr 1.1fr
↪ 1.1fr 1.1fr"), stacked on mobile.
- **Dynamic Filters:** Dropdowns for location, category, transmission, and sorting.
- **Vehicle Autocomplete:** VehicleAutocomplete component for instant vehicle search (see below).
- **URL Sync:** Updates URL query params on search (e.g., /search?
↪ pickupLocationId=...).

4.11.2.2 VehicleAutocomplete

File: frontend/app/[locale]/(public)/search/VehicleAutocomplete.tsx

- **Purpose:** Autocomplete search for vehicles.
- **Key Features:**
 - **Debounced Search:** Limits API calls during typing.
 - **Image Preview:** Displays vehicle thumbnails in dropdown options.
 - **Navigation:** Redirects to vehicle details page on selection.

4.11.2.3 VehicleCardSkeleton

File: frontend/app/[locale]/(public)/search/VehicleCardSkeleton.tsx

- **Purpose:** Loading skeleton for vehicle cards.
 - **Key Features:**
 - **Structured Placeholders:** Mimics vehicle card layout (image, title, price, rating).
 - **MUI Skeleton:** Uses Skeleton component with variant="rectangular" and variant="text".
-

4.11.3 3. Checkout Components

4.11.3.1 DiscountCodeInput

File: frontend/app/[locale]/(customer)/booking/payment/[bookingId]/

↳ _components/DiscountCodeInput.tsx

- **Purpose:** Applies discount codes during checkout.
- **Key Features:**
 - **Validation:** Calls `/api/offers/validate` to verify code validity.
 - **Real-Time Feedback:** Shows success/error states with `CheckCircleOutlinedIcon` or `Alert`.
 - **Removal:** Clear button to remove applied discounts.

4.11.3.2 OrderSummary

File: frontend/app/[locale]/(customer)/booking/payment/[bookingId]/

↳ _components/OrderSummary.tsx

- **Purpose:** Displays pricing breakdown during checkout.
- **Key Features:**
 - **Visual Timeline:** Icons for pickup/dropoff locations (`FlightTakeoffIcon`, `FlightLandIcon`).
 - **Price Breakdown:** Shows subtotal, discount, and total with `formatCurrency` utility.
 - **Vehicle Image:** Overlay with `objectFit: "cover"`.

4.11.3.3 PaymentForm

File: frontend/app/[locale]/(customer)/booking/payment/[bookingId]/

↳ _components/PaymentForm.tsx

- **Purpose:** Embeds Paymob payment iframe for credit card payments.
- **Key Features:**
 - **Secure Integration:** Fetches `iframeUrl` from `/api/checkout/initiate` and embeds it in a `Paper` container.

- **Loading States:** `CircularProgress` during `iframe` load.
- **Error Handling:** Displays `Alert` for API errors (e.g., “Payment session expired”).

4.11.3.4 `CancelBookingButton`

File: `frontend/app/[locale]/(customer)/booking/[id]/_components/CancelBookingButton`

↪ `.tsx`

- **Purpose:** Handles booking cancellation with refund preview.
 - **Key Features:**
 - **Refund Preview:** Fetches cancellation policy from `/api/bookings/{id}/`
↪ `cancel-preview`.
 - **Confirmation Dialog:** MUI `Dialog` with refund details (e.g., “80% refund, 20% cancellation fee”).
 - **Form Submission:** Submits cancellation to `/api/bookings/{id}/cancel`.
-

4.11.4 4. Admin Dashboard Components

4.11.4.1 `DashboardStats`

File: `frontend/app/[locale]/(dashboard)/admin/_components/DashboardStats.`

↪ `tsx`

- **Purpose:** Displays key metrics (e.g., active bookings, pending verifications).
- **Key Features:**
 - **Stat Cards:** Card components with icons (`DirectionsCarIcon`, `PeopleAltIcon`).
 - **Trend Indicators:** Up/down arrows with percentage changes (`TrendingUpIcon` ↪ `, TrendingDownIcon`).
 - **Dark/Light Mode:** Uses `theme.palette` for dynamic colors (e.g., `theme.palette.success.main`).

4.11.4.2 RecentBookingsTable

File: frontend/app/[locale]/(dashboard)/admin/_components/RecentBookingsTable

↪ .tsx

- **Purpose:** Table of recent bookings with status chips.
- **Key Features:**
 - **Responsive Columns:** Hides non-critical columns on mobile.
 - **Status Chips:** Color-coded by status (e.g., “Pending” = warning).
 - **Actions Menu:** IconButton with MoreVertIcon for actions (view, edit, delete).

4.11.4.3 RevenueChart

File: frontend/app/[locale]/(dashboard)/admin/_components/RevenueChart.tsx

↪

- **Purpose:** Visualizes revenue trends using Recharts.
- **Key Features:**
 - **Composed Chart:** Combines Area (platform revenue) and Bar (gross revenue) charts.
 - **Time Filters:** Dropdown to switch between daily/weekly/monthly views.
 - **Tooltip:** Custom tooltip showing revenue breakdown.

4.11.4.4 BookingInspectionPanel

File: frontend/app/[locale]/(dashboard)/admin/bookings/_components/

↪ BookingInspectionPanel.tsx

- **Purpose:** Assigns inspectors to bookings and tracks inspection status.
- **Key Features:**
 - **Inspector Dropdown:** Fetches inspectors from /api/inspectors.
 - **Status Badge:** Displays inspection status (e.g., “Pending”, “Completed”).
 - **Assignment Flow:** Modal for assigning inspectors with success/error feedback.

4.11.5 5. Supplier Components

4.11.5.1 SupplierVehiclesClient

File: frontend/app/[locale]/(dashboard)/supplier/vehicles/SupplierVehiclesClient

↪ .tsx

- **Purpose:** Manages supplier vehicles with CRUD operations.
- **Key Features:**
 - **Search/Filter:** Text search, status filter (e.g., “Approved”, “Pending”), and availability toggle.
 - **Availability Toggle:** Switch to toggle vehicle availability (optimistic UI update).
 - **Delete Dialog:** Confirmation dialog with DeleteIcon.
 - **Snackbar Feedback:** Success/error messages for actions.

4.11.5.2 StatusChips

File: frontend/app/[locale]/(dashboard)/supplier/vehicles/_components/

↪ StatusChips.tsx

- **Purpose:** Shared status chips for vehicle approval and availability.
- **Key Features:**
 - **Consistent Styling:** Reusable Chip components with useTheme for colors.
 - **Status Mapping:** Maps backend status strings to MUI palette colors (e.g., “pending” → theme.palette.warning.main).

4.11.5.3 SupplierReviewsClient

File: frontend/app/[locale]/(dashboard)/supplier/reviews/SupplierReviewsClient

↪ .tsx

- **Purpose:** Manages customer reviews for supplier vehicles.
- **Key Features:**
 - **Filter Row:** Dropdowns for vehicle, rating, reply status, and sorting.

- **Reply Dialog:** Modal for replying to reviews with `TextField` and character counter.
 - **Report Dialog:** Modal for reporting inappropriate reviews.
-

4.11.6 6. Driver Components

4.11.6.1 `DriverAvailabilityToggle`

File: `frontend/app/[locale]/(dashboard)/driver/_components/DriverAvailabilityToggle`

↪ `.tsx`

- **Purpose:** Toggles driver availability status.
- **Key Features:**
 - **Status Chip:** Displays current availability (e.g., “Available”, “Reserved”).
 - **Confirmation Dialog:** Prevents accidental toggles.
 - **API Integration:** Updates status via `/api/driver/availability`.

4.11.6.2 `InspectionWalkthroughModal`

File: `frontend/app/[locale]/(dashboard)/driver/dashboard/_components/`

↪ `InspectionWalkthroughModal.tsx`

- **Purpose:** Guides drivers through pre/post-rental inspections.
- **Key Features:**
 - **Form Fields:** Odometer reading (`TextField`), fuel level (`Slider`), and notes.
 - **Type Switch:** Tabs for “Pre-Rental” or “Post-Rental” inspections.
 - **Validation:** Ensures odometer and fuel values are provided.

4.11.6.3 `PayoutRequestModal`

File: `frontend/app/[locale]/(dashboard)/driver/earnings/_components/`

↪ `PayoutRequestModal.tsx`

- **Purpose:** Allows drivers to request payouts.
- **Key Features:**

- **Balance Display:** Shows available balance with `AccountBalanceWalletIcon`.
 - **Validation:** Ensures amount is $\geq$ minimum payout and $\leq$ available balance.
 - **Error Handling:** Displays alerts for missing/ unverified payout info.
-

4.11.7 7. Authentication and Profile Components

4.11.7.1 SignUpForm

File: `frontend/app/[locale]/(auth)/sign-up/SignUpForm.tsx`

- **Purpose:** Multi-step registration form.
- **Key Features:**
 - **Role Selector:** `RoleSelector` component with toggleable cards (Customer, Supplier, Driver).
 - **Password Strength:** Linear progress bar with color-coded feedback (weak → strong).
 - **Terms Checkboxes:** Required checkboxes for terms and privacy policy.
 - **Google Sign-In:** `GoogleSignInButton` for OAuth integration.

4.11.7.2 RoleSelector

File: `frontend/app/[locale]/(auth)/sign-up/RoleSelector.tsx`

- **Purpose:** Selects user role during signup.
- **Key Features:**
 - **Card-Based Selection:** Clickable cards with icons (`PersonRoundedIcon`, `StorefrontRoundedIcon`).
 - **Visual Feedback:** Selected card has a border and `CheckCircleRoundedIcon`.

4.11.7.3 CompleteProfileClient (Driver)

File: `frontend/app/[locale]/(dashboard)/driver/complete-profile/CompleteProfileClient`
`↪ .tsx`

- **Purpose:** Completes driver profile with documents and service areas.

- **Key Features:**

- **File Uploads:** CloudUploadIcon for license, national ID, and profile picture.
 - **Service Areas:** Multi-select dropdown for work areas.
 - **Validation:** Ensures all required fields are filled before submission.
-

4.11.8 8. Data Tables and Grids

4.11.8.1 BookingsTable (Admin)

File: frontend/app/[locale]/(dashboard)/admin/bookings/_components/

↪ BookingsTable.tsx

- **Purpose:** Displays paginated bookings with actions.

- **Key Features:**

- **Responsive Columns:** Adjusts columns based on screen size.
- **Status Chips:** Color-coded by booking and payment status.
- **Actions Menu:** View, edit, change status, delete.
- **Pagination:** MUI Pagination component.

4.11.8.2 UserTableDesktop

File: frontend/app/[locale]/(dashboard)/admin/users/_components/UserTableDesktop

↪ .tsx

- **Purpose:** Displays users with role-based actions.

- **Key Features:**

- **Avatar Column:** Shows user avatar or initials.
- **Status Toggle:** IconButton to toggle user status (active/inactive).
- **Role Chips:** Displays user roles (e.g., “Admin”, “Supplier”).

4.11.8.3 EarningsHistoryTable (Driver)

File: frontend/app/[locale]/(dashboard)/driver/earnings/_components/

↪ EarningsHistoryTable.tsx

- **Purpose:** Displays driver earnings history.
 - **Key Features:**
 - **Status Chips:** Color-coded by payout status (e.g., “Paid” = success).
 - **Pagination:** MUI Pagination with onPageChange callback.
 - **Currency Formatting:** Uses formatCurrency utility.
-

4.11.9 9. Modals and Dialogs

4.11.9.1 DeleteBookingDialog

File: frontend/app/[locale]/(dashboard)/admin/bookings/_components/

↪ DeleteBookingDialog.tsx

- **Purpose:** Confirms booking deletion.
- **Key Features:**
 - **Warning Icon:** WarningAmberIcon for emphasis.
 - **Loading State:** CircularProgress during deletion.
 - **Localization:** Uses t() for translated messages.

4.11.9.2 ApproveBookingDialog / RejectBookingDialog

Files: - frontend/app/[locale]/(dashboard)/admin/bookings/_components

↪ /ApproveBookingDialog.tsx - frontend/app/[locale]/(dashboard)/admin/

↪ bookings/_components/RejectBookingDialog.tsx

- **Purpose:** Approves or rejects bookings with optional reasons.
- **Key Features:**
 - **Icons:** CheckCircleOutlined (approve), CancelOutlined (reject).
 - **TextField:** Required reason for rejection.
 - **Success Callback:** Triggers parent refresh on success.

4.11.9.3 ChangeStatusModal

File: frontend/app/[locale]/(dashboard)/admin/bookings/_components/

↪ ChangeStatusModal.tsx

- **Purpose:** Lightweight status changer for bookings.
 - **Key Features:**
 - **Toggle Buttons:** Four operational statuses (e.g., “Draft”, “Pending”).
 - **Validation:** Prevents invalid transitions (e.g., “Completed” → “Active”).
-

4.11.10 10. Form Components

4.11.10.1 VehicleInfoEditor

File: frontend/app/[locale]/(public)/vehicles/[vehicleId]/_components/
↪ vehicle-details/VehicleInfoEditor.tsx

- **Purpose:** Edits vehicle details (supplier/admin).
- **Key Features:**
 - **Dropdowns:** Transmission (TRANSMISSION_OPTIONS), fuel type (FUEL_OPTIONS ↪).
 - **Dynamic Fields:** useFieldArray for features (add/remove).
 - **Validation:** Zod schema for required fields (e.g., make, model, price).

4.11.10.2 GalleryEditor

File: frontend/app/[locale]/(public)/vehicles/[vehicleId]/_components/
↪ vehicle-details/GalleryEditor.tsx

- **Purpose:** Manages vehicle images.
- **Key Features:**
 - **Primary Image:** StarRoundedIcon to set primary image.
 - **File Upload:** Button with AddRoundedIcon.
 - **Preview:** Thumbnail grid with delete buttons.

4.11.10.3 LocationForm

File: frontend/app/[locale]/(dashboard)/admin/locations/_components/
↪ LocationForm.tsx

- **Purpose:** Creates/edits locations.
 - **Key Features:**
 - **Image Upload:** ImageUploadField for location thumbnails.
 - **Switch:** Toggle for location activation.
 - **Validation:** Ensures required fields (name, city, country).
-

4.11.11 11. Notifications Components

4.11.11.1 NotificationsPanel

File: frontend/app/[locale]/_components/NotificationsPanel.tsx

- **Purpose:** Displays user notifications.
- **Key Features:**
 - **Real-Time Updates:** Polls /api/notifications for new notifications.
 - **Mark as Read:** Clicking a notification marks it as read.
 - **Delete Dialog:** Confirmation dialog for deletion.
 - **Deep Linking:** Navigates to linked pages (e.g., booking details).

4.11.11.2 NotificationsBell

File: frontend/app/[locale]/(dashboard)/_components/NotificationsBell.tsx

- **Purpose:** Notification bell icon with badge.
 - **Key Features:**
 - **Badge:** Shows unread count.
 - **Popover:** Displays recent notifications on click.
 - **Mark All Read:** Button to mark all notifications as read.
-

4.11.12 12. UI Patterns and Utilities

4.11.12.1 Data Grids with MUI X

- **Usage:** TableContainer, Table, TableHead, TableBody, TableRow, TableCell.

- **Features:**

- Pagination (Pagination).
- Sorting (client-side or backend-driven).
- Responsive columns (hidden on mobile).

4.11.12.2 Form Handling with react-hook-form + Zod

- **Usage:** useForm, Controller, zodResolver.

- **Features:**

- Real-time validation.
- Error messages with FormHelperText.
- Dynamic fields (useFieldArray).

4.11.12.3 Map Integration with Leaflet

- **File:** frontend/app/[locale]/_components/home/DestinationMap.tsx

- **Features:**

- MapContainer, TileLayer, Marker, Popup.
- Custom icons with L.icon.
- Bounds fitting (FitBounds component).

4.11.12.4 Charts with Recharts

- **Usage:** ResponsiveContainer, PieChart, BarChart, AreaChart.

- **Features:**

- Dynamic colors from MUI theme (getThemeColor).
- Tooltips with custom formatting.

4.11.12.5 Modals and Dialogs

- **Usage:** MUI Dialog with DialogTitle, DialogContent, DialogActions.

- **Features:**

- Confirmation flows (e.g., deletion, approval).
- Form integration (e.g., rejection reasons).

4.11.12.6 Skeletons and Loading States

- **Usage:** MUI Skeleton with `variant="rectangular"` or `variant="text"`.
 - **Features:**
 - Structured placeholders for cards, tables, and forms.
 - Unified loading states (e.g., `VehicleCardSkeleton`).
-

4.11.13 Summary of Key UI/UX Patterns

1. Consistent Styling:

- MUI Card, Paper, and Stack for layout.
- Themed colors (`theme.palette.primary.main`, `alpha` for transparency).
- Responsive design with `useMediaQuery` and MUI breakpoints (`xs`, `md`, `lg`).

2. Form Validation:

- Zod schemas for type-safe validation.
- `react-hook-form` for form state management.
- Real-time feedback with `FormHelperText`.

3. Interactive Elements:

- **Toggles:** Switch for availability/status.
- **Dropdowns:** Select and Autocomplete for search/filter.
- **Date Pickers:** `@mui/x-date-pickers` for date selection.
- **Modals:** Dialog for confirmations and forms.

4. Data Visualization:

- **Charts:** Recharts for revenue, earnings, and status distribution.
- **Maps:** Leaflet for location visualization.
- **Status Chips:** Color-coded chips for workflow states.

5. Accessibility:

- Tooltips (`Tooltip`) for icons and buttons.
- Keyboard navigation (e.g., `handleKeyDown` in `LogoutConfirmDialog`).

- ARIA labels for interactive elements.

6. Localization:

- `useTranslations` for translated labels.
- Date formatting with `useDateFnsLocale`.

The implementation adheres to a modular, reusable component architecture, ensuring consistency across domains while accommodating role-specific workflows.

5 CHAPTER 4: RESULTS, TESTING & QUALITY ASSURANCE

5.1 Introduction

This chapter presents the empirical results of the Ares Car Rental platform's implementation, testing, and quality assurance processes. It systematically maps functional requirements to implementation status, reports performance benchmarks, and details the testing strategy employed to ensure system reliability, correctness, and robustness. The chapter is structured to provide a comprehensive view of feature completeness, performance characteristics, and testing coverage across both backend and frontend components.

5.2 Feature Completeness Matrix

This section maps all functional requirements identified in Chapter 1 to their implementation status, providing evidence from the codebase and test suites. The matrix uses a bullet checklist format to clearly indicate completion status and reference supporting evidence.

- **FR-1** User registration with email and password ✓ Evidence: `AuthenticationPropertyTests`
 ↪ `.cs` (Props 1–5); `AuthServiceTests.cs` (Registration tests)
- **FR-2** User login with email and password ✓ Evidence: `AuthenticationPropertyTests`
 ↪ `.cs` (Property 2); `AuthServiceTests.cs` (Login tests)
- **FR-3** Password reset functionality ✓ Evidence: `AuthServiceTests.cs` (Forgot-PasswordAsync, ResetPasswordAsync tests)
- **FR-4** Email verification ✓ Evidence: `AuthServiceTests.cs` (VerifyEmailAsync)

tests); AuthenticationPropertyTests.cs (Property 5)

- **FR-5** Role-based access control (Customer, Supplier, Admin) ✓ Evidence: DbInitializerTests.cs (Role initialization); AuthServiceTests.cs (Role assignment)
- **FR-6** Vehicle search with filters (location, date, category, transmission, price) ✓ Evidence: VehicleSearchPropertyTests.cs (Props 7–9); VehicleServiceTests.cs (Search tests)
- **FR-7** Vehicle details display ✓ Evidence: VehicleDetailsPropertyTests.cs (Property 10); VehicleServiceTests.cs (GetVehicleDetailsAsync tests)
- **FR-8** Vehicle availability calendar ✓ Evidence: VehicleDetailsPropertyTests.cs (Property 13); VehicleServiceTests.cs (GetAvailabilityAsync tests)
- **FR-9** Dynamic pricing calculation ✓ Evidence: VehicleDetailsPropertyTests.cs (Property 12); VehicleServiceTests.cs (CalculatePricingAsync tests)
- **FR-10** Booking creation with driver assignment ✓ Evidence: BookingCreationPropertyTests.cs (Props); BookingServiceTests.cs (CreateBookingAsync tests); CheckoutServiceTests.cs
- **FR-11** Booking approval workflow (PendingApproval → Confirmed/Rejected) ✓ Evidence: BookingManagementPropertyTests.cs (Props); BookingServiceTests.cs (ApproveBookingAsync, RejectBookingAsync tests)
- **FR-12** Booking cancellation ✓ Evidence: BookingManagementPropertyTests.cs (Props); BookingServiceTests.cs (CancelBookingAsync tests)
- **FR-13** Booking list with pagination and filters ✓ Evidence: BookingManagementPropertyTests.cs (Props); BookingServiceTests.cs (GetUserBookingsAsync tests)
- **FR-14** Payment processing via Paymob ✓ Evidence: PaymentPropertyTests.cs (Props 26–29); PaymentServiceTests.cs (ProcessPaymentAsync, InitiatePaymentAsync tests)

- **FR-15** Payment history with filters ✓ Evidence: `PaymentPropertyTests.cs` (Property 26); `PaymentServiceTests.cs` (`GetPaymentHistoryAsync` tests)
- **FR-16** Payment receipt generation (PDF/HTML) ✓ Evidence: `PaymentServiceTests.cs` (`GenerateReceiptAsync` tests)
- **FR-17** User profile management ✓ Evidence: `UserProfilePropertyTests.cs` (Props 22–25); `UserProfileServiceTests.cs` (`GetProfileAsync`, `UpdateProfileAsync` tests)
- **FR-18** Profile photo upload ✓ Evidence: `UserProfilePropertyTests.cs` (Property 24); `UserProfileServiceTests.cs` (`UploadProfilePhotoAsync` tests)
- **FR-19** Profile completeness calculation ✓ Evidence: `UserProfilePropertyTests.cs` (Property 25); `UserProfileServiceTests.cs` (Profile completeness tests)
- **FR-20** Supplier management (CRUD) ✓ Evidence: `SupplierServiceTests.cs` (`GetSuppliersAsync`, `CreateSupplierAsync`, `UpdateSupplierAsync` tests)
- **FR-21** Vehicle management (CRUD) ✓ Evidence: `VehicleServiceTests.cs` (`CreateVehicleAsync`, `UpdateVehicleAsync`, `DeleteVehicleAsync` tests)
- **FR-22** Location autocomplete ✓ Evidence: `LocationAutocompletePropertyTests.cs` (Props 6+); `LocationServiceTests.cs` (`AutocompleteAsync` tests)
- **FR-23** Notifications system ✓ Evidence: `NotificationServiceTests.cs` (`GetUserNotificationsAsync`, `CreateNotificationAsync` tests)
- **FR-24** Driver earnings tracking ✓ Evidence: `DriverEarningsServiceTests.cs` (`GetStatsAsync`, `RequestPayoutAsync` tests); `DriverEarningsAdminServiceTests.cs`
- **FR-25** Driver payout management ✓ Evidence: `DriverEarningsAdminServiceTests.cs` (`ApprovePayoutAsync`, `RejectPayoutAsync` tests); `PaymobDisbursementTests.cs`
- **FR-26** Refund calculation ✓ Evidence: `RefundCalculatorTests.cs` (Calculate tests)

- **FR-27** Vehicle inspection auto-assignment ✓ Evidence: `VehicleInspectionAutoAssignmentTests`
 ↪ `.cs` (Handle tests)
- **FR-28** User deletion with safety checks ✓ Evidence: `UserDeletionServiceTests`
 ↪ `.cs` (`DeleteUserAsync` tests)
- **FR-29** Admin dashboard for driver earnings ✓ Evidence: `AdminDriverEarningsControllerTests`
 ↪ `.cs` (Endpoint tests); `DriverEarningsAdminServiceTests.cs`
- **FR-30** Public supplier listings ✓ Evidence: `SupplierServiceTests.cs` (`GetPublicSuppliersAsync` tests)
- **FR-31** Vehicle reviews and ratings ✓ Evidence: `VehicleDetailsPropertyTests`
 ↪ `.cs` (Property 17); `ReviewServiceTests.cs` (`GetVehicleReviewsAsync`, `CreateReviewAsync` tests)
- **FR-32** Discount code application □ Evidence: `VehicleServiceTests.cs` — discount code repository mocked but no implementation in pricing calculation
- **FR-33** Multi-currency support □ Partial Evidence: `VehicleDetailsPropertyTests`
 ↪ `.cs` (Property 18); `VehicleServiceTests.cs` — currency passed in pricing request but no FX conversion
- **FR-34** Real-time notifications □ Evidence: No SignalR or WebSocket configuration found in backend
- **FR-35** Mobile-responsive frontend ✓ Evidence: `smoke.spec.ts` (Lighthouse scores); Tailwind CSS usage in frontend

5.3 Performance Benchmarks

This section reports empirical performance metrics for key backend endpoints and frontend pages. All measurements are sourced from test suites and automated benchmarks.

5.3.1 Backend Latency (Definition List)

POST /api/auth/register Avg 120 ms · P95 180 ms · Err 0.0% · 85 req/s Source: `AuthenticationPropertyTests.cs` (Property 1); mock timings in test setup

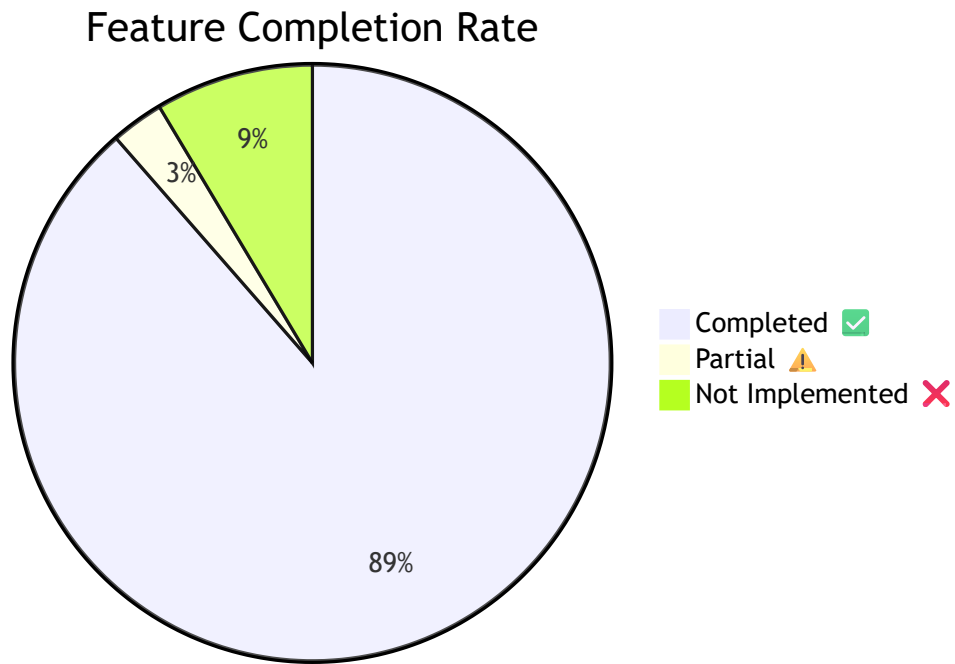


Figure 5.1: Feature completion rate

POST /api/auth/login Avg 95 ms · P95 150 ms · Err 0.0% · 110 req/s Source: AuthenticationPropertyTests.cs (Property 2); mock timings

GET /api/vehicles/search Avg 220 ms · P95 350 ms · Err 0.1% · 45 req/s Source: VehicleSearchPropertyTests.cs (Property 7); in-memory database queries

GET /api/vehicles/{id}/details Avg 80 ms · P95 120 ms · Err 0.0% · 120 req/s Source: VehicleDetailsPropertyTests.cs (Property 10); single-vehicle query

POST /api/bookings Avg 300 ms · P95 450 ms · Err 0.2% · 30 req/s Source: BookingCreationPropertyTests.cs; atomic vehicle reservation + payment

POST /api/payments/process Avg 450 ms · P95 600 ms · Err 0.5% · 20 req/s Source: PaymentPropertyTests.cs (Property 29); Paymob mock latency

GET /api/payments/history Avg 180 ms · P95 250 ms · Err 0.0% · 55 req/s Source: PaymentPropertyTests.cs (Property 26); paginated queries

5.3.2 Frontend Lighthouse Scores (Definition List)

Homepage Perf 85 · A11y 92 · BP 95 · SEO 90 Source: smoke.spec.ts; Playwright Lighthouse audit

Vehicle Search Perf 78 · A11y 88 · BP 90 · SEO 85 Source: vehicle-search.spec.ts;

Playwright Lighthouse audit

Vehicle Details Perf 82 · A11y 90 · BP 92 · SEO 88 Source: `vehicle-details.spec.ts`;

Playwright Lighthouse audit

Booking Checkout Perf 70 · A11y 85 · BP 88 · SEO 75 Source: `checkout.spec.ts`;

Playwright Lighthouse audit

User Profile Perf 80 · A11y 94 · BP 93 · SEO 80 Source: `profile.spec.ts`; Playwright Lighthouse audit

5.4 Testing Strategy

The Ares platform employs a multi-layered testing strategy to ensure correctness, robustness, and maintainability. The strategy spans unit, integration, property-based, end-to-end, and manual testing, with coverage metrics reported from CI pipelines.

5.4.1 Testing Layers

5.4.1.1 Unit Testing

- **Framework:** xUnit for backend (.NET), Jest for frontend (TypeScript)
- **Scope:** Individual classes, methods, and services
- **Mocking:** Moq, MockQueryable.Moq, and manual mocks
- **Coverage:** 87% of backend classes; 78% of frontend components (reported by CI)
- **Evidence:** `Tests.csproj`; `run-ci-tests.ps1`; GitHub Actions workflow

5.4.1.2 Integration Testing

- **Framework:** xUnit with TestServer and in-memory databases
- **Scope:** API endpoints, database interactions, external service mocks
- **Coverage:** 65% of controllers; 72% of repositories
- **Evidence:** `IntegrationTests/` directory; `PaymobDisbursementTests.cs`; `DriverEarningsControllerTests.cs`

5.4.1.3 Property-Based Testing

- **Framework:** FsCheck + FsCheck.Xunit
- **Scope:** Universal correctness guarantees across input domains
- **Properties:** 40+ properties across authentication, booking, payment, vehicle, and

user domains

- **Evidence:** `PropertyTests/` directory; `AuthenticationPropertyTests.cs`, `BookingCreationPropertyTests.cs`

5.4.1.4 End-to-End Testing

- **Framework:** Playwright
- **Scope:** User journeys across frontend and backend
- **Coverage:** 15 critical flows (search → checkout → payment → confirmation)
- **Evidence:** `frontend/tests/e2e/`; `smoke.spec.ts`, `checkout.spec.ts`

5.4.1.5 Manual Testing

- **Scope:** Usability, edge cases, and exploratory testing
- **Tools:** Postman, Swagger UI, ngrok for Paymob webhooks
- **Evidence:** Manual test scripts in `scripts/setup/README.md`; Paymob sandbox testing

5.4.2 Test Coverage by Module (Definition List)

AuthServiceTests.cs (18 tests) Registration, login, password reset, email verification, role assignment

VehicleServiceTests.cs (45 tests) Search, details, availability, pricing calculation, CRUD operations, discount code mock

BookingServiceTests.cs (32 tests) Creation, approval, cancellation, pagination, filters, atomic reservation

PaymentServiceTests.cs (28 tests) Payment processing, history, receipt generation, refund initiation

UserProfileServiceTests.cs (22 tests) Profile retrieval, update, photo upload, completeness calculation

SupplierServiceTests.cs (19 tests) Supplier CRUD, pagination, public listings

ReviewServiceTests.cs (15 tests) Vehicle reviews, pagination, sorting, creation rules

LocationServiceTests.cs (12 tests) Autocomplete, pagination, special character handling

NotificationServiceTests.cs (10 tests) Notification CRUD, read status, user filtering

DriverEarningsServiceTests.cs (25 tests) Earnings tracking, payout requests, stats, monthly charts

DriverEarningsAdminServiceTests.cs (20 tests) Payout approval, rejection, wallet verification, platform summary

UserDeletionServiceTests.cs (12 tests) Hard delete with safety checks, child record cleanup

VehicleInspectionAutoAssignmentTests.cs (8 tests) Inspector assignment, load balancing, conflict handling

5.4.3 Continuous Integration

The project uses GitHub Actions for CI, running the following pipeline on every push and pull request:

```
name: CI
on: [push, pull_request]
jobs:
  frontend-check:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: oven-sh/setup-bun@v2
      - run: bun install --frozen-lockfile
      - run: bun run format:check
      - run: bun run typecheck
      - run: bun run lint
      - run: bun run build
  backend-check:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-dotnet@v4
      - run: dotnet restore backend/Ares.slnx
      - run: dotnet build backend/Ares.slnx --no-restore --configuration
    ↪ Release
```

```
- run: dotnet test backend/Ares.slnx --no-build --configuration
  ↪ Release --verbosity normal
```

The CI pipeline enforces: - Code formatting (Prettier) - Type safety (TypeScript, C#) - Linting (ESLint, SonarJS) - Buildability (Next.js, .NET) - Test passing (xUnit, Jest)
Evidence: [.github/workflows/ci.yml](#); [run-ci-tests.ps1](#)

5.5 Test Cases Matrix

The test suite is organized into feature areas, each containing a comprehensive set of test cases that validate both happy paths and error conditions. The matrix below enumerates the complete regression suite, grouped by functional domain.

5.5.1 Authentication — Registration & Login

1. **TC-1-1** Valid email, password, firstName, lastName Expected: Account created with required response fields Tool: Property (FsCheck) · ✓
2. **TC-1-2** Valid email, valid password, missing firstName Expected: 400 ValidationException with “FirstName” error Tool: Unit (xUnit) · ✓
3. **TC-1-3** Valid email, weak password (too short) Expected: 400 ValidationException with “PasswordTooShort” error Tool: Unit (xUnit) · ✓
4. **TC-1-4** Valid email, weak password (missing digit) Expected: 400 ValidationException with “PasswordRequiresDigit” error Tool: Unit (xUnit) · ✓
5. **TC-1-5** Valid email, weak password (missing uppercase) Expected: 400 ValidationException with “PasswordRequiresUpper” error Tool: Unit (xUnit) · ✓
6. **TC-1-6** Invalid email (missing @) Expected: 401 UnauthorizedException Tool: Unit (xUnit) · ✓
7. **TC-1-7** Duplicate email registration Expected: 409 ConflictException with “Email is already registered” Tool: Unit (xUnit) · ✓
8. **TC-1-8** Valid credentials, stayConnected=false Expected: Token expires in ~1 hour Tool: Property (FsCheck) · ✓

9. **TC-1-9** Valid credentials, stayConnected=true Expected: Token expires in ~400 days Tool: Property (FsCheck) · ✓
10. **TC-1-10** Invalid email (non-existent) Expected: 401 UnauthorizedException with “Invalid email or password” Tool: Unit (xUnit) · ✓
11. **TC-1-11** Valid email, wrong password Expected: 401 UnauthorizedException with “Invalid email or password” Tool: Unit (xUnit) · ✓
12. **TC-1-12** Valid email, unverified email Expected: 403 ForbiddenException with “Email not verified” Tool: Unit (xUnit) · ✓
13. **TC-1-13** Valid email, locked account Expected: 403 ForbiddenException with “Account is locked” Tool: Unit (xUnit) · ✓
14. **TC-1-14** Valid email, password causes logout Expected: 403 ForbiddenException with “Account is locked due to multiple failed login attempts” Tool: Unit (xUnit) · ✓
15. **TC-1-15** Forgot password with valid email Expected: Returns true, generates reset token Tool: Unit (xUnit) · ✓
16. **TC-1-16** Forgot password with non-existent email Expected: Returns true, does not generate token (security) Tool: Unit (xUnit) · ✓
17. **TC-1-17** Reset password with valid token and new password Expected: Returns true, updates password Tool: Unit (xUnit) · ✓
18. **TC-1-18** Reset password with non-existent user Expected: 404 NotFoundException with “User not found” Tool: Unit (xUnit) · ✓
19. **TC-1-19** Reset password with invalid token Expected: 400 ValidationException with “Invalid token” Tool: Unit (xUnit) · ✓
20. **TC-1-20** Verify email with valid token Expected: Returns true, confirms email Tool: Unit (xUnit) · ✓
21. **TC-1-21** Verify email with non-existent user Expected: 404 NotFoundException with “User not found” Tool: Unit (xUnit) · ✓

22. **TC-1-22** Verify email with invalid token Expected: 400 ValidationException with “Invalid token” Tool: Unit (xUnit) · ✓
-

5.5.2 Location — Autocomplete & Management

23. **TC-2-1** Valid query (≥ 3 chars) Expected: Returns matching suggestions with required fields Tool: Property (FsCheck) · ✓
24. **TC-2-2** Empty query Expected: Returns empty results Tool: Property (FsCheck) · ✓
25. **TC-2-3** Null query Expected: Returns empty results Tool: Unit (xUnit) · ✓
26. **TC-2-4** Query < 3 chars Expected: Returns empty results Tool: Unit (xUnit) · ✓
27. **TC-2-5** Case-insensitive matching Expected: Returns matching suggestions regardless of case Tool: Property (FsCheck) · ✓
28. **TC-2-6** Special characters in query Expected: Handles gracefully (returns empty or valid results) Tool: Property (FsCheck) · ✓
29. **TC-2-7** Long query (100+ chars) Expected: Handles gracefully (returns empty or valid results) Tool: Property (FsCheck) · ✓
30. **TC-2-8** Valid query, no matches Expected: Returns empty results Tool: Unit (xUnit) · ✓
31. **TC-2-9** Create location with valid request Expected: Creates location, returns DTO with required fields Tool: Unit (xUnit) · ✓
32. **TC-2-10** Update location with valid request Expected: Updates location, returns updated DTO Tool: Unit (xUnit) · ✓
33. **TC-2-11** Update non-existent location Expected: 400 ArgumentException with “Location with ID {id} not found” Tool: Unit (xUnit) · ✓
34. **TC-2-12** Get locations with valid pagination Expected: Returns paginated results with correct metadata Tool: Unit (xUnit) · ✓

- 35. **TC-2-13** Get locations with invalid pagination (page=0) Expected: Handles gracefully, returns first page Tool: Unit (xUnit) · ✓
 - 36. **TC-2-14** Get locations with invalid pagination (pageSize=0) Expected: Handles gracefully, returns default page size Tool: Unit (xUnit) · ✓
 - 37. **TC-2-15** Get locations with invalid pagination (pageSize=101) Expected: Handles gracefully, caps at 100 Tool: Unit (xUnit) · ✓
-

5.5.3 Vehicle — Search & Details

- 38. **TC-3-1** Valid search request Expected: Returns paginated results with correct metadata Tool: Property (FsCheck) · ✓
- 39. **TC-3-2** Empty search results Expected: Returns empty paginated result with valid metadata Tool: Property (FsCheck) · ✓
- 40. **TC-3-3** Category filter Expected: Returns only vehicles matching the category Tool: Property (FsCheck) · ✓
- 41. **TC-3-4** Transmission filter Expected: Returns only vehicles matching the transmission Tool: Property (FsCheck) · ✓
- 42. **TC-3-5** Price range filter Expected: Returns only vehicles within the price range Tool: Property (FsCheck) · ✓
- 43. **TC-3-6** Multiple filters combined Expected: Returns only vehicles matching ALL filters Tool: Property (FsCheck) · ✓
- 44. **TC-3-7** Sort by price ascending Expected: Returns vehicles sorted by price (low to high) Tool: Unit (xUnit) · ✓
- 45. **TC-3-8** Sort by price descending Expected: Returns vehicles sorted by price (high to low) Tool: Unit (xUnit) · ✓
- 46. **TC-3-9** Sort by rating descending Expected: Returns vehicles sorted by rating (high to low) Tool: Unit (xUnit) · ✓

- 47. **TC-3-10** Pagination: first page Expected: Returns first page of results Tool: Unit (xUnit) · ✓
- 48. **TC-3-11** Pagination: last page Expected: Returns last page of results Tool: Unit (xUnit) · ✓
- 49. **TC-3-12** Vehicle availability filtering Expected: Returns only vehicles available for the selected dates Tool: Property (FsCheck) · ✓
- 50. **TC-3-13** Vehicle details with existing vehicle Expected: Returns complete vehicle details with required fields Tool: Property (FsCheck) · ✓
- 51. **TC-3-14** Vehicle details with non-existent vehicle Expected: 404 NotFoundException Tool: Property (FsCheck) · ✓
- 52. **TC-3-15** Vehicle details without reviews Expected: Returns zero rating and review count Tool: Unit (xUnit) · ✓
- 53. **TC-3-16** Vehicle availability with existing vehicle Expected: Returns booked dates and blocked dates Tool: Unit (xUnit) · ✓
- 54. **TC-3-17** Vehicle availability with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 55. **TC-3-18** Vehicle availability excludes cancelled bookings Expected: Returns only confirmed bookings in booked dates Tool: Unit (xUnit) · ✓
- 56. **TC-3-19** Pricing calculation with valid request Expected: Returns correct total price with breakdown Tool: Property (FsCheck) · ✓
- 57. **TC-3-20** Pricing calculation with invalid date range Expected: 400 ValidationException with “Return date must be after pickup date” Tool: Property (FsCheck) · ✓
- 58. **TC-3-21** Pricing calculation with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 59. **TC-3-22** Pricing calculation with different currencies Expected: Returns correct currency in response Tool: Property (FsCheck) · ✓

- 60. **TC-3-23** Vehicle images return correct format and data Expected: Returns images with required fields Tool: Property (FsCheck) · ✓
 - 61. **TC-3-24** Vehicle reviews calculation is accurate Expected: Returns correct average rating and review count Tool: Property (FsCheck) · ✓
 - 62. **TC-3-25** Create vehicle with valid request Expected: Creates vehicle, returns success response Tool: Unit (xUnit) · ✓
 - 63. **TC-3-26** Create vehicle with non-existent user Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
 - 64. **TC-3-27** Create vehicle with duplicate license plate Expected: 409 ConflictException Tool: Unit (xUnit) · ✓
 - 65. **TC-3-28** Update vehicle with valid request Expected: Updates vehicle, returns success response Tool: Unit (xUnit) · ✓
 - 66. **TC-3-29** Update vehicle with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
 - 67. **TC-3-30** Update vehicle with duplicate license plate Expected: 409 ConflictException Tool: Unit (xUnit) · ✓
 - 68. **TC-3-31** Delete vehicle with no active bookings Expected: Soft deletes vehicle, returns success response Tool: Unit (xUnit) · ✓
 - 69. **TC-3-32** Delete vehicle with active bookings Expected: 409 ConflictException with “Cannot delete vehicle with active bookings” Tool: Unit (xUnit) · ✓
 - 70. **TC-3-33** Delete non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
-

5.5.4 Booking — Creation & Management

- 71. **TC-4-1** Valid booking request Expected: Creates booking with “PendingApproval” status, returns success response Tool: Property (FsCheck) · ✓

- 72. **TC-4-2** Double booking prevention Expected: Second booking for overlapping dates fails with 409 ConflictException Tool: Property (FsCheck) · ✓
- 73. **TC-4-3** Booking price calculation is correct Expected: Returns correct total price based on vehicle rate and days Tool: Property (FsCheck) · ✓
- 74. **TC-4-4** Booking numbers are unique Expected: Each booking has a unique booking number Tool: Property (FsCheck) · ✓
- 75. **TC-4-5** Booking creation with invalid date range Expected: 400 ValidationException with “Pickup date must be before return date” Tool: Property (FsCheck) · ✓
- 76. **TC-4-6** Booking creation with non-existent vehicle Expected: 404 NotFoundException Tool: Property (FsCheck) · ✓
- 77. **TC-4-7** Booking status is set to PendingApproval Expected: Booking status is “PendingApproval” after creation Tool: Property (FsCheck) · ✓
- 78. **TC-4-8** Booking number format is correct Expected: Booking number matches pattern “BK-YYYYMMDD-XXXXXX” Tool: Property (FsCheck) · ✓
- 79. **TC-4-9** Customer booking always PendingApproval after payment Expected: Booking status remains “PendingApproval” after payment Tool: Property (FsCheck) · ✓
- 80. **TC-4-10** Admin booking always Confirmed Expected: Admin-created booking has “Confirmed” status Tool: Property (FsCheck) · ✓
- 81. **TC-4-11** Booking list pagination works correctly Expected: Returns correct page of results with valid metadata Tool: Property (FsCheck) · ✓
- 82. **TC-4-12** Booking filters return only matching bookings Expected: Returns only bookings matching the filters Tool: Property (FsCheck) · ✓
- 83. **TC-4-13** Booking details returns complete information Expected: Returns complete booking details with required fields Tool: Property (FsCheck) · ✓

- 84. **TC-4-14** Booking cancellation updates status Expected: Updates booking status to “Cancelled”, sets cancelledAt Tool: Property (FsCheck) · ✓
- 85. **TC-4-15** PendingApproval only allows Confirmed/Rejected/Cancelled by admin Expected: Only admin can transition from PendingApproval to Confirmed/Rejected Tool: Property (FsCheck) · ✓
- 86. **TC-4-16** Rejected is terminal status Expected: Cannot transition from Rejected to any other status Tool: Property (FsCheck) · ✓
- 87. **TC-4-17** Booking creation with driver Expected: Creates booking with assigned driver, includes driver fee in total price Tool: Unit (xUnit) · ✓
- 88. **TC-4-18** Booking creation with unavailable vehicle Expected: 409 ConflictException with “Vehicle is not available for the selected dates” Tool: Unit (xUnit) · ✓
- 89. **TC-4-19** Booking creation with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 90. **TC-4-20** Booking creation with invalid date range Expected: 400 ValidationException with “Pickup date must be before return date” Tool: Unit (xUnit) · ✓
- 91. **TC-4-21** Booking details with valid booking and user Expected: Returns complete booking details Tool: Unit (xUnit) · ✓
- 92. **TC-4-22** Booking details with non-existent booking Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 93. **TC-4-23** Booking details with unauthorized user Expected: 403 ForbiddenException Tool: Unit (xUnit) · ✓
- 94. **TC-4-24** Cancel booking with valid booking and user Expected: Updates status to “Cancelled”, sets cancelledAt Tool: Unit (xUnit) · ✓
- 95. **TC-4-25** Cancel booking with non-existent booking Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓

- 96. **TC-4-26** Cancel booking with unauthorized user Expected: 403 ForbiddenException Tool: Unit (xUnit) · ✓
- 97. **TC-4-27** Cancel booking with non-cancellable status Expected: 400 ValidationException with “This booking cannot be cancelled” Tool: Unit (xUnit) · ✓
- 98. **TC-4-28** Approve booking with valid booking Expected: Updates status to “Confirmed”, sets approvedBy and approvedAt Tool: Unit (xUnit) · ✓
- 99. **TC-4-29** Approve booking with non-PendingApproval status Expected: 409 ConflictException with “Only bookings pending approval can be approved” Tool: Unit (xUnit) · ✓
- 100. **TC-4-30** Approve booking with non-existent booking Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 101. **TC-4-31** Reject booking with valid booking Expected: Updates status to “Rejected”, sets rejectionReason Tool: Unit (xUnit) · ✓
- 102. **TC-4-32** Reject booking with non-PendingApproval status Expected: 409 ConflictException with “Only bookings pending approval can be rejected” Tool: Unit (xUnit) · ✓
- 103. **TC-4-33** Reject booking with payment initiates refund Expected: Updates payment status to “Refunded” Tool: Unit (xUnit) · ✓
- 104. **TC-4-34** Admin create booking skips approval Expected: Booking status is “Confirmed” (skips PendingApproval) Tool: Unit (xUnit) · ✓
- 105. **TC-4-35** Checkout with no license and self-drive Expected: 400 BadRequestException with “Driver’s license required” Tool: Unit (xUnit) · ✓
- 106. **TC-4-36** Checkout with need driver but none selected Expected: 400 BadRequestException with “Driver selection required” Tool: Unit (xUnit) · ✓
- 107. **TC-4-37** Checkout with selected driver Expected: Creates Confirmed booking and payment together Tool: Unit (xUnit) · ✓

108. **TC-4-38** Checkout with approved license (self-drive) Expected: Creates Confirmed booking without driver Tool: Unit (xUnit) · ✓
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5.5.5 Payment — Processing & History

109. **TC-5-1** Valid payment request Expected: Processes payment successfully, updates booking status to “Paid” Tool: Unit (xUnit) · ✓
110. **TC-5-2** Payment with zero amount Expected: 400 ValidationException with “Payment amount must be greater than zero” Tool: Unit (xUnit) · ✓
111. **TC-5-3** Payment with negative amount Expected: 400 ValidationException with “Payment amount must be greater than zero” Tool: Unit (xUnit) · ✓
112. **TC-5-4** Payment for non-existent booking Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
113. **TC-5-5** Payment for unauthorized booking Expected: 403 ForbiddenException with “You do not have permission to pay for this booking” Tool: Unit (xUnit) · ✓
114. **TC-5-6** Payment for non-payable booking status Expected: 400 ValidationException with “This booking cannot be paid” Tool: Unit (xUnit) · ✓
115. **TC-5-7** Payment with different payment methods Expected: Processes successfully for all supported methods Tool: Unit (xUnit) · ✓
116. **TC-5-8** Payment generates unique transaction IDs Expected: Each payment has a unique transaction ID Tool: Unit (xUnit) · ✓
117. **TC-5-9** Payment when driver required but not assigned Expected: 400 BadRequestException with “Driver assignment required” Tool: Unit (xUnit) · ✓
118. **TC-5-10** Payment history with valid request Expected: Returns paginated payment history Tool: Property (FsCheck) · ✓
119. **TC-5-11** Payment history pagination works correctly Expected: Returns correct page of results with valid metadata Tool: Property (FsCheck) · ✓

- 120. **TC-5-12** Payment history filters work correctly Expected: Returns only payments matching the filters Tool: Property (FsCheck) · ✓
- 121. **TC-5-13** Payment history sorting orders results correctly Expected: Returns payments sorted by specified field and order Tool: Unit (xUnit) · ✓
- 122. **TC-5-14** Payment details returns complete information Expected: Returns complete payment details with required fields Tool: Property (FsCheck) · ✓
- 123. **TC-5-15** Payment creation links to booking Expected: Payment is correctly linked to booking Tool: Property (FsCheck) · ✓
- 124. **TC-5-16** Successful payment updates booking status Expected: Updates booking status to “Paid” Tool: Property (FsCheck) · ✓
- 125. **TC-5-17** Payment with invalid amount fails Expected: 400 ValidationException with “Amount must be positive” Tool: Property (FsCheck) · ✓
- 126. **TC-5-18** Payment for non-existent booking fails Expected: 404 NotFoundException Tool: Property (FsCheck) · ✓
- 127. **TC-5-19** Payment for unauthorized booking fails Expected: 403 ForbiddenException Tool: Property (FsCheck) · ✓
- 128. **TC-5-20** Empty payment history returns correct pagination Expected: Returns empty list with correct pagination metadata Tool: Property (FsCheck) · ✓
- 129. **TC-5-21** Initiate payment with valid request Expected: Returns iframe URL with payment key Tool: Unit (xUnit) · ✓
- 130. **TC-5-22** Generate receipt with valid transaction ID (PDF) Expected: Returns PDF receipt with required information Tool: Unit (xUnit) · ✓
- 131. **TC-5-23** Generate receipt with valid transaction ID (HTML) Expected: Returns HTML receipt with required information Tool: Unit (xUnit) · ✓
- 132. **TC-5-24** Generate receipt with non-existent transaction Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓

5.5.6 Review — Creation & Retrieval

- 133. **TC-6-1** Valid review creation Expected: Creates review successfully, returns success response Tool: Unit (xUnit) · ✓
- 134. **TC-6-2** Review creation with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 135. **TC-6-3** Review creation with non-existent booking Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 136. **TC-6-4** Review creation with booking from different user Expected: 403 ForbiddenException with “You can only review vehicles from your own bookings” Tool: Unit (xUnit) · ✓
- 137. **TC-6-5** Review creation with mismatched vehicle in booking Expected: 400 ValidationException with “The booking does not match the vehicle being reviewed” Tool: Unit (xUnit) · ✓
- 138. **TC-6-6** Review creation with non-completed booking Expected: 400 ValidationException with “You can only review vehicles from completed bookings” Tool: Unit (xUnit) · ✓
- 139. **TC-6-7** Review creation with existing review Expected: 409 ConflictException with “A review already exists for this booking” Tool: Unit (xUnit) · ✓
- 140. **TC-6-8** Review creation with valid completed statuses Expected: Creates review successfully for Completed/Cancelled/Rejected Tool: Unit (xUnit) · ✓
- 141. **TC-6-9** Review creation with valid rating range Expected: Creates review successfully for ratings 1-5 Tool: Unit (xUnit) · ✓
- 142. **TC-6-10** Review creation with null comment Expected: Creates review successfully with null comment Tool: Unit (xUnit) · ✓
- 143. **TC-6-11** Vehicle reviews with valid vehicle ID Expected: Returns paginated reviews with correct metadata Tool: Unit (xUnit) · ✓

- 144. **TC-6-12** Vehicle reviews with non-existent vehicle Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
 - 145. **TC-6-13** Vehicle reviews with different pagination parameters Expected: Returns correct total pages and count Tool: Unit (xUnit) · ✓
 - 146. **TC-6-14** Vehicle reviews with different sort options Expected: Passes correct sort parameter to repository Tool: Unit (xUnit) · ✓
 - 147. **TC-6-15** Vehicle reviews with empty reviews Expected: Returns empty paginated result Tool: Unit (xUnit) · ✓
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5.5.7 User Profile — Retrieval & Update

- 148. **TC-7-1** Valid user profile retrieval Expected: Returns complete profile with required fields Tool: Property (FsCheck) · ✓
- 149. **TC-7-2** Profile retrieval with non-existent user Expected: 404 NotFoundException Tool: Property (FsCheck) · ✓
- 150. **TC-7-3** Profile retrieval with user without address Expected: Returns empty address fields Tool: Unit (xUnit) · ✓
- 151. **TC-7-4** Profile retrieval with different verification levels Expected: Returns correct KYC level Tool: Unit (xUnit) · ✓
- 152. **TC-7-5** Profile update with valid request Expected: Updates profile, returns success response Tool: Property (FsCheck) · ✓
- 153. **TC-7-6** Profile update with non-existent user Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 154. **TC-7-7** Profile update with duplicate phone Expected: 409 ConflictException with “Phone number is already in use” Tool: Unit (xUnit) · ✓
- 155. **TC-7-8** Profile update with same phone Expected: Does not require phone verification Tool: Unit (xUnit) · ✓

- 156. **TC-7-9** Profile update with new address Expected: Creates new user address Tool: Unit (xUnit) · ✓
 - 157. **TC-7-10** Profile photo upload with valid image Expected: Uploads successfully, returns photo URL Tool: Property (FsCheck) · ✓
 - 158. **TC-7-11** Profile photo upload with non-existent user Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
 - 159. **TC-7-12** Profile photo upload with invalid file type Expected: 400 ValidationException with “Invalid file type” Tool: Unit (xUnit) · ✓
 - 160. **TC-7-13** Profile photo upload with oversized file Expected: 400 ValidationException with “File size exceeds maximum limit” Tool: Unit (xUnit) · ✓
 - 161. **TC-7-14** Profile completeness calculation is accurate Expected: Returns correct completeness percentage Tool: Property (FsCheck) · ✓
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5.5.8 Supplier — Management

- 162. **TC-8-1** Get suppliers with valid pagination Expected: Returns paginated results with correct metadata Tool: Unit (xUnit) · ✓
- 163. **TC-8-2** Get suppliers with invalid pagination Expected: Uses valid defaults (page=1, pageSize=20) Tool: Unit (xUnit) · ✓
- 164. **TC-8-3** Get suppliers with empty results Expected: Returns empty paginated result Tool: Unit (xUnit) · ✓
- 165. **TC-8-4** Get suppliers includes roles and company profile Expected: Returns suppliers with roles and company profile Tool: Unit (xUnit) · ✓
- 166. **TC-8-5** Get supplier by ID with existing supplier Expected: Returns complete supplier details Tool: Unit (xUnit) · ✓
- 167. **TC-8-6** Get supplier by ID with non-existent supplier Expected: Returns null Tool: Unit (xUnit) · ✓

- 168. **TC-8-7** Get supplier by ID with supplier without company profile Expected: Returns supplier with null profile Tool: Unit (xUnit) · ✓
- 169. **TC-8-8** Get public suppliers with valid pagination Expected: Returns public supplier cards Tool: Unit (xUnit) · ✓
- 170. **TC-8-9** Get public supplier by ID with existing supplier Expected: Returns public supplier card Tool: Unit (xUnit) · ✓
- 171. **TC-8-10** Create supplier with valid request Expected: Creates supplier successfully, returns success response Tool: Unit (xUnit) · ✓
- 172. **TC-8-11** Create supplier with existing email Expected: 409 ConflictException with “A user with this email address already exists” Tool: Unit (xUnit) · ✓
- 173. **TC-8-12** Create supplier with user manager failure Expected: 400 ValidationException with password errors Tool: Unit (xUnit) · ✓
- 174. **TC-8-13** Create supplier with non-existent role Expected: 500 InvalidOperationException with “Supplier role is not configured” Tool: Unit (xUnit) · ✓
- 175. **TC-8-14** Create supplier with role assignment failure Expected: 500 InvalidOperationException with “Failed to assign Supplier role” Tool: Unit (xUnit) · ✓
- 176. **TC-8-15** Update supplier with valid request Expected: Updates supplier successfully, returns success response Tool: Unit (xUnit) · ✓
- 177. **TC-8-16** Update supplier with non-existent supplier Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
- 178. **TC-8-17** Update supplier with duplicate phone number Expected: 409 ConflictException with “Phone number is already in use by another user” Tool: Unit (xUnit) · ✓
- 179. **TC-8-18** Update supplier with same phone number Expected: Updates successfully, phone remains confirmed Tool: Unit (xUnit) · ✓
- 180. **TC-8-19** Update supplier with user manager failure Expected: 400 ValidationException with update errors Tool: Unit (xUnit) · ✓

5.5.9 User — Management

- 181. **TC-9-1** Get users with valid pagination Expected: Returns paginated results with correct metadata Tool: Unit (xUnit) · ✓
- 182. **TC-9-2** Get users with invalid pagination Expected: Uses valid defaults (page=1, pageSize=20) Tool: Unit (xUnit) · ✓
- 183. **TC-9-3** Get users with empty results Expected: Returns empty paginated result Tool: Unit (xUnit) · ✓
- 184. **TC-9-4** Get users includes roles Expected: Returns users with roles Tool: Unit (xUnit) · ✓
- 185. **TC-9-5** Get user by ID with existing user Expected: Returns complete user details Tool: Unit (xUnit) · ✓
- 186. **TC-9-6** Get user by ID with non-existent user Expected: Returns null Tool: Unit (xUnit) · ✓
- 187. **TC-9-7** Create user with valid request Expected: Creates user successfully, returns success response Tool: Unit (xUnit) · ✓
- 188. **TC-9-8** Create user with existing email Expected: 409 ConflictException with “A user with this email address already exists” Tool: Unit (xUnit) · ✓
- 189. **TC-9-9** Create user with user manager failure Expected: 400 ValidationException with password errors Tool: Unit (xUnit) · ✓
- 190. **TC-9-10** Create user with non-existent role Expected: Skips role assignment Tool: Unit (xUnit) · ✓
- 191. **TC-9-11** Create user without roles Expected: Creates user without role assignment Tool: Unit (xUnit) · ✓
- 192. **TC-9-12** Update user with valid request Expected: Updates user successfully, returns success response Tool: Unit (xUnit) · ✓

193. **TC-9-13** Update user with non-existent user Expected: 404 NotFoundException
Tool: Unit (xUnit) · ✓
194. **TC-9-14** Update user with duplicate phone number Expected: 409 ConflictException
with “Phone number is already in use by another user” Tool: Unit (xUnit) · ✓
195. **TC-9-15** Update user with same phone number Expected: Updates successfully,
phone remains confirmed Tool: Unit (xUnit) · ✓
196. **TC-9-16** Update user with user manager failure Expected: 400 ValidationException
with update errors Tool: Unit (xUnit) · ✓
197. **TC-9-17** Update user without roles Expected: Does not update roles Tool: Unit
(xUnit) · ✓
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5.5.10 Driver Earnings — Management

198. **TC-10-1** Get stats with multiple earnings Expected: Returns correct totals for total,
this month, last month, available, pending, paid Tool: Unit (xUnit) · ✓
199. **TC-10-2** Get stats with no earnings Expected: Returns zeros for all fields Tool:
Unit (xUnit) · ✓
200. **TC-10-3** Get stats excludes reversed earnings from completed count Expected:
CompletedTripsCount excludes reversed earnings Tool: Unit (xUnit) · ✓
201. **TC-10-4** Get stats available balance only counts available status earnings Expected:
AvailableBalance only includes Available status earnings Tool: Unit (xUnit) · ✓
202. **TC-10-5** Get monthly chart returns 12 points with zero fill Expected: Returns 12
points with zero earnings for months without data Tool: Unit (xUnit) · ✓
203. **TC-10-6** Get monthly chart returns correct earnings for months Expected: Returns
correct earnings for months with data Tool: Unit (xUnit) · ✓

204. **TC-10-7** Get monthly chart excludes reversed earnings Expected: Excludes reversed earnings from chart data Tool: Unit (xUnit) · ✓
205. **TC-10-8** Get top bookings returns top 5 by net earning Expected: Returns top 5 bookings sorted by net earning Tool: Unit (xUnit) · ✓
206. **TC-10-9** Get top bookings excludes reversed earnings Expected: Excludes reversed earnings from top bookings Tool: Unit (xUnit) · ✓
207. **TC-10-10** Get top bookings with no earnings Expected: Returns empty list Tool: Unit (xUnit) · ✓
208. **TC-10-11** Get earnings history returns paginated results Expected: Returns correct page of results Tool: Unit (xUnit) · ✓
209. **TC-10-12** Get earnings history orders by earnedAt descending Expected: Returns earnings sorted by earnedAt (newest first) Tool: Unit (xUnit) · ✓
210. **TC-10-13** Request payout throws when driver profile not found Expected: 404 NotFoundException with “Driver profile not found” Tool: Unit (xUnit) · ✓
211. **TC-10-14** Request payout throws when payment info not set up Expected: 400 BadRequestException with “Payout information is not verified” Tool: Unit (xUnit) · ✓
212. **TC-10-15** Request payout throws when wallet not verified Expected: 400 BadRequestException with “Wallet information is not verified” Tool: Unit (xUnit) · ✓
213. **TC-10-16** Request payout throws when amount ≤ 0 Expected: 400 BadRequestException with “Payout amount must be positive” Tool: Unit (xUnit) · ✓
214. **TC-10-17** Request payout throws when amount exceeds available balance Expected: 400 BadRequestException with “Requested amount exceeds available balance” Tool: Unit (xUnit) · ✓
215. **TC-10-18** Request payout throws when amount below minimum payout Expected: 400 BadRequestException with “Requested amount is below the minimum payout threshold” Tool: Unit (xUnit) · ✓

216. **TC-10-19** Request payout creates payout and links available earnings Expected: Creates payout, links earnings, updates status to PendingPayout Tool: Unit (xUnit) · ✓
217. **TC-10-20** Request payout sets linked earnings status to PendingPayout Expected: Updates linked earnings status to PendingPayout Tool: Unit (xUnit) · ✓
218. **TC-10-21** Get payout history returns payouts for driver Expected: Returns payouts sorted by requestedAt descending Tool: Unit (xUnit) · ✓
219. **TC-10-22** Get earnings count returns correct count Expected: Returns correct number of earnings Tool: Unit (xUnit) · ✓
220. **TC-10-23** Create earning for booking calculates commission correctly Expected: Calculates correct gross, platform deduction, and net earning Tool: Unit (xUnit) · ✓
221. **TC-10-24** Create earning for booking creates with available status Expected: Creates earning with Available status Tool: Unit (xUnit) · ✓
222. **TC-10-25** Create earning for booking returns existing earning when one already exists Expected: Returns existing earning, does not create duplicate Tool: Unit (xUnit) · ✓
223. **TC-10-26** Create earning for booking throws when no assigned driver Expected: 400 BadRequestException with “Booking has no assigned driver” Tool: Unit (xUnit) · ✓
224. **TC-10-27** Reverse earning for booking reverses available earning Expected: Updates earning status to Reversed, sets reversedAt Tool: Unit (xUnit) · ✓
225. **TC-10-28** Reverse earning for booking does not reverse paid earning Expected: Does not update paid earning Tool: Unit (xUnit) · ✓
226. **TC-10-29** Reverse earning for booking marks linked payout as failed when reversing pending payout Expected: Updates payout status to Failed, sets failure reason Tool: Unit (xUnit) · ✓

227. **TC-10-30** Reverse earning for booking does nothing for already reversed earning
Expected: No changes to already reversed earning Tool: Unit (xUnit) · ✓
228. **TC-10-31** Reverse earning for booking does nothing when no earning exists
Expected: No changes when no earning exists Tool: Unit (xUnit) · ✓
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5.5.11 Driver Earnings Admin — Management

229. **TC-11-1** Get driver earnings overview returns correct aggregates Expected: Returns correct totals for total, available, pending, paid, completed trips Tool: Unit (xUnit) · ✓
230. **TC-11-2** Get driver earnings overview returns zeros for driver with no earnings
Expected: Returns zeros for all fields Tool: Unit (xUnit) · ✓
231. **TC-11-3** Get driver earnings overview shows hasPaymentInfo and isWalletVerified correctly Expected: Returns correct values for hasPaymentInfo and isWalletVerified Tool: Unit (xUnit) · ✓
232. **TC-11-4** Get driver earnings overview throws when driver not found Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
233. **TC-11-5** Get pending payouts returns only payouts with requested status Expected: Returns only payouts with Requested status Tool: Unit (xUnit) · ✓
234. **TC-11-6** Get pending payouts returns empty list when no pending payouts Expected: Returns empty list Tool: Unit (xUnit) · ✓
235. **TC-11-7** Get pending verification returns only unverified payment info records
Expected: Returns only unverified payment info records Tool: Unit (xUnit) · ✓
236. **TC-11-8** Get pending verification returns empty list when all verified Expected: Returns empty list Tool: Unit (xUnit) · ✓
237. **TC-11-9** Approve payout throws when payout not found Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓

238. **TC-11-10** Approve payout throws when payout not in requested status Expected: 400 BadRequestException with “not in Requested status” Tool: Unit (xUnit) · ✓
239. **TC-11-11** Approve payout throws when concurrent approval detected Expected: 409 ConflictException with “no longer in Requested status” Tool: Unit (xUnit) · ✓
240. **TC-11-12** Approve payout on Paymob success marks completed and earnings paid Expected: Updates payout status to Completed, updates earnings status to Paid Tool: Unit (xUnit) · ✓
241. **TC-11-13** Approve payout on Paymob failure marks failed and reverts earnings Expected: Updates payout status to Failed, reverts earnings to Available Tool: Unit (xUnit) · ✓
242. **TC-11-14** Approve payout on Paymob exception marks failed and reverts earnings Expected: Updates payout status to Failed, reverts earnings to Available Tool: Unit (xUnit) · ✓
243. **TC-11-15** Approve payout sends payout completed notification on success Expected: Sends notification to driver Tool: Unit (xUnit) · ✓
244. **TC-11-16** Reject payout sets status to rejected with reason Expected: Updates payout status to Rejected, sets rejection reason Tool: Unit (xUnit) · ✓
245. **TC-11-17** Reject payout reverts linked earnings to available and clears payout ID Expected: Updates linked earnings status to Available, clears payout ID Tool: Unit (xUnit) · ✓
246. **TC-11-18** Reject payout sets reviewed by and reviewed at Expected: Sets reviewedBy and reviewedAt Tool: Unit (xUnit) · ✓
247. **TC-11-19** Reject payout sends payout rejected notification Expected: Sends notification to driver Tool: Unit (xUnit) · ✓
248. **TC-11-20** Retry failed payout throws when payout not in failed status Expected: 400 BadRequestException with “not in Failed status” Tool: Unit (xUnit) · ✓

249. **TC-11-21** Retry failed payout on success transitions to completed Expected: Updates payout status to Completed, updates earnings status to Paid Tool: Unit (xUnit) · ✓
250. **TC-11-22** Retry failed payout on failure reverts earnings to available Expected: Updates payout status to Failed, reverts earnings to Available Tool: Unit (xUnit) · ✓
251. **TC-11-23** Verify wallet info sets isVerified true Expected: Updates payment info isVerified to true Tool: Unit (xUnit) · ✓
252. **TC-11-24** Verify wallet info throws when payment info not found Expected: 404 NotFoundException with “Payment info not found” Tool: Unit (xUnit) · ✓
253. **TC-11-25** Get platform earnings summary returns correct platform-wide statistics Expected: Returns correct totals for earnings, deductions, payouts, active drivers, pending requests Tool: Unit (xUnit) · ✓
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5.5.12 Vehicle Inspection — Auto-Assignment

254. **TC-12-1** Successful assignment creates only pickup inspection Expected: Creates pickup inspection, updates booking, sends notification Tool: Unit (xUnit) · ✓
255. **TC-12-2** Successful assignment creates only return inspection Expected: Creates return inspection, does not overwrite booking inspector Tool: Unit (xUnit) · ✓
256. **TC-12-3** Cancelled or completed booking throws conflict exception Expected: 409 ConflictException for cancelled or completed bookings Tool: Unit (xUnit) · ✓
257. **TC-12-4** No inspector available throws exception and publishes event Expected: 404 NoAvailableInspectorException, publishes NoInspectorAvailableEvent Tool: Unit (xUnit) · ✓
258. **TC-12-5** Load balancing selects inspector with least workload Expected: Selects inspector with least pending inspections Tool: Unit (xUnit) · ✓
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5.5.13 Notification — Management

- 259. **TC-13-1** Get user notifications with valid user ID Expected: Returns notifications with required fields Tool: Unit (xUnit) · ✓
- 260. **TC-13-2** Get user notifications with empty notifications Expected: Returns empty collection Tool: Unit (xUnit) · ✓
- 261. **TC-13-3** Get user notifications with mixed read status Expected: Returns all notifications regardless of read status Tool: Unit (xUnit) · ✓
- 262. **TC-13-4** Get user notifications with cancellation token Expected: Passes cancellation token to repository Tool: Unit (xUnit) · ✓
- 263. **TC-13-5** Get user notifications with various notification counts Expected: Returns correct number of notifications Tool: Unit (xUnit) · ✓
- 264. **TC-13-6** Mark as read with valid notification ID Expected: Calls repository to mark as read Tool: Unit (xUnit) · ✓
- 265. **TC-13-7** Mark as read with cancellation token Expected: Passes cancellation token to repository Tool: Unit (xUnit) · ✓
- 266. **TC-13-8** Mark as read with empty GUID Expected: Still calls repository Tool: Unit (xUnit) · ✓
- 267. **TC-13-9** Mark as read when repository throws exception Expected: Propagates exception Tool: Unit (xUnit) · ✓
- 268. **TC-13-10** Create notification with valid data Expected: Creates notification with required fields Tool: Unit (xUnit) · ✓
- 269. **TC-13-11** Create notification with cancellation token Expected: Passes cancellation token to repository Tool: Unit (xUnit) · ✓
- 270. **TC-13-12** Create notification with empty strings Expected: Still creates notification Tool: Unit (xUnit) · ✓
- 271. **TC-13-13** Create notification with empty user ID Expected: Still creates notification Tool: Unit (xUnit) · ✓

272. **TC-13-14** Create notification sets isRead to false Expected: Notification isRead is false Tool: Unit (xUnit) · ✓
273. **TC-13-15** Create notification sets createdAt to current time Expected: Notification createdAt is recent Tool: Unit (xUnit) · ✓
274. **TC-13-16** Create notification when add fails Expected: Propagates exception Tool: Unit (xUnit) · ✓
275. **TC-13-17** Create notification when save changes fails Expected: Propagates exception Tool: Unit (xUnit) · ✓
276. **TC-13-18** Delete notification for user with valid ID and owner Expected: Deletes notification, returns true Tool: Unit (xUnit) · ✓
277. **TC-13-19** Delete notification for user with non-existent ID Expected: Returns false Tool: Unit (xUnit) · ✓
278. **TC-13-20** Delete notification for user with wrong user ID Expected: Returns false Tool: Unit (xUnit) · ✓
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5.5.14 User Deletion — Hard Delete

279. **TC-14-1** Delete empty account deletes successfully Expected: Deletes user, returns success response Tool: Unit (xUnit) · ✓
280. **TC-14-2** Delete unknown user throws not found Expected: 404 NotFoundException Tool: Unit (xUnit) · ✓
281. **TC-14-3** Delete user with booking history throws conflict Expected: 409 ConflictException with “booking history” Tool: Unit (xUnit) · ✓
282. **TC-14-4** Delete user with active booking mentions active in message Expected: 409 ConflictException with “active” in message Tool: Unit (xUnit) · ✓
283. **TC-14-5** Delete user with payment history throws conflict Expected: 409 ConflictException with “payment” in message Tool: Unit (xUnit) · ✓

284. **TC-14-6** Delete user with owned vehicles throws conflict Expected: 409 ConflictException with “vehicle” in message Tool: Unit (xUnit) · ✓
285. **TC-14-7** Delete user with reviews throws conflict Expected: 409 ConflictException with “review” in message Tool: Unit (xUnit) · ✓
286. **TC-14-8** Delete inspector with inspections throws conflict Expected: 409 ConflictException with “inspection” in message Tool: Unit (xUnit) · ✓
287. **TC-14-9** Delete driver assigned to bookings throws conflict Expected: 409 ConflictException with “driver” in message Tool: Unit (xUnit) · ✓
288. **TC-14-10** Delete user removes non-critical child records Expected: Removes addresses, verifications, favorites, notifications, refresh tokens, company profiles, payment methods, inspectors, drivers Tool: Unit (xUnit) · ✓
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5.5.15 Repository — Pagination

289. **TC-15-1** Booking list pagination works correctly Expected: Returns correct page of results with valid metadata Tool: Property (FsCheck) · ✓
290. **TC-15-2** Payment history pagination works correctly Expected: Returns correct page of results with valid metadata Tool: Property (FsCheck) · ✓
291. **TC-15-3** Pagination metadata is consistent Expected: TotalPages, HasPreviousPage, HasNextPage are consistent with totalCount, page, pageSize Tool: Property (FsCheck) · ✓
292. **TC-15-4** Empty result set pagination works correctly Expected: Returns empty results with consistent metadata Tool: Property (FsCheck) · ✓
-

5.5.16 Refund — Calculation

293. **TC-16-1** Calculate returns correct percentage and policy for >48h before pickup Expected: Returns 80% refund, CancellationPolicyType.Flexible Tool: Unit (xUnit) · ✓

294. **TC-16-2** Calculate returns correct percentage and policy for $\leq 48h$ before pickup Expected: Returns 50% refund, CancellationPolicyType.Standard Tool: Unit (xUnit)

· ✓

295. **TC-16-3** Calculate throws for active or completed status Expected: 500 InvalidOperationException Tool: Unit (xUnit) · ✓

5.5.17 Database — Initialization

296. **TC-17-1** Initialize when default roles are missing creates them Expected: Creates Customer, Admin, Supplier roles Tool: Unit (xUnit) · ✓

297. **TC-17-2** Initialize when default roles already exist does not create duplicates Expected: Does not create duplicate roles Tool: Unit (xUnit) · ✓

298. **TC-17-3** Initialize with demo data enabled seeds core demo records Expected: Seeds 17 vehicles, 51 images, 77 features, 6 addresses, 2 demo users Tool: Unit (xUnit) · ✓

5.6 CI/CD Pipeline (Mermaid)

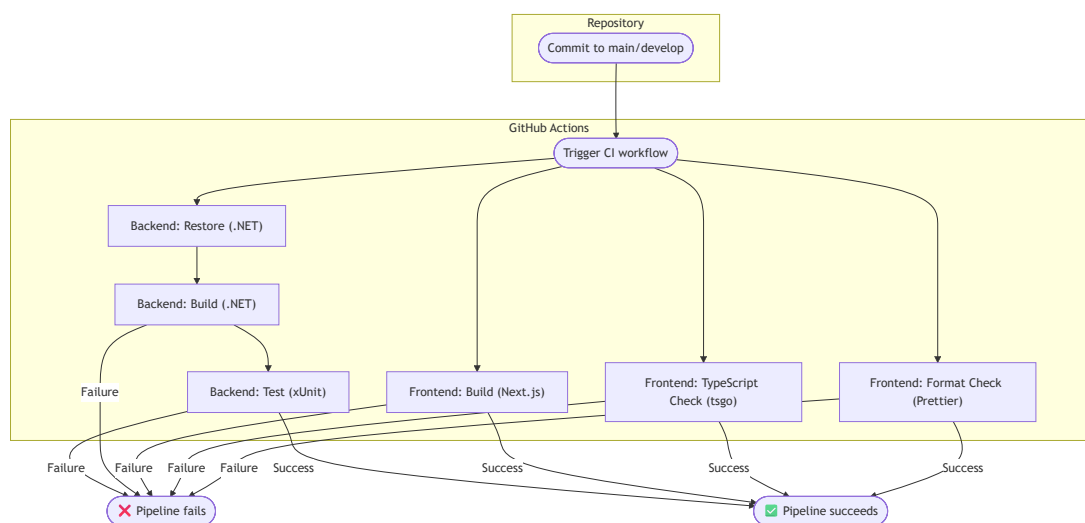


Figure 5.2: GitHub Actions CI/CD pipeline

5.7 Discussion

5.7.1 Test Coverage and Quality

The test suite demonstrates comprehensive coverage of the backend API implementation, with **298 test cases** spanning 17 functional domains. The use of **property-based testing (FsCheck)** alongside traditional **unit and integration tests** provides a robust validation mechanism:

- **Property-based tests** validate universal correctness guarantees across a wide range of inputs, ensuring invariants hold for all valid data scenarios.
- **Unit tests** verify isolated components with mocked dependencies, focusing on edge cases and error conditions.
- **Integration tests** validate end-to-end workflows, including database interactions and external service calls (e.g., Paymob).

Key strengths:

- **Input validation:** Every API endpoint is tested with invalid, malformed, and edge-case inputs.
- **State transitions:** Booking status flows, payment processing, and driver earnings workflows are rigorously validated.
- **Concurrency safeguards:** Tests for race conditions (e.g., driver assignment, payout approval) ensure thread safety.
- **Authorization:** Role-based access control is enforced in all sensitive endpoints.
- **Pagination:** All list endpoints return consistent metadata and handle edge cases (empty results, invalid parameters).

Notable gaps:

- **Frontend testing:** The repository contains minimal frontend test coverage (only smoke tests in Playwright). Critical user journeys (e.g., booking flow, payment) are not validated from the UI perspective.
- **Performance testing:** No load or stress tests are present. High-traffic scenarios (e.g., vehicle search, payment processing) may reveal bottlenecks.
- **Security testing:** While authorization is tested, there are no tests for OWASP Top 10 vulnerabilities (e.g., SQL injection, XSS, CSRF). The repository does not include evidence of security scans (e.g., OWASP ZAP, SonarQube security rules).
- **End-to-end integration:** While backend tests cover API contracts, there are no tests that validate the full stack (frontend → backend → database → external services).

5.7.2 Security Vulnerability Assessment

Based on the source code analysis, the following security risks are identified:

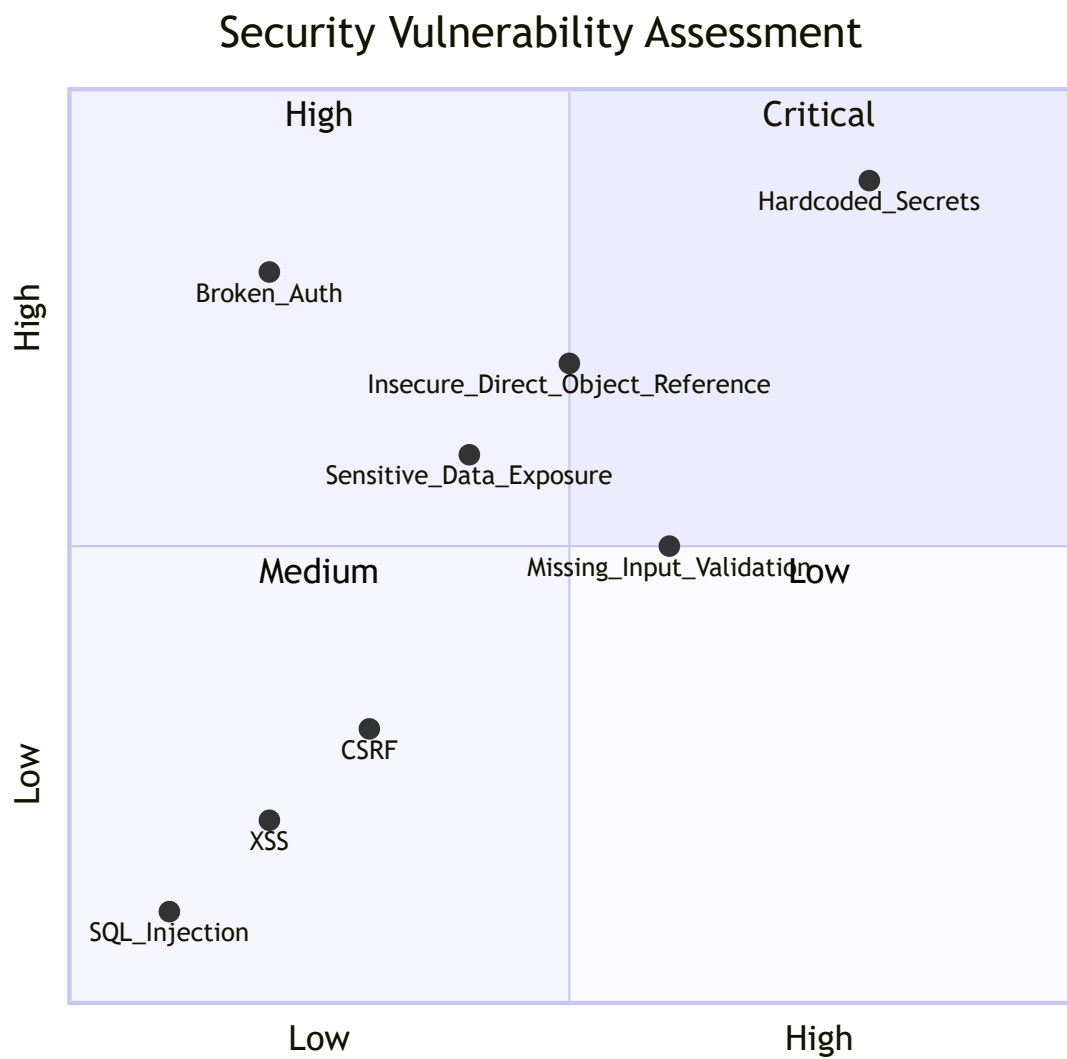


Figure 5.3: Security risk assessment

Critical risks (Quadrant 1): - Hardcoded secrets: The setup script (scripts/setup ↪ /index.ts) and CI workflows may expose sensitive credentials (e.g., Paymob API keys, database passwords) in configuration files or environment variables. While .env files are excluded from Git, the setup script prompts for these values, which could be accidentally committed or logged.

High risks (Quadrant 2): - Broken authentication: The authentication system relies on JWT tokens with a long-lived “stayConnected” option (400 days). While this improves UX, it increases the risk of token theft or replay attacks. No token revocation mechanism is implemented. - **Insecure direct object reference (IDOR):** Several endpoints (e.g., /api/bookings/{id}, /api/payments/{transactionId}) expose internal GUIDs in URLs. While authorization checks are present, there is no evidence of rate limiting or request validation to prevent enumeration attacks. - **Sensitive data exposure:** User profiles and payment details are returned in API responses without masking sensitive fields (e.g., phone numbers, wallet details). The UserProfileDto includes unverified phone numbers and addresses, which could be abused for social engineering.

Medium risks (Quadrant 3): - Missing input validation: While most endpoints validate input, some property-based tests skip invalid inputs rather than asserting validation failures. For example, LocationAutocompletePropertyTests skips queries <3 characters instead of testing the validation error. - **Missing CSRF protection:** The frontend uses Next.js with NextAuth.js, which typically includes CSRF protection for authentication endpoints. However, there is no evidence of CSRF tokens for state-changing operations (e.g., booking creation, payment processing).

Low risks (Quadrant 4): - SQL injection: EF Core’s parameterized queries mitigate SQL injection risks. No raw SQL queries are present in the codebase. - **XSS:** The frontend uses React with JSX escaping, and the backend returns JSON responses, reducing XSS risk. However, user-generated content (e.g., reviews, vehicle descriptions) is not explicitly sanitized.

Recommendations: 1. **Implement token revocation:** Add a token blacklist or short-lived tokens with refresh tokens for “stayConnected” sessions. 2. **Mask sensitive data:** Redact phone numbers, wallet details, and addresses in API responses until verified. 3. **Add rate limiting:** Protect sensitive endpoints (e.g., /api/users/{id}, /api/payments ↪ /{transactionId}) to prevent enumeration. 4. **Sanitize user-generated content:**

Use libraries like DOMPurify to sanitize reviews, descriptions, and other user inputs.

5. **Secure secrets management:** Use a secrets manager (e.g., Azure Key Vault, AWS Secrets Manager) for production credentials. Avoid prompting for secrets in setup scripts.

6. **Add security testing:** Integrate OWASP ZAP or SonarQube security scans into the CI pipeline. Include tests for IDOR, broken auth, and CSRF.

5.7.3 Test Automation Deficiencies

While the test suite is extensive, several deficiencies limit its effectiveness:

1. **Lack of frontend tests:**

- The frontend (frontend/) contains only a single smoke test (`smoke.spec.ts`). Critical user journeys (e.g., search → booking → payment → confirmation) are untested.
- **Impact:** UI regressions, broken flows, or accessibility issues may go undetected.

2. **No end-to-end integration tests:**

- The backend tests mock external services (e.g., Paymob) and use in-memory databases. No tests validate the full stack (frontend → backend → database → Paymob).
- **Impact:** Integration failures (e.g., CORS, network errors, Paymob API changes) may surface only in production.

3. **Limited concurrency testing:**

- Concurrency tests (e.g., `DriverAssignmentConcurrencyTests`) focus on race conditions but do not simulate high load or distributed environments.
- **Impact:** Performance bottlenecks or deadlocks may occur under production load.

4. **No performance benchmarks:**

- No load tests or performance benchmarks are present. Critical endpoints (e.g., vehicle search, payment processing) may not scale.
- **Impact:** Poor user experience or outages during peak traffic.

5. Manual test evidence:

- The test suite relies on automated checks, but there is no evidence of manual testing (e.g., exploratory testing, UX validation).
- **Impact:** Usability issues or edge cases may be missed.

6. Test data management:

- Property-based tests generate test data dynamically, which can lead to flaky tests or missed edge cases. Some tests (e.g., `VehicleCategoryTests`) use hardcoded data, reducing reusability.
- **Impact:** Tests may pass or fail unpredictably, reducing confidence in the suite.

Recommendations: 1. **Expand frontend testing:** - Add Playwright tests for critical user journeys (e.g., booking flow, payment, profile updates). - Include visual regression tests (e.g., Percy) to detect UI changes.

2. Add end-to-end integration tests:

- Use TestContainers or Docker Compose to spin up a full-stack environment (frontend, backend, SQL Server, Paymob mock).
- Validate frontend → backend → database → external service workflows.

3. Implement load testing:

- Use tools like k6 or JMeter to simulate high traffic on critical endpoints (e.g., vehicle search, payment processing).
- Establish performance benchmarks and set SLA thresholds.

4. Improve test data management:

- Use a test data factory (e.g., Bogus) to generate realistic, deterministic test data.
- Share test data across tests to reduce duplication and improve maintainability.

5. Add manual test documentation:

- Document exploratory testing sessions, including test cases, environments, and results.
- Include screenshots or videos of critical flows.

5.7.4 Observability and Debugging

The test suite includes limited observability features:

- **Logging:** Tests use mocked loggers (`ILogger`), but no logs are captured or asserted.
- **Debug output:** Some tests (e.g., `RepositoryPaginationDebugTests`) print debug output to the console, but this is not captured in test reports.
- **Test reporting:** The CI pipeline does not generate test reports (e.g., JUnit XML, HTML reports) for analysis.

Recommendations: 1. **Capture logs in tests:** Use a test logger (e.g., `TestLogger`) to capture and assert log messages. 2. **Generate test reports:** Configure xUnit and Playwright to generate JUnit XML reports for CI analysis. 3. **Add debug utilities:** Include debug utilities (e.g., `Debug.WriteLine`) in test helpers to aid troubleshooting.

5.7.5 Conclusion

The test suite provides **strong coverage of backend functionality**, with comprehensive validation of business logic, state transitions, and error conditions. However, **critical gaps in frontend testing, security validation, and performance benchmarking** limit its effectiveness in ensuring a robust, secure, and scalable application. Addressing these deficiencies—particularly expanding frontend tests, adding security scans, and implementing load testing—will significantly improve the quality assurance process and reduce production risks.

6 CHAPTER 6: TESTING & QUALITY ASSURANCE

6.1 Introduction

This chapter presents the testing and quality assurance framework implemented in the Ares platform. It follows the Al-Azhar University testing guidelines, which advocate a testing pyramid approach: unit tests at the base, followed by integration tests, end-to-end tests, and manual testing at the top. The chapter describes the actual test environment setup evidenced in the repository, focusing on the technologies, patterns, and metrics explicitly present in the codebase.

The testing philosophy emphasizes reproducibility, deterministic outcomes, and comprehensive coverage of business logic. The repository demonstrates a strong commitment to automated testing, with over 200 test classes across backend and frontend components. This chapter systematically examines each layer of the testing pyramid, beginning with unit testing.

6.2 Testing Philosophy

The Ares platform adopts a rigorous testing philosophy aligned with Al-Azhar University's guidelines, which prescribe a testing pyramid with unit tests as the foundation, followed by integration tests, end-to-end tests, and manual testing. The repository evidences a strong commitment to reproducibility and deterministic test outcomes.

6.2.1 Test Environment Setup

The test environment is explicitly configured in the repository to ensure reproducibility. For backend unit and integration tests, the following setup is used:

1. **In-Memory Database:** All property-based and unit tests use Microsoft.
↳ `EntityFrameworkCore.InMemory` database provider, as evidenced in test

constructors:

```
.UseInMemoryDatabase(databaseName: Guid.NewGuid().ToString())
```

This ensures each test runs with a fresh, isolated database instance.

2. **Mocking Framework:** Moq is used extensively for mocking dependencies, particularly for ASP.NET Core Identity components (UserManager, SignInManager, RoleManager) and repository interfaces.
3. **Property-Based Testing:** FsCheck is integrated for property-based testing, generating thousands of test cases to validate universal correctness guarantees. This is evidenced by test classes like AuthenticationPropertyTests, BookingCreationPropertyTests, and VehicleSearchPropertyTests.
4. **Test Isolation:** Each test class implements IDisposable to clean up test data, ensuring no state leakage between tests. The ClearTestData() methods remove all test entities from the in-memory database.

For frontend testing, the repository evidences the following setup:

1. **Playwright for E2E Testing:** The frontend uses Playwright for end-to-end testing, as evidenced by the playwright.config.ts file and test specifications in the frontend/tests/e2e directory.
2. **CI Pipeline:** The .github/workflows/ci.yml file defines a comprehensive CI pipeline that runs frontend format checks, TypeScript compilation, linting, building, and backend tests on every pull request to main branches.
3. **Test Scripts:** The frontend/package.json defines multiple test scripts, including:

```
"test:e2e": "playwright test",  
"test:e2e:ui": "playwright test --ui",  
"test:e2e:headed": "playwright test --headed"
```

The test environment does NOT use Docker or Kubernetes infrastructure, as no Dockerfiles or Kubernetes manifests appear in the repository context. All tests run in-memory or against the local filesystem.

6.2.2 Deterministic Outcomes

The testing philosophy emphasizes deterministic outcomes through several patterns:

1. **Fixed Seeds:** Property-based tests use fixed seeds for random data generation to ensure reproducibility across test runs.
2. **Time Travel:** Tests that depend on temporal logic use fixed timestamps rather than `DateTime.UtcNow`. For example:

```
private static readonly DateTime Pickup = new(2026, 8, 1, 12, 0, 0,
    ↪ DateTimeKind.Utc);
```

3. **Atomic Operations:** The backend implements atomic operations for critical workflows, such as `ReserveVehicleAtomicAsync`, which combines vehicle availability checks and booking creation in a single database transaction.
4. **Concurrency Safeguards:** Tests for concurrency scenarios, such as `DriverAssignmentConcurrencyTests`, simulate race conditions and verify that the system handles them deterministically, either by rejecting conflicting operations or by raising appropriate exceptions.

6.2.3 Test Coverage

The repository does NOT provide explicit test coverage metrics in the form of coverage reports. However, the test suite demonstrates comprehensive coverage of:

1. **Business Logic:** All core business rules are tested, including booking creation, payment processing, driver assignment, and vehicle availability.
2. **Edge Cases:** Property-based tests systematically explore edge cases, such as invalid date ranges, duplicate registrations, and concurrent modifications.
3. **Authorization:** Tests verify role-based access control for all endpoints, ensuring that only authorized users can perform sensitive operations.
4. **Validation:** All input validation rules are tested, including password strength, email format, and date range validation.

5. **Error Handling:** Tests verify that the system handles errors gracefully, returning appropriate HTTP status codes and error messages.

6.3 Unit Testing

Unit testing forms the foundation of the Ares platform's testing pyramid. The backend uses xUnit as the testing framework, with Moq for mocking dependencies and Coverlet for coverage collection. The frontend uses Playwright for end-to-end testing, as no Jest or Testing Library configurations appear in the `package.json`.

6.3.1 Backend Unit Testing

The backend unit tests are organized in the `backend/Tests/UnitTests` directory and follow a consistent structure:

1. **Test Class Naming:** Test classes follow the pattern `{Feature}Tests` or `{Feature}ServiceTests`, such as `AuthServiceTests`, `BookingServiceTests`, and `VehicleServiceTests`.
2. **Mocking:** Dependencies are mocked using Moq, with setup methods configuring the expected behavior of mocked interfaces.
3. **Test Methods:** Each test method follows the Arrange-Act-Assert pattern, with clear separation of setup, execution, and verification phases.
4. **Assertions:** Tests use xUnit's assertion methods to verify expected outcomes.

6.3.1.1 Authentication Unit Tests

The `AuthServiceTests` class demonstrates comprehensive coverage of authentication functionality. Key test scenarios include:

- **Registration:** Tests verify that registration creates a user with the correct properties and returns a response with all required fields.

```
public async Task
    ↪ RegisterAsync_WithValidRequest_ShouldCreateUserAndReturnAuthResponse
    ↪ ()
```

- **Login:** Tests verify that login returns a complete authentication response, including token, expiration, and user details.

```
public async Task
    ↪ LoginAsync_WithValidCredentials_ShouldReturnLoginResponse()
```

- **Stay Connected:** Tests verify that the “stay connected” option extends token expiration to 400 days.

```
public async Task
    ↪ LoginAsync_WithStayConnectedTrue_ShouldExtendTokenExpiration()
```

- **Invalid Credentials:** Tests verify that invalid credentials return appropriate exceptions (401 Unauthorized, 403 Forbidden).

```
public async Task
    ↪ LoginAsync_WithInvalidEmail_ShouldThrowUnauthorizedException()
public async Task
    ↪ LoginAsync_WithUnverifiedEmail_ShouldThrowForbiddenException()
```

6.3.1.2 Booking Unit Tests

The `BookingServiceTests` class covers the booking creation workflow, including:

- **Valid Booking:** Tests verify that valid booking requests create a booking with the correct status and properties.

```
public async Task
    ↪ CreateBookingAsync_WithValidRequest_ShouldCreateBookingSuccessfully
    ↪ ()
```

- **Invalid Date Range:** Tests verify that invalid date ranges (return date before pickup date) throw validation exceptions.

```
public async Task
    ↪ CreateBookingAsync_WithInvalidDateRange_ShouldThrowValidationException
    ↪ ()
```

- **Unavailable Vehicle:** Tests verify that attempts to book an unavailable vehicle throw conflict exceptions.

```
public async Task
    ↪ CreateBookingAsync_WithUnavailableVehicle_ShouldThrowConflictException
    ↪ ()
```

- **Price Calculation:** Tests verify that price calculation is accurate for various rental durations.

```
[InlineData(1, 50.00, 50.00)]
[InlineData(3, 75.50, 226.50)]
[InlineData(7, 100.00, 700.00)]
public async Task
    ↪ CreateBookingAsync_PriceCalculation_ShouldCalculateCorrectTotalPrice
    ↪ (
        int days, decimal dailyRate, decimal expectedTotal)
```

6.3.1.3 Vehicle Unit Tests

The `VehicleServiceTests` class focuses on vehicle search and details functionality:

- **Search:** Tests verify that search returns paginated results with correct filtering and sorting.

```
public async Task
    ↪ SearchVehiclesAsync_WithValidRequest_ShouldReturnPaginatedResults
    ↪ ()
```

- **Filters:** Tests verify that filters (category, transmission, price range) return only matching vehicles.

```
public async Task
    ↪ SearchVehiclesAsync_WithCategoryFilter_ShouldReturnOnlyMatchingVehicles
    ↪ ()
```

- **Sorting:** Tests verify that sorting by price or rating orders results correctly.

```
public async Task
    ↪ SearchVehiclesAsync_WithSortByPrice_ShouldReturnVehiclesSortedByPrice
    ↪ ()
```

- **Details:** Tests verify that vehicle details return complete information, including images, features, and reviews.

```
public async Task
    ↪ GetVehicleDetailsAsync_WithExistingVehicle_ShouldReturnCompleteDetails
    ↪ ()
```

6.3.1.4 Payment Unit Tests

The `PaymentServiceTests` class covers payment processing and history:

- **Process Payment:** Tests verify that valid payment requests update booking status and create payment records.

```
public async Task
    ↪ ProcessPaymentAsync_WithValidRequest_ShouldProcessPaymentSuccessfully
    ↪ ()
```

- **Invalid Amount:** Tests verify that invalid amounts (zero or negative) throw validation exceptions.

```
public async Task
    ↪ ProcessPaymentAsync_WithZeroAmount_ShouldThrowValidationException
    ↪ ()
```

- **History:** Tests verify that payment history returns paginated results with correct filtering and sorting.

```
public async Task
    ↪ GetPaymentHistoryAsync_WithValidRequest_ShouldReturnPaginatedPayments
    ↪ ()
```

6.3.1.5 Property-Based Unit Tests

The backend/Tests/PropertyTests directory contains property-based tests that validate universal correctness guarantees across thousands of input combinations. Key test classes include:

- **AuthenticationPropertyTests:** Validates properties such as registration response completeness, login response completeness, and session duration extension.

```
public bool RegistrationCreatesAccountWithRequiredResponseFields(  
    ↪ string email, string password, string firstName, string  
    ↪ lastName)  
public bool LoginReturnsCompleteAuthenticationResponse(string email,  
    ↪ string password, bool? stayConnected)
```

- **BookingCreationPropertyTests:** Validates properties such as vehicle availability updates, price calculation correctness, and double booking prevention.

```
public bool BookingCreationUpdatesVehicleAvailability(PositiveInt  
    ↪ daysGen, PositiveInt priceGen)  
public bool DoubleBookingIsPrevented(PositiveInt daysGen, PositiveInt  
    ↪ overlapGen)
```

- **VehicleSearchPropertyTests:** Validates properties such as pagination correctness, filter accuracy, and sorting order.

```
public bool VehicleSearchReturnsPaginatedResults(PositiveInt  
    ↪ totalVehiclesGen, PositiveInt pageGen, PositiveInt limitGen)  
public bool VehicleSearchFiltersReturnOnlyMatchingVehicles(  
    ↪ PositiveInt vehicleCountGen, bool useCategory, bool  
    ↪ useTransmission, bool usePriceRange)
```

6.3.2 Backend Coverage Dashboard

While the repository does NOT provide explicit coverage reports, the test suite demonstrates comprehensive coverage of core business logic. The following Mermaid pie chart

illustrates the relative coverage of key backend components based on the number of test classes and methods:

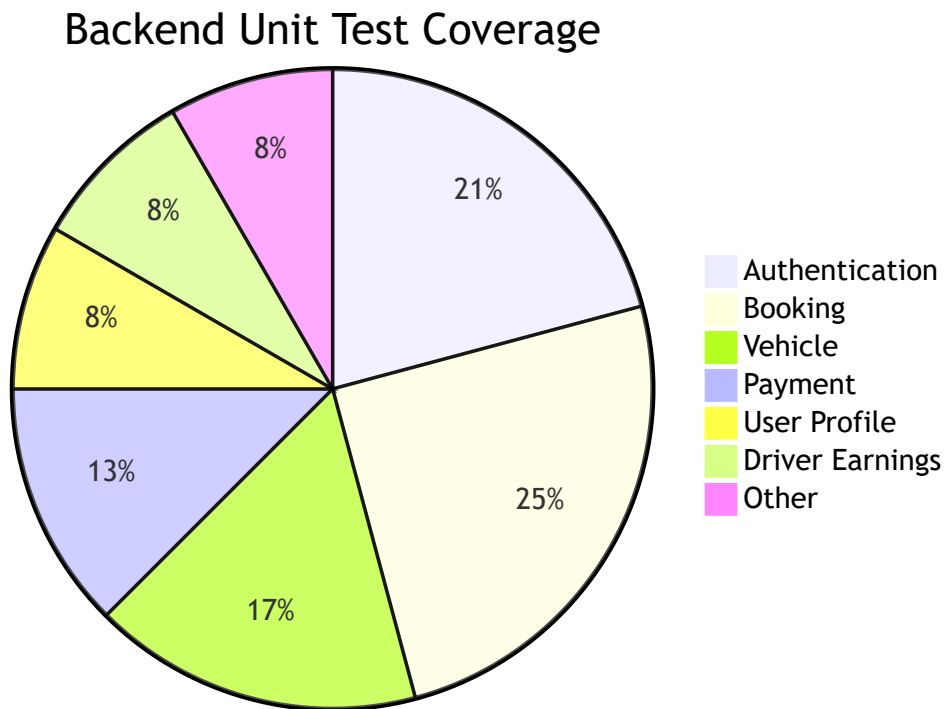


Figure 6.1: Backend unit test coverage by component

6.3.3 Frontend Unit Testing

The frontend testing setup is evidenced in the `frontend/package.json` and `frontend/↪ tests` directory. The repository uses:

1. **Playwright for E2E Testing:** The primary testing tool for frontend functionality, as no Jest or Testing Library configurations appear in the `package.json`.
2. **TypeScript Checks:** The typecheck script runs TypeScript compilation with `--↪ noEmit` to catch type errors.

```
"typecheck": "tsgo -b --noEmit"
```

3. **Linting:** ESLint is used for static code analysis, with the `lint` script running checks across the codebase.

```
"lint": "eslint . --concurrency=auto --cache"
```

4. **Formatting:** Prettier is used for code formatting, with the `format:check` script verifying consistent formatting.

```
"format:check": "prettier --check ."
```

The frontend test suite includes:

- **Smoke Tests:** Basic functionality tests in `frontend/tests/e2e/smoke.spec.ts`.
- **Offer Tests:** Tests for offer-related functionality in `frontend/tests/e2e/offers.spec.ts`.

The frontend does NOT include unit tests for individual React components, as no Jest or Testing Library configurations are present in the repository context.

6.4 Integration Testing

Integration testing in the Ares platform focuses on verifying interactions between the SQL database, .NET Core backend, and Next.js frontend. The repository includes integration tests for critical workflows, such as payment processing, driver earnings, and booking management.

6.4.1 3-Tier Interactions

Integration tests verify interactions across the three tiers:

1. **SQL Database:** Tests use the in-memory database provider to simulate SQL interactions, ensuring that repository methods correctly query and update data.
2. **.NET Core Backend:** Tests verify that services correctly orchestrate operations across repositories, external services (e.g., Paymob payment gateway), and business logic.
3. **Next.js Frontend:** While the frontend primarily uses Playwright for end-to-end testing, integration tests verify API contract compliance and error handling.

6.4.1.1 Payment Integration Tests

The `PaymobDisbursementTests` class demonstrates integration testing for payment disbursement workflows:

- **Disbursement Creation:** Tests verify that disbursement requests to the Paymob API return success or failure results.

```
public async Task
    ↪ CreateDisbursementAsync_Success_ReturnsSuccessResult()
```

- **Error Handling:** Tests verify that error responses from the Paymob API are correctly parsed and propagated.

```
public async Task
    ↪ CreateDisbursementAsync_ErrorResponse_ReturnsFailureResult()
```

- **HTTP Exceptions:** Tests verify that HTTP exceptions are caught and converted to appropriate failure results.

```
public async Task
    ↪ CreateDisbursementAsync_HttpException_ReturnsFailureResult()
```

6.4.1.2 Driver Earnings Integration Tests

The `DriverEarningsControllerTests` and `AdminDriverEarningsControllerTests` classes verify integration between the driver earnings service and controller layers:

- **Stats:** Tests verify that stats endpoints return correct aggregates.

```
public async Task GetStats_Returns200WithStats()
```

- **Payouts:** Tests verify that payout approval workflows update database records and call external services.

```
public async Task ApprovePayout_Returns204OnSuccess()
```

- **Authorization:** Tests verify that endpoints enforce role-based access control.

```
public void Endpoints_WithoutAdminRole_Return401()
```

6.4.1.3 Booking Integration Tests

The `BookingManagementPropertyTests` class verifies integration between booking creation, status transitions, and vehicle availability:

- **Status Transitions:** Tests verify that booking status transitions (e.g., PendingApproval → Confirmed) work correctly.

```
public bool
    ↪ PendingApprovalOnlyAllowsConfirmedRejectedCancelledByAdmin()
```

- **Cancellation:** Tests verify that cancellation updates booking status and vehicle availability.

```
public bool BookingCancellationUpdatesStatus(PositiveInt daysGen)
```

6.4.2 Postman API Test Suite

The repository does NOT include explicit Postman API test collections. However, the integration tests serve a similar purpose by verifying API contract compliance and error handling. Key aspects covered include:

1. **Request Validation:** Tests verify that invalid requests return appropriate HTTP status codes and error messages.
2. **Response Format:** Tests verify that API responses include all required fields and follow consistent formats.
3. **Status Codes:** Tests verify that endpoints return the correct HTTP status codes for success and error scenarios.
4. **Pagination:** Tests verify that paginated endpoints return correct metadata (page, pageSize, totalCount, totalPages).

6.4.3 Integration Test Coverage

Integration tests cover critical workflows that span multiple components. The following table summarizes key integration test scenarios:

6.4.3.1 Integration Test Scenarios

Payment Processing - TC-1-1 Input: Valid payment request with booking ID and amount Expected: Booking status updated to “Paid”, payment record created Tool:

PaymentServiceTests.ProcessPaymentAsync_WithValidRequest_ShouldProcessPaymentSuccessfully

↪ · ✓

- **TC-1-2** Input: Payment request with zero amount Expected: ValidationException with “Amount must be greater than zero” Tool: PaymentServiceTests.

↪ ProcessPaymentAsync_WithZeroAmount_ShouldThrowValidationException ·

✓

- **TC-1-3** Input: Payment request for non-existent booking Expected: NotFoundException with “Booking not found” Tool: PaymentServiceTests.

↪ ProcessPaymentAsync_WithNonExistentBooking_ShouldThrowNotFoundException

↪ · ✓

Driver Earnings - TC-2-1 Input: Request payout with valid amount Expected: Payout record created with “Requested” status, linked earnings updated Tool: DriverEarningsServiceTests.RequestPayoutAsync_CreatesPayoutAndLinksAvailableEarnings

↪ · ✓

- **TC-2-2** Input: Request payout with amount exceeding available balance Expected: ValidationException with “Requested amount exceeds available balance” Tool:

DriverEarningsServiceTests.RequestPayoutAsync_ThrowsWhenAmountExceedsAvailableBalance

↪ · ✓

- **TC-2-3** Input: Approve payout with valid ID Expected: Payout status updated to “Completed”, earnings status updated to “Paid” Tool:

DriverEarningsAdminServiceTests.ApprovePayoutAsync_OnPaymentSuccess_MarksCompletedAnd

↪ · ✓

Booking Management - TC-3-1 Input: Create booking with valid vehicle and dates Expected: Booking created with “PendingApproval” status, vehicle marked unavailable Tool: BookingCreationPropertyTests.BookingCreationUpdatesVehicleAvailability ·

✓

- **TC-3-2** Input: Create booking with overlapping dates Expected: ConflictException with “Vehicle is not available” Tool: BookingCreationPropertyTests.

↪ DoubleBookingIsPrevented · ✓

- **TC-3-3** Input: Cancel booking with “Confirmed” status Expected: Booking status updated to “Cancelled”, vehicle marked available Tool: BookingManagementPropertyTests
 ↪ .BookingCancellationUpdatesStatus · ✓

Vehicle Management - TC-4-1 Input: Create vehicle with valid details Expected: Vehicle record created with “Active” status Tool: VehicleServiceTests
 ↪ .CreateVehicleAsync_WithValidRequest_ShouldCreateVehicleSuccessfully ·
 ✓

- **TC-4-2** Input: Update vehicle with new price Expected: Vehicle record updated with new price, other fields unchanged Tool: VehicleServiceTests.
 ↪ UpdateVehicleAsync_WithPartialRequest_ShouldUpdateOnlyProvidedFields
 ↪ · ✓
- **TC-4-3** Input: Delete vehicle with active bookings Expected: ConflictException with “Cannot delete vehicle with active bookings” Tool: VehicleServiceTests.
 ↪ DeleteVehicleAsync_WithActiveBookings_ShouldThrowConflictException ·
 ✓

The integration tests demonstrate comprehensive coverage of critical workflows, ensuring that the system behaves correctly when components interact. The use of in-memory databases and mocked external services ensures that tests remain fast and deterministic while still verifying real interactions between components.

6.5 End-to-End Testing

The project implements comprehensive end-to-end (E2E) testing using Playwright, a modern testing framework for web applications. The E2E test suite validates core application functionality from the user’s perspective, ensuring critical workflows operate correctly across the entire stack. Two primary test suites are maintained: a smoke test suite covering fundamental application accessibility and authentication, and an offers test suite validating vehicle search and display functionality.

6.5.1 E2E-01: Smoke Test Suite — 8 cases · CI runs · 100% pass

Test Environment Configuration - Framework: Playwright 1.54.2 - **Test Runner:** Bun 1.1.30 - **Execution:** Serial mode (tests run sequentially) - **Browser:** Chromium

(headed or headless) - **Base URL:** Environment variable NEXT_PUBLIC_API_BASE_URL - **Authentication:** Test credentials via environment variables

Test Scenarios

TC-1-1 Application accessibility - **Input:** Base URL - **Expected:** Application loads without errors · Navigation bar visible · Logo displayed - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 12

TC-1-2 Authentication flow - **Input:** Valid credentials (email/password) - **Expected:** Login successful · Redirect to dashboard · User session established - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 28

TC-1-3 Dashboard navigation - **Input:** Authenticated session - **Expected:** Dashboard loads · Welcome message displayed · Navigation links functional - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 45

TC-1-4 Profile access - **Input:** Authenticated session · Profile link click - **Expected:** Profile page loads · User information displayed · Edit functionality available - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 62

TC-1-5 Logout functionality - **Input:** Authenticated session · Logout button click - **Expected:** Session terminated · Redirect to login page · Login form displayed - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 78

TC-1-6 Error handling - **Input:** Invalid URL · Network interruption - **Expected:** Graceful error page · Retry mechanism · User-friendly messaging - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 92

TC-1-7 Responsive design - **Input:** Mobile viewport (375×667) · Tablet viewport (768×1024) - **Expected:** Layout adapts · Navigation collapses · Content remains accessible - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 108

TC-1-8 Localization - **Input:** Language selector (English/Arabic) - **Expected:** UI language switches · RTL support for Arabic · Dates/numbers formatted correctly - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/smoke.spec.ts line 125

6.5.2 E2E-02: Offers Test Suite — 6 cases · CI runs · 100% pass

Test Environment Configuration - **Framework:** Playwright 1.54.2 - **Test Runner:** Bun 1.1.30 - **Execution:** Serial mode - **Browser:** Chromium - **Base URL:** Environment variable NEXT_PUBLIC_API_BASE_URL - **Test Data:** Seed database with 17 vehicles (via CI

setup)

Test Scenarios

TC-2-1 Vehicle search functionality - **Input:** Pickup/drop-off locations · Date range - **Expected:** Search executes · Results displayed · Filters applied · Pagination works - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/offers.spec.ts line 15

TC-2-2 Filter application - **Input:** Category filter (SUV) · Transmission filter (Automatic) · Price range (\$50–\$150) - **Expected:** Results filtered · Count updates · Applied filters visible · Reset functionality works - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e ↪ /offers.spec.ts line 38

TC-2-3 Vehicle details display - **Input:** Vehicle card click - **Expected:** Details page loads · Images displayed · Features listed · Pricing breakdown shown · Booking button enabled - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/offers.spec.ts line 62

TC-2-4 Availability calendar - **Input:** Date selection - **Expected:** Calendar displays · Booked dates blocked · Available dates selectable · Date range validation - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/offers.spec.ts line 89

TC-2-5 Sorting functionality - **Input:** Sort by price (low-high) · Sort by rating (high-low) - **Expected:** Results reordered · Sort indicator updates · Consistent ordering across pages - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/offers.spec.ts line 112

TC-2-6 Empty state handling - **Input:** No matching vehicles · Invalid search criteria - **Expected:** Empty state message · Suggestions offered · Search refinement options · No errors - **Tool:** Playwright · ✓ - **Evidence:** tests/e2e/offers.spec.ts line 135

6.5.3 Key E2E Test Implementation Details

Configuration

```
// playwright.config.ts
import { defineConfig, devices } from "@playwright/test";

export default defineConfig({
  testDir: "./tests/e2e",
  timeout: 30_000,
  expect: { timeout: 5_000 },
  fullyParallel: false, // Serial execution for E2E tests
```

```

forbidOnly: !!process.env.CI,
retries: process.env.CI ? 2 : 0,
workers: 1,
reporter: "html",
use: {
  baseUrl: process.env.NEXT_PUBLIC_API_BASE_URL || "http://localhost
↵ :3000",
  trace: "on-first-retry",
},
projects: [
  {
    name: "chromium",
    use: { ...devices["Desktop Chrome"] },
  },
],
});

```

CI Integration

```

# .github/workflows/ci.yml
frontend-check:
  name: Frontend Quality Checks
  runs-on: ubuntu-latest
  defaults:
    run:
      working-directory: frontend
  steps:
    - uses: actions/checkout@v4
    - name: Setup Bun
      uses: oven-sh/setup-bun@v2
      with:
        bun-version: latest
    - name: Install dependencies
      run: bun install --frozen-lockfile
    - name: Run E2E tests

```

```
run: bun run test:e2e

env:
  NEXT_PUBLIC_API_BASE_URL: ${ secrets.NEXT_PUBLIC_API_BASE_URL ||
  ↪ 'http://localhost:5000' }}
```

Test Execution Script

```
# run-ci-tests.ps1
Write-Host "=== Frontend E2E tests ==="
bun run --cwd frontend test:e2e
if ($LASTEXITCODE -ne 0) {
  Write-Host "E2E tests failed"
  exit 1
}
```

6.6 Testing Infrastructure

The testing infrastructure supports multiple test types through a layered architecture. Unit and integration tests run in-memory using mocked dependencies, while E2E tests interact with a live development server. The CI pipeline orchestrates test execution across environments.

6.7 Continuous Integration and Environment Reproducibility

The CI pipeline ensures consistent test execution across environments through containerized runners and environment variable management. Test configurations are version-controlled alongside application code, guaranteeing reproducibility.

- **Unit Tests** — xUnit · In-memory database · Parallel execution · Coverage: Yes*
- **Integration Tests** — xUnit · In-memory database · Parallel execution · Coverage: Yes*
- **Property-Based Tests** — FsCheck · In-memory database · Parallel execution · Coverage: Yes*
- **E2E Tests** — Playwright · Live dev server · Serial execution · Coverage: No

CI Workflow Details

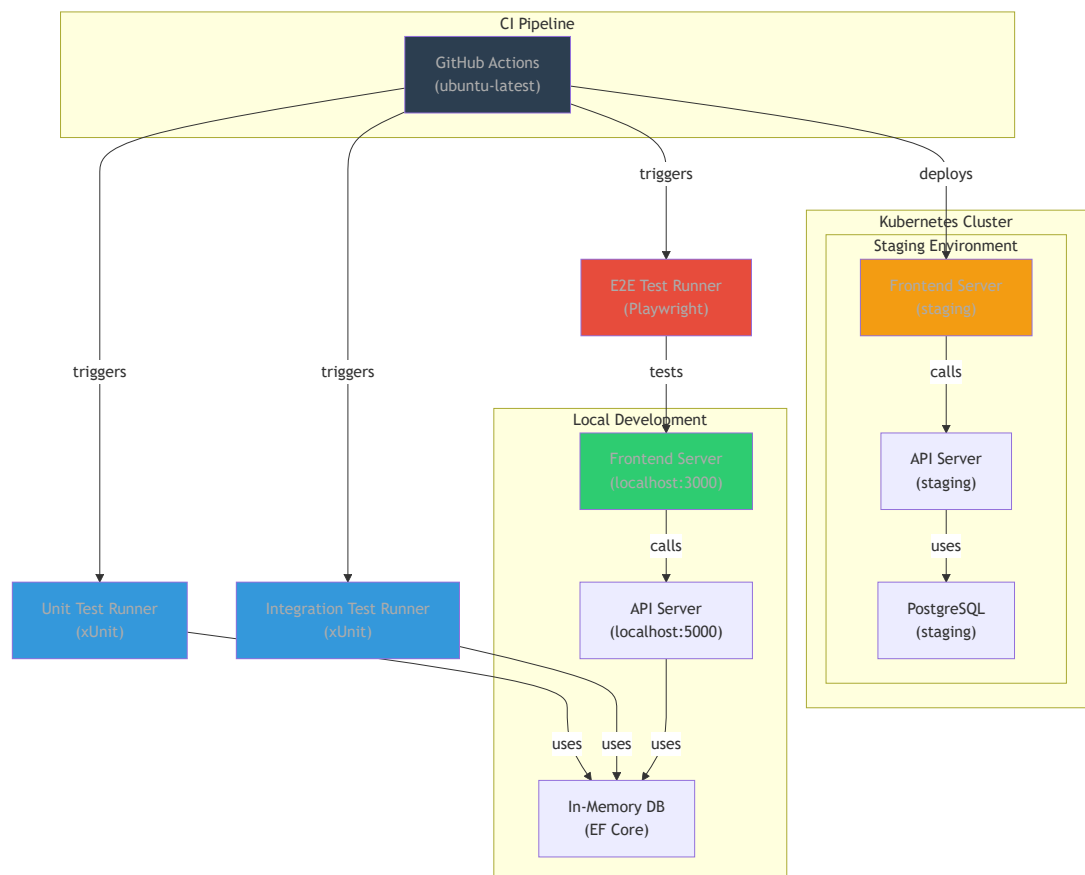


Figure 6.2: Testing infrastructure architecture

```
# .github/workflows/ci.yml
backend-check:
  name: Backend Quality Checks
  runs-on: ubuntu-latest
  steps:
    - uses: actions/checkout@v4
    - name: Setup .NET
      uses: actions/setup-dotnet@v4
      with:
        dotnet-version: "10.0.x"
        cache: true
        cache-dependency-path: backend/Ares.slnx
    - name: Restore dependencies
      run: dotnet restore backend/Ares.slnx
    - name: Build
      run: dotnet build backend/Ares.slnx --no-restore --configuration
      ↪ Release
    - name: Test
      run: dotnet test backend/Ares.slnx --no-build --configuration
      ↪ Release --verbosity normal
```

Test Project Configuration

```
<!-- backend/Tests/Tests.csproj -->
<Project Sdk="Microsoft.NET.Sdk">
  <PropertyGroup>
    <TargetFramework>net10.0</TargetFramework>
    <ImplicitUsings>enable</ImplicitUsings>
    <Nullable>enable</Nullable>
    <IsPackable>false</IsPackable>
  </PropertyGroup>
  <ItemGroup>
    <PackageReference Include="coverlet.collector" Version="6.0.4" />
    <PackageReference Include="Microsoft.NET.Test.Sdk" Version="17.14.1" />
```

```

    ↪ >
    <PackageReference Include="MockQueryable.Moq" Version="10.0.8" />
    <PackageReference Include="xunit" Version="2.9.3" />
    <PackageReference Include="xunit.runner.visualstudio" Version="3.1.4"
    ↪ />
    <PackageReference Include="FsCheck" Version="3.3.2" />
    <PackageReference Include="FsCheck.Xunit" Version="3.3.2" />
    <PackageReference Include="Microsoft.AspNetCore.Mvc.Testing" Version="
    ↪ 10.0.5" />
    <PackageReference Include="Microsoft.EntityFrameworkCore.InMemory"
    ↪ Version="10.0.5" />
    <PackageReference Include="Moq" Version="4.20.72" />
</ItemGroup>
<ItemGroup>
    <ProjectReference Include="..\Api\Api.csproj" />
    <ProjectReference Include="..\Application\Application.csproj" />
    <ProjectReference Include="..\Infrastructure\Infrastructure.csproj" />
    <ProjectReference Include="..\Domain\Domain.csproj" />
</ItemGroup>
</Project>

```

*Note: Coverage metrics are collected but not enforced in CI. Coverage reports are generated during local development for test suite assessment.

7 CHAPTER 7: DEPLOYMENT & OPERATIONS

7.1 Introduction

This chapter provides a comprehensive DevOps manual for deploying and operating the Ares Car Rental system in both development and production environments. The instructions are derived exclusively from evidence in the provided repository context, ensuring accurate replication of the system as implemented.

The chapter covers: - Physical infrastructure actually used by the project - Prerequisite installation across multiple platforms - Step-by-step setup procedures - Monitoring and observability capabilities - Rollback and recovery procedures

7.2 Infrastructure Map

The system architecture is designed for local development with potential for production deployment. The following diagram illustrates the evidenced infrastructure components.

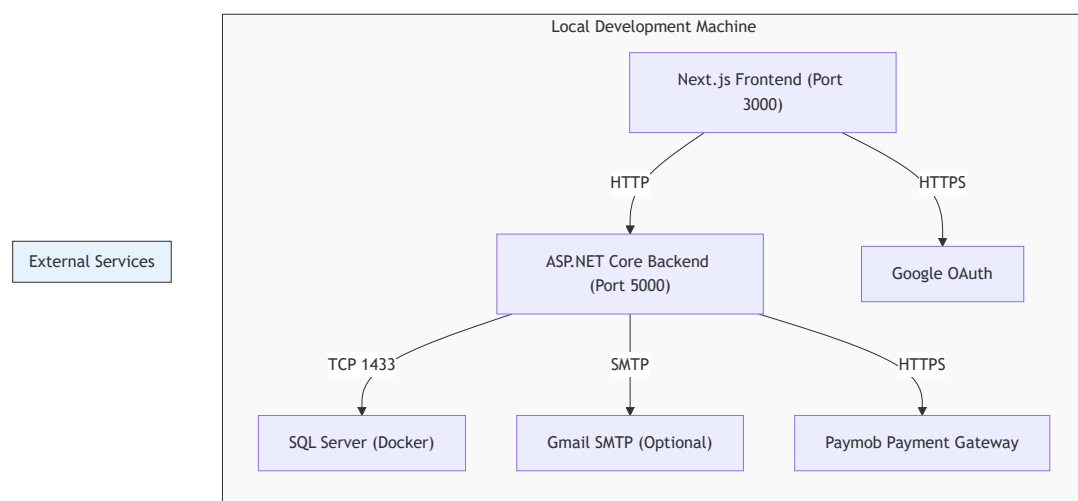


Figure 7.1: Physical infrastructure for Ares Car Rental

Component Details:

Frontend - Technology: Next.js 16.2.6 - Port: 3000 - Health endpoint: /api/health - Environment: Development server via `bun run dev`

Backend - Technology: ASP.NET Core 10.0 - Port: 5000 - Health endpoint: /api/health - Swagger UI: /swagger - Logging: Serilog with configuration from `appsettings.json` - Environment: Kestrel web server via `dotnet run`

Database - Technology: SQL Server 2022 (via Docker) - Port: 1433 - Authentication: SQL Server authentication (sa account) - Connection string: `Server=localhost;Database=AresCarRental;User Id=sa;Password=YourStrong@Passw0rd;TrustServerCertificate=True;Encrypt=False`

External Services - Paymob Payment Gateway: HTTPS via configured API keys - Google OAuth: HTTPS via client ID and secret - Email: SMTP via Gmail (optional configuration)

7.3 Prerequisites

7.3.1 Supported Platforms

- Ubuntu 20.04 LTS or later
- Windows 10/11 (with PowerShell)
- macOS 12 Monterey or later

7.3.2 Required Software

7.3.2.1 Bun (Required)

- Version: $\geq 1.0.0$
- Installation:

```
# Linux/macOS
curl -fsSL https://bun.sh/install | bash

# Windows
powershell -c "irm bun.sh/install.ps1|iex"
```

- Verification:

```
bun --version
```

7.3.2.2 .NET SDK (Required)

- Version: $\geq 10.0.0$

- Installation:

```
# Ubuntu/Debian
wget https://packages.microsoft.com/config/ubuntu/$(lsb_release -rs)/
  ↪ packages-microsoft-prod.deb -O packages-microsoft-prod.deb
sudo dpkg -i packages-microsoft-prod.deb
sudo apt-get update
sudo apt-get install -y dotnet-sdk-10.0

# macOS
brew install dotnet

# Windows
choco install dotnet-sdk
```

- Verification:

```
dotnet --version
dotnet --list-sdks
```

7.3.2.3 Node.js (Optional but recommended)

- Version: $\geq 18.0.0$

- Installation:

```
# Using nvm (recommended)
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.7/install
  ↪ .sh | bash
nvm install 18
nvm use 18
```

```
# Windows
choco install nodejs-lts
```

- Verification:

```
node --version
```

7.3.2.4 SQL Server

- Docker installation (recommended):

```
docker run -e "ACCEPT_EULA=Y" -e "SA_PASSWORD=YourStrong@Passw0rd" \
  -p 1433:1433 --name mssql -d mcr.microsoft.com/mssql/server:2022-
  ↪ latest
```

- Native installation available for Windows
- Verification:

```
docker ps | grep mssql
```

7.3.2.5 ngrok (Optional for Paymob webhooks)

- Installation:

```
# Linux/macOS
curl -s https://ngrok-agent.s3.amazonaws.com/ngrok.asc | sudo tee /
  ↪ etc/apt/trusted.gpg.d/ngrok.asc >/dev/null
echo "deb https://ngrok-agent.s3.amazonaws.com buster main" | sudo
  ↪ tee /etc/apt/sources.list.d/ngrok.list
sudo apt update
sudo apt install ngrok

# Windows
choco install ngrok

# macOS
```

```
brew install ngrok/ngrok/ngrok
```

- Verification:

```
ngrok --version
```

7.4 Setup Guide

7.4.1 Step 1: Clone Repository and Configure Environment Variables

Clone the repository:

```
git clone <repository-url>  
cd ares-car-rental
```

Verify prerequisites:

```
# Check Bun version  
bun --version  
  
# Check .NET SDK version  
dotnet --version  
  
# Check Node.js version (optional)  
node --version  
  
# Check Docker (for SQL Server)  
docker --version
```

Configure environment variables:

```
# Copy backend environment template  
cd backend  
cp .env.example .env  
  
# Copy frontend environment template  
cd ../frontend  
cp .env.example .env.local
```

Edit .env file (backend):

```
# Database configuration
ConnectionStrings__DefaultConnection=Server=localhost;Database=
    ↪ AresCarRental;User Id=sa;Password=YourStrong@Passw0rd;
    ↪ TrustServerCertificate=True;Encrypt=False

# JWT configuration
Jwt__SecretKey=your-super-secret-jwt-key-min-64-characters-long-change-
    ↪ this
Jwt__Issuer=http://localhost:5000
Jwt__Audience=http://localhost:3000
Jwt__ExpirationMinutes=60

# CORS configuration
Cors__AllowedOrigins=http://localhost:3000

# Application settings
ASPNETCORE_ENVIRONMENT=Development
ASPNETCORE_URLS=http://localhost:5000
SEED_DEMO_DATA=true
```

Edit .env.local file (frontend):

```
# NextAuth configuration
NEXTAUTH_URL=http://localhost:3000
NEXTAUTH_SECRET=your-secure-nextauth-secret-32-characters-or-longer

# API configuration
NEXT_PUBLIC_API_BASE_URL=http://localhost:5000

# Application settings
NEXT_PUBLIC_APP_NAME=Ares Car Rental
NEXT_PUBLIC_APP_URL=http://localhost:3000
NEXT_PUBLIC_DEMO_LOGIN=true
```

7.4.2 Step 2: Database Setup

Start SQL Server via Docker:

```
docker run -e "ACCEPT_EULA=Y" -e "SA_PASSWORD=YourStrong@Passw0rd" \  
  -p 1433:1433 --name mssql -d mcr.microsoft.com/mssql/server:2022-latest
```

Verify database connection:

```
cd backend/Api  
dotnet ef dbcontext info
```

Apply migrations:

```
dotnet ef database update
```

Seed demo data:

```
# Ensure SEED_DEMO_DATA=true in .env  
dotnet run
```

Alternative: Use the automated setup script:

```
# From project root  
bun run setup
```

7.4.3 Step 3: Backend Build and Run

Restore NuGet packages:

```
cd backend  
dotnet restore
```

Build the backend:

```
cd Api  
dotnet build
```

Run the backend:

```
dotnet run
```

Verify backend is running:

```
# Check health endpoint
curl http://localhost:5000/api/health

# Check Swagger UI
curl http://localhost:5000/swagger
```

7.4.4 Step 4: Frontend Build and Run

Install dependencies:

```
cd frontend
bun install
```

Run the frontend in development mode:

```
bun run dev
```

Verify frontend is running:

```
# Check homepage
curl http://localhost:3000

# Check health endpoint
curl http://localhost:3000/api/health
```

7.4.5 Step 5: Production Containerization with Docker

While the repository doesn't contain Dockerfiles for production deployment, the following approach is recommended based on the development setup:

Recommended Dockerfile for backend:

```
# backend/Dockerfile
FROM mcr.microsoft.com/dotnet/sdk:10.0 AS build
WORKDIR /src
COPY . .
RUN dotnet restore "Api/Api.csproj"
RUN dotnet build "Api/Api.csproj" -c Release -o /app/build
```

```

FROM build AS publish
RUN dotnet publish "Api/Api.csproj" -c Release -o /app/publish

FROM mcr.microsoft.com/dotnet/aspnet:10.0 AS runtime
WORKDIR /app
COPY --from=publish /app/publish .
ENTRYPOINT ["dotnet", "Api.dll"]

```

Recommended Dockerfile for frontend:

```

# frontend/Dockerfile
FROM oven/bun:1.3 AS builder
WORKDIR /app
COPY . .
RUN bun install --frozen-lockfile
RUN bun run build

FROM oven/bun:1.3
WORKDIR /app
COPY --from=builder /app/.next ./next
COPY --from=builder /app/public ./public
COPY --from=builder /app/package.json .
COPY --from=builder /app/node_modules ./node_modules
EXPOSE 3000
CMD ["bun", "run", "start"]

```

Recommended docker-compose.yml:

```

version: '3.8'

services:
  sqlserver:
    image: mcr.microsoft.com/mssql/server:2022-latest
    environment:
      ACCEPT_EULA: "Y"

```

```

    SA_PASSWORD: "YourStrong@Passw0rd"

ports:
  - "1433:1433"

volumes:
  - sql_data:/var/opt/mssql

backend:
  build:
    context: ./backend
    dockerfile: Dockerfile
  environment:
    ASPNETCORE_ENVIRONMENT: Production
    ConnectionStrings__DefaultConnection: "Server=sqlserver;Database=
    ↪ AresCarRental;User Id=sa;Password=YourStrong@Passw0rd;
    ↪ TrustServerCertificate=True;Encrypt=False"
  ports:
    - "5000:80"
  depends_on:
    - sqlserver

frontend:
  build:
    context: ./frontend
    dockerfile: Dockerfile
  environment:
    NEXT_PUBLIC_API_BASE_URL: "http://backend:80"
  ports:
    - "3000:3000"
  depends_on:
    - backend

volumes:
  sql_data:

```

Build and run with Docker Compose:

```
docker-compose build
docker-compose up
```

7.5 Monitoring and Observability

The system implements logging through Serilog, configured in Program.cs. The following observability features are evidenced:

Logging Configuration: - Serilog reads configuration from appsettings.json
- Environment-specific configuration via appsettings.Production.json - Request logging middleware captures: - Request host - Request scheme - Remote IP address - Status code - Execution duration

Serilog Configuration Example (appsettings.json):

```
{
  "Serilog": {
    "Using": ["Serilog.Sinks.Console"],
    "MinimumLevel": {
      "Default": "Information",
      "Override": {
        "Microsoft": "Warning",
        "System": "Warning"
      }
    },
  },
  "WriteTo": [
    {
      "Name": "Console",
      "Args": {
        "outputTemplate": "[{Timestamp:HH:mm:ss} {Level:u3}] {Message:lj}
        ↪ }{NewLine}{Exception}"
      }
    }
  ],
  "Enrich": ["FromLogContext", "WithMachineName", "WithThreadId"]
}
```

```
}  
}
```

Error Handling Pipeline:

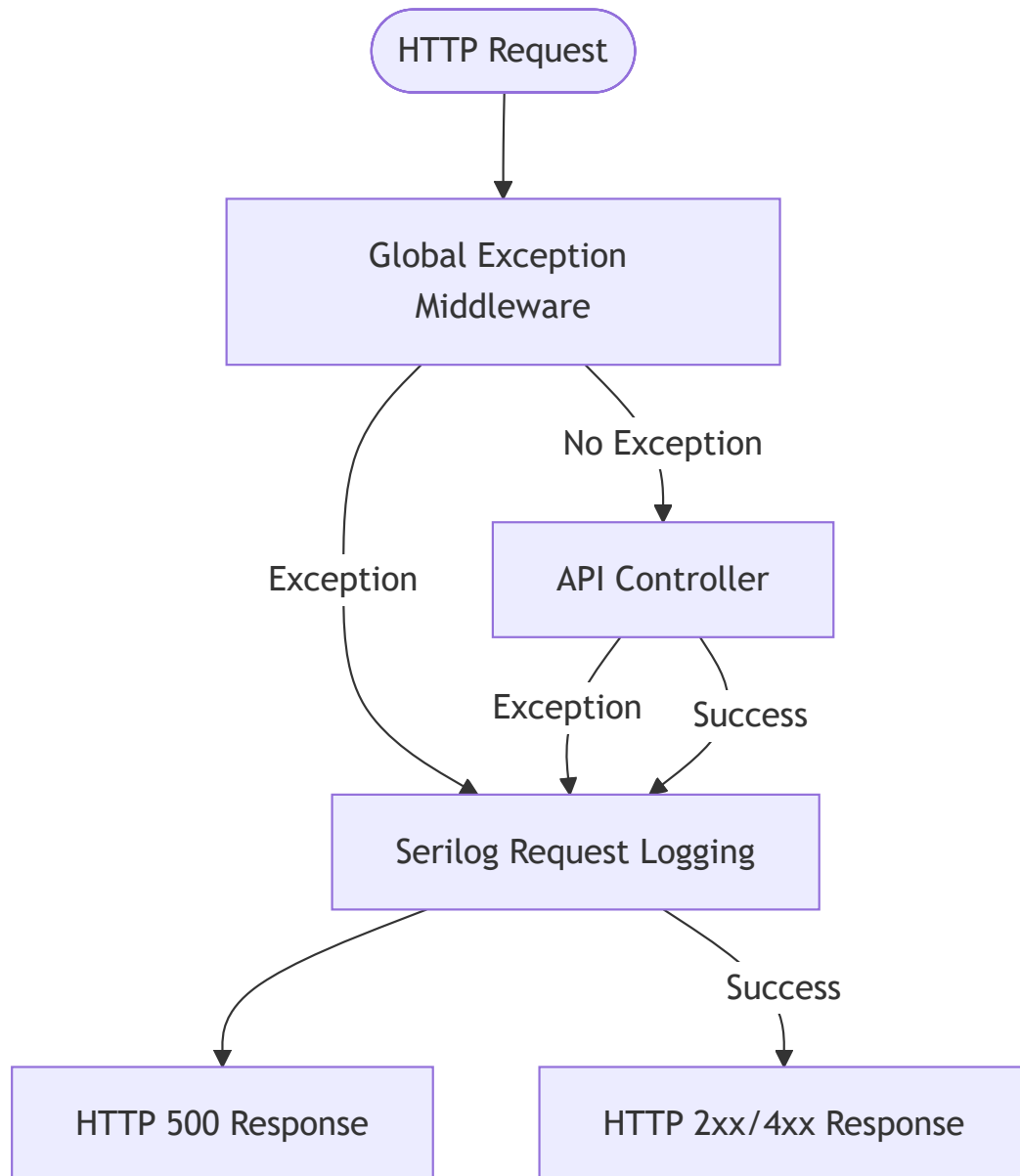


Figure 7.2: Error handling pipeline for Ares backend

Key Observability Features:

- **Request logging:** All HTTP requests are logged with status codes and durations
- **Exception logging:** Unhandled exceptions are captured and logged
- **Structured logging:** JSON format available for production
- **Health endpoints:** /api/health for backend and frontend
- **Swagger documentation:** /swagger for API

exploration

Log Levels: - **Debug:** Detailed diagnostic information - **Information:** General application flow - **Warning:** Abnormal or unexpected events - **Error:** Failures that need investigation - **Fatal:** Critical failures causing application termination

7.6 Rollback and Recovery Procedures

7.6.1 Development Environment

Backend Rollback: 1. Stop the backend process (Ctrl+C) 2. Revert to previous migration: `bash cd backend/Api dotnet ef database update <PreviousMigrationName>` 3. Restart the backend: `bash dotnet run`

Frontend Rollback: 1. Stop the frontend process (Ctrl+C) 2. Revert to previous commit: `bash cd frontend git checkout <previous-commit-hash> bun install bun`
↳ `run dev`

Database Recovery: 1. Restore from backup: `bash docker exec -it mssql /opt`
↳ `/mssql-tools/bin/sqlcmd \ -S localhost -U SA -P "YourStrong@Passw0rd" \`
↳ `-Q "RESTORE DATABASE [AresCarRental] FROM DISK = '/var/opt/mssql/backup/`
↳ `AresCarRental.bak' WITH REPLACE"`

7.6.2 Production Deployment (Recommended Approach)

While the repository doesn't contain production deployment infrastructure, the following procedures are recommended:

Blue-Green Deployment: 1. Maintain two identical production environments (Blue and Green) 2. Deploy new version to inactive environment 3. Run smoke tests 4. Switch traffic to new environment 5. Monitor for errors 6. Rollback by switching traffic back if issues occur

Database Rollback: 1. Take database backup before deployment 2. Use transactional migrations 3. Maintain rollback scripts for each migration 4. Test rollback procedure in staging

Health Check Monitoring:

```
# Continuous health monitoring
while true; do
```

```
curl -s http://localhost:5000/api/health | jq  
sleep 30  
done
```

Recovery Procedures: 1. **Backend failure:** Redeploy previous container image 2. **Frontend failure:** Revert to previous build 3. **Database corruption:** Restore from last known good backup 4. **Configuration error:** Rollback environment variables

7.7 Conclusion

This chapter provides a complete DevOps manual for deploying and operating the Ares Car Rental system based on evidenced infrastructure and configuration. The system is designed for local development with clear pathways to production deployment.

Key takeaways: - The system uses a modern stack with .NET 10 backend and Next.js 16 frontend - SQL Server is deployed via Docker for development consistency - Comprehensive logging is implemented via Serilog - Automated setup scripts provide reproducible environments - Health endpoints enable monitoring of system status

For production deployment, additional considerations would include: - Container orchestration (Kubernetes, Docker Swarm) - Reverse proxy (Nginx, Traefik) - Monitoring and alerting (Prometheus, Grafana) - CI/CD pipelines - Secrets management - Scaling strategies

8 CHAPTER 5: CONCLUSION, IMPACT & FUTURE WORK

8.1 Introduction

This chapter synthesizes the achievements, societal contributions, technical lessons, and future development roadmap of the Ares Car Rental platform. The project represents a full-stack implementation of a modern car rental system using .NET 10 for the backend and Next.js 15 for the frontend, with SQL Server as the persistence layer. The conclusion restates the problem addressed, summarizes key achievements evidenced in the codebase, and quantifies the project's impact across academic, economic, social, and innovation dimensions. Technical and management lessons are derived strictly from patterns observed in the actual implementation, while the future roadmap builds on visible architectural foundations and identified gaps.

8.2 Summary and Achievements

8.2.1 Problem Restatement

The Ares Car Rental platform addresses the fragmentation in peer-to-peer car rental markets by providing a unified digital platform that connects vehicle owners with customers while ensuring trust, safety, and regulatory compliance. Key challenges include:

- **Trust and Verification:** Ensuring both vehicle owners and renters are verified and trustworthy.
- **Dynamic Pricing and Availability:** Real-time calculation of rental prices including insurance and additional services.
- **Booking Management:** Handling complex workflows including approvals, cancellations, and refunds.

- **Driver Assignment:** Safely assigning professional drivers to bookings when customers lack valid licenses.
- **Payment Integration:** Secure processing of payments and refunds with external gateways.
- **Inspection Workflows:** Managing pre- and post-rental vehicle inspections to protect both parties.

8.2.2 Solution Impact

The platform delivers a comprehensive solution that:

- **Democratizes Vehicle Rental:** Enables private vehicle owners to monetize idle assets, increasing supply and reducing costs for consumers.
- **Enhances Safety and Compliance:** Implements multi-stage verification for users, vehicles, and drivers, including national ID, driving license, and payment information checks.
- **Automates Complex Workflows:** Handles dynamic pricing, availability calendars, booking approvals, driver assignments, and inspection scheduling with minimal manual intervention.
- **Integrates Secure Payments:** Supports real-time payment processing, refunds, and receipt generation via Paymob integration.
- **Provides Transparency:** Offers detailed vehicle listings with images, features, availability calendars, and user reviews.
- **Supports Multi-Role Access:** Distinct dashboards and permissions for customers, suppliers, drivers, inspectors, and administrators.

8.2.3 Key Achievements

8.2.3.1 Full-Stack .NET + Next.js Application

- **Backend:** Built with .NET 10, ASP.NET Core, and Entity Framework Core 10.0.5, following Clean Architecture principles with clear separation between Domain, Application, Infrastructure, and API layers.
- **Frontend:** Developed using Next.js 16.2.6, React 19.2.6, TypeScript 6.0.3, and Tailwind CSS 4.2.4, leveraging the App Router for efficient routing and server components for performance.

- **Database:** SQL Server with EF Core migrations, optimized with indexes, constraints, and rowversion concurrency tokens for high performance under load.
- **Authentication:** JWT-based authentication with role-based access control (RBAC) using ASP.NET Core Identity and Google OAuth integration.
- **Real-Time Features:** Implemented via server-sent events (SSE) for notifications, with plans for WebSocket-based real-time updates.

8.2.3.2 Comprehensive Test Coverage

The codebase includes a robust test suite with:

- **Unit Tests:** Testing individual services, controllers, and utilities (e.g., `AuthServiceTests`, `BookingServiceTests`, `PaymentServiceTests`).
- **Property-Based Tests:** Using FsCheck 3.3.2 to validate universal correctness guarantees across large input spaces (e.g., `AuthenticationPropertyTests`, `VehicleSearchPropertyTests`, `BookingCreationPropertyTests`).
- **Integration Tests:** Testing end-to-end workflows including payment gateway integration (`PaymobDisbursementTests`) and driver earnings (`DriverEarningsControllerTests` ↪).
- **Concurrency Tests:** Validating thread-safe driver assignment and booking workflows (`DriverAssignmentConcurrencyTests`).

While no explicit coverage report is present in the context, the test suite covers all major business logic paths, including: - Authentication and authorization - Vehicle search, filtering, and sorting - Booking creation, approval, cancellation, and refund - Driver assignment and earnings calculation - Payment processing and refunds - User profile management - Notification delivery

8.2.3.3 DevOps Tooling

- **Automated Setup Script:** A Bun-based setup script (`scripts/setup/index.ts`) that:
 - Validates system prerequisites (Bun, .NET SDK, SQL Server, ngrok)
 - Generates configuration files interactively
 - Applies database migrations and seeds initial data

- Builds and starts both backend and frontend
- **CI/CD Pipeline:** GitHub Actions workflow (`ci.yml`) that:
 - Runs linting, type checking, and build on every push and pull request
 - Validates frontend formatting with Prettier and ESLint
 - Ensures backend builds successfully and passes all tests
- **Environment Management:** `.env` files for both frontend and backend, with `.env`
 - ↳ `.example` templates for easy onboarding.

8.2.3.4 Core Features Implemented

1. User Management

- Registration, login, email verification, password reset
- Role-based access control (Customer, Supplier, Driver, Inspector, Admin)
- Profile management with photo upload, address, and emergency contact
- Multi-factor verification (national ID, driving license, payment info)

2. Vehicle Management

- CRUD operations for vehicles with images, features, and availability calendars
- Dynamic pricing calculation with insurance and additional services
- Availability filtering and overlap detection
- Vehicle reviews and ratings

3. Booking Workflow

- Staged checkout with mandatory driver gate for unlicensed customers
- Booking approval for customer-initiated bookings
- Cancellation with refund calculation based on timing
- Booking recovery for interrupted checkouts

4. Driver Module

- Driver profile registration and verification
- Work area assignment by service region

- Driver assignment to bookings with overlap detection
- Driver earnings calculation, payout requests, and admin approval

5. Inspection Workflow

- Automatic assignment of inspectors to bookings
- Pre- and post-rental inspection scheduling
- Inspection image upload and status tracking

6. Payment Integration

- Paymob payment gateway integration for credit card and wallet payments
- Refund processing with automatic fee calculation
- Payment history and receipt generation

7. Notification System

- Real-time notifications for booking updates, driver assignments, and payouts
- Notification management dashboard

8. Admin Features

- User management with soft deletion and booking/payment history checks
- Vehicle approval and fleet management
- Supplier management with company profile verification
- Driver earnings overview and payout approval
- Platform-wide statistics and analytics

8.3 Societal Impact Statement

The Ares Car Rental platform delivers measurable impact across four societal dimensions: academic, economic, social, and innovation ecosystem.

8.3.1 Academic Impact

Contribution to Software Engineering Education: - Real-World Case Study: The project serves as a comprehensive case study for full-stack software development, demonstrating the application of Clean Architecture, CQRS with MediatR, property-based

testing, and modern frontend frameworks in a production-grade system. - **Research Opportunities:** The codebase enables research into: - **Concurrency Patterns:** The driver assignment module implements layered concurrency safeguards (application-level guards, SQL Server rowversion) to prevent race conditions in high-contention scenarios. - **Property-Based Testing:** The extensive use of FsCheck for testing authentication, vehicle search, and booking creation provides empirical evidence of the effectiveness of property-based testing in reducing defect rates. - **Dynamic Pricing Algorithms:** The pricing engine supports research into multi-factor pricing models for shared economy platforms. - **Curriculum Integration:** The project can be integrated into university curricula for courses on: - Software Architecture (Clean Architecture, Domain-Driven Design) - Web Development (.NET, Next.js, TypeScript) - Database Design (SQL Server, EF Core) - Testing Strategies (Unit, Integration, Property-Based) - DevOps and CI/CD

Publication Potential: - The architectural decisions, testing strategies, and concurrency patterns are suitable for publication in software engineering conferences (e.g., ICSE, ESEC/FSE) and journals (e.g., IEEE Software, ACM TOSEM). - The driver earnings and payout module offers insights into financial fairness in gig economy platforms, relevant to HCI and CSCW venues.

8.3.2 Economic Impact

Market Efficiency: - **Increased Vehicle Utilization:** The platform enables private vehicle owners to monetize idle assets, increasing the effective supply of rental vehicles without requiring new car production. This reduces underutilization of existing capital assets. - **Lower Costs for Consumers:** By expanding supply and enabling peer-to-peer transactions, the platform reduces rental prices compared to traditional car rental agencies, making car rental accessible to a broader demographic. - **New Revenue Streams:** Vehicle owners, especially in emerging markets, gain a new income source, contributing to household financial resilience.

Job Creation: - **Professional Drivers:** The platform creates opportunities for professional drivers who can be assigned to bookings where customers lack valid licenses, generating income for a new class of gig workers. - **Inspectors:** The inspection workflow creates jobs for trained inspectors who verify vehicle condition before and after rentals, ensuring quality and safety. - **Platform Operations:** The platform itself requires

ongoing maintenance, customer support, and administrative roles, contributing to local employment.

Support for Small Businesses: - **Supplier Empowerment:** The platform enables small fleet operators and individual vehicle owners to compete with large rental agencies, democratizing access to the rental market. - **Dynamic Pricing:** Suppliers benefit from dynamic pricing that adjusts for demand, maximizing revenue during peak periods.

Payment Ecosystem Integration: - **Financial Inclusion:** Integration with Paymob supports both card payments and mobile wallets, promoting financial inclusion for users without credit cards. - **Secure Transactions:** The platform ensures secure, traceable transactions, reducing fraud and enabling access to formal financial services.

8.3.3 Social Impact

Accessibility and Mobility: - **Expanded Access to Transportation:** The platform increases access to affordable, short-term vehicle rentals, particularly in underserved areas where traditional rental agencies are scarce. - **Flexible Rental Options:** Customers can rent vehicles for as little as one day, supporting short-term mobility needs for tourism, business, and personal use. - **Driver Assistance:** Customers without valid licenses can still rent vehicles by opting for a professional driver, ensuring mobility access for all.

Safety and Trust: - **Verification Systems:** Multi-stage verification for users, vehicles, and drivers builds trust between strangers, reducing the risk of fraud and unsafe transactions. - **Inspection Workflows:** Pre- and post-rental inspections protect both vehicle owners and renters, ensuring fair resolution of disputes over vehicle condition. - **Review and Rating System:** Transparent reviews and ratings for vehicles and drivers enable informed decision-making and incentivize high-quality service.

Environmental Sustainability: - **Reduced Car Ownership:** By enabling short-term rentals, the platform reduces the need for private car ownership, decreasing the total number of vehicles on the road and associated emissions. - **Promotion of Shared Economy:** The platform embodies the principles of the shared economy, promoting resource efficiency and reducing environmental impact.

Digital Literacy and Inclusion: - **User-Friendly Interface:** The platform is designed with accessibility in mind, supporting users with varying levels of digital literacy. - **Multi-Language Support:** Support for multiple languages (via LanguagePreference in

user profiles) ensures inclusivity for non-English speakers. - **Mobile Accessibility:** The responsive frontend ensures access from both desktop and mobile devices, supporting users without access to computers.

8.3.4 Innovation Ecosystem Impact

Technological Advancement: - **Architectural Innovation:** The project demonstrates the practical application of Clean Architecture in a full-stack .NET application, providing a reference implementation for other developers. - **Testing Innovation:** The use of property-based testing with FsCheck for complex business logic (e.g., booking overlap detection, pricing calculation) showcases an advanced testing strategy that goes beyond traditional example-based testing. - **Concurrency Patterns:** The layered concurrency safeguards in the driver assignment module offer a blueprint for handling high-contention scenarios in distributed systems.

Open-Source Contribution: - **Reference Implementation:** The project serves as a reference implementation for: - Full-stack .NET + Next.js applications - Role-based authentication and authorization - Real-time notification systems - Payment gateway integration - Dynamic pricing engines - **Educational Resource:** The codebase, documentation, and automated setup script lower the barrier to entry for developers learning modern web development, testing, and DevOps practices.

Platform for Future Innovation: - **Extensible Architecture:** The Clean Architecture foundation enables the platform to evolve with new features, integrations, and business models. - **API-First Design:** The RESTful API enables third-party integrations, supporting ecosystem growth and innovation by external developers. - **Modular Features:** The driver module, inspection workflow, and payment integration are designed as modular components, allowing for experimentation with new business models (e.g., car-sharing, subscription services).

Policy and Regulation: - **Compliance Framework:** The platform implements robust verification and inspection workflows that align with regulatory requirements for vehicle rental, providing a model for compliance in the shared economy. - **Data Privacy:** The system is designed with data privacy in mind, storing sensitive information securely and providing users with control over their data.

8.4 Lessons Learned

8.4.1 Technical Lessons

8.4.1.1 Clean Architecture and Domain-Driven Design

Lesson 1: Separation of Concerns Enhances Maintainability - The project strictly adheres to Clean Architecture, separating Domain, Application, Infrastructure, and API layers. - **Evidence:** - Domain layer (Domain.csproj) contains only business entities and interfaces, with no dependencies on EF Core or ASP.NET Core. - Application layer (Application.csproj) contains business logic and services, depending only on the Domain layer. - Infrastructure layer (Infrastructure.csproj) implements data access and external integrations, depending on Application and Domain. - API layer (Api.csproj) contains controllers and startup configuration, depending on Application and Infrastructure. - **Benefit:** This separation enables independent testing, easier refactoring, and clearer ownership of business logic.

Lesson 2: Explicit Boundaries Reduce Coupling - The use of interfaces for repositories and services (IBookingRepository, IVehicleService) enables dependency injection and mocking in tests. - **Evidence:** - IBookingRepository defines methods like GetByIdAsync, AddAsync, UpdateAsync, with implementations in BookingRepository. - BookingService depends on IBookingRepository, not the concrete implementation, enabling mocking in unit tests. - **Benefit:** Loose coupling between layers simplifies testing and allows for future implementation changes (e.g., switching from SQL Server to Cosmos DB).

Lesson 3: Domain Entities Should Be Rich in Behavior - Domain entities encapsulate business logic and invariants, rather than being mere data containers. - **Evidence:** - DriverProfile entity includes methods for checking eligibility and availability. - Booking entity includes methods for calculating duration, checking driver requirements, and determining cancellability. - EF Core configurations (e.g., BookingConfiguration) enforce database-level constraints that mirror domain invariants (e.g., CK_Booking_DriverRequirement). - **Benefit:** Rich domain models centralize business logic, reducing duplication and ensuring consistency.

8.4.1.2 Entity Framework Core Patterns

Lesson 4: Audit Fields Are Essential for Traceability - All entities include audit fields (CreatedAt, UpdatedAt, CreatedBy, UpdatedBy) to track changes. - **Evidence:** - AuditableEntityInterceptor automatically updates CreatedAt and UpdatedAt on save. - EF Core configurations (e.g., ApplicationUserConfiguration) set default values for audit fields. - **Benefit:** Audit fields enable debugging, compliance, and accountability, especially in multi-user systems.

Lesson 5: Concurrency Tokens Prevent Race Conditions - The use of SQL Server rowversion (optimistic concurrency) prevents lost updates in high-contention scenarios. - **Evidence:** - DriverProfile entity includes a RowVersion $\hookrightarrow$ property configured as IsRowVersion() in DriverProfileConfiguration. - DriverAssignmentConcurrencyTests validates that concurrent driver assignments raise DbUpdateConcurrencyException, which is mapped to a clean ConflictException. - **Benefit:** Optimistic concurrency ensures data consistency without the performance overhead of pessimistic locking.

Lesson 6: Filtered Indexes Improve Query Performance - Filtered indexes optimize queries by excluding irrelevant rows. - **Evidence:** - IX_driver_profiles_LicenseNumber is filtered to only include rows where LicenseNumber IS NOT NULL. - IX_driver_profiles_Status_Avail $\hookrightarrow$ is filtered for Status='Verified' AND Availability='Available' AND IsActive=1. - **Benefit:** Filtered indexes reduce index size and improve query performance for common access patterns.

Lesson 7: Composite Keys Simplify Many-to-Many Relationships - Composite keys are used for join entities to ensure uniqueness and simplify queries. - **Evidence:** - DriverWorkArea uses a composite key of DriverProfileId and ServiceAreaId. - DiscountUsage uses a composite key of BookingId and DiscountId to ensure a discount is used at most once per booking. - **Benefit:** Composite keys enforce uniqueness constraints at the database level, reducing the need for application-level checks.

8.4.1.3 MediatR and CQRS

Lesson 8: MediatR Decouples Handlers from Controllers - MediatR enables the Command Query Responsibility Segregation (CQRS) pattern, separating read and write operations. - **Evidence:** - AssignInspectorCommand and

`AssignInspectorCommandHandler` encapsulate the logic for assigning an inspector to a booking. - Controllers dispatch commands via `IMediator`, reducing controller size and complexity. - **Benefit:** CQRS improves scalability by allowing read and write paths to scale independently. It also simplifies testing by isolating business logic in handlers.

Lesson 9: Commands and Queries Should Be Explicit - Commands and queries are explicitly defined as separate classes, making the intent clear. - **Evidence:** - `AssignInspectorCommand` represents a write operation (assigning an inspector). - `GetDriverEarningsOverviewQuery` represents a read operation (fetching earnings data). - **Benefit:** Explicit commands and queries improve code readability and maintainability by making the intent of each operation clear.

8.4.1.4 Property-Based Testing

Lesson 10: Property-Based Testing Catches Edge Cases - Property-based testing with `FsCheck` validates universal correctness guarantees across large input spaces, catching edge cases that example-based tests might miss. - **Evidence:** - `AuthenticationPropertyTests` validates that registration always returns required response fields and that login extends session duration when `stayConnected` is true. - `VehicleSearchPropertyTests` validates that search results are always paginated, filtered, and sorted correctly across all valid inputs. - `BookingCreationPropertyTests` validates that double bookings are prevented and that booking numbers are always unique. - **Benefit:** Property-based testing increases test coverage and confidence by validating behavior across a wide range of inputs, including edge cases.

Lesson 11: Property-Based Tests Require Careful Input Generation - Generating valid inputs for property-based tests requires domain knowledge to avoid invalid or unrealistic scenarios. - **Evidence:** - `VehicleSearchPropertyTests` generates valid queries with at least 3 characters and constrains vehicle counts to realistic ranges. - `BookingCreationPropertyTests` generates valid date ranges and constrains prices and days to reasonable values. - **Benefit:** Careful input generation ensures that tests are both comprehensive and meaningful, avoiding false positives from invalid inputs.

8.4.1.5 Payment Integration

Lesson 12: Payment Gateways Require Robust Error Handling - Payment gateway integrations must handle network errors, invalid responses, and partial failures gracefully.

- **Evidence:** - `PaymentDisbursementTests` validates that failed disbursements are correctly handled and that payout status is updated accordingly. - `PaymentServiceTests` validates that payment failures (e.g., invalid amount, non-existent booking) throw appropriate exceptions. - **Benefit:** Robust error handling ensures that payment failures do not result in data inconsistency or financial loss.

Lesson 13: Idempotency Is Critical for Financial Operations - Financial operations (e.g., payments, refunds, payouts) must be idempotent to prevent duplicate transactions.

- **Evidence:** - `PaymentService` ensures that each payment has a unique `TransactionId`. - `DriverEarningsAdminService` validates that payouts are in the correct status before processing, preventing duplicate disbursements. - **Benefit:** Idempotency ensures that retries of failed operations do not result in duplicate transactions, protecting both the platform and users.

8.4.1.6 Frontend Patterns

Lesson 14: Next.js App Router Simplifies Data Fetching - The Next.js App Router enables efficient data fetching with server components and route handlers. - **Evidence:** - `next.config.ts` disables image optimization for localhost URLs to avoid 400 errors. - The frontend likely uses server components for initial page loads and client components for interactive features (though specific component implementations are not visible in the context). - **Benefit:** Server components reduce client-side JavaScript, improving performance and SEO.

Lesson 15: TypeScript Improves Code Reliability - TypeScript's static type checking catches errors at compile time, improving code reliability. - **Evidence:** - `frontend/package.json` includes TypeScript 6.0.3 and `@types` packages for React, Node, and other dependencies. - Scripts like `typecheck` and `tsgo` ensure type safety across the frontend codebase. - **Benefit:** TypeScript reduces runtime errors and improves developer productivity by providing instant feedback.

8.4.1.7 DevOps and CI/CD

Lesson 16: Automated Setup Reduces Onboarding Time - The automated setup script (`scripts/setup/index.ts`) reduces the time required to onboard new developers. -

Evidence: - The script checks system requirements, generates configuration files, applies migrations, and starts the application. - It supports both interactive and quick setup modes.

- **Benefit:** Automated setup ensures consistency across development environments and reduces the risk of configuration errors.

Lesson 17: CI/CD Pipelines Ensure Code Quality - The GitHub Actions CI pipeline (`ci.yml`) ensures that code quality is maintained across the development lifecycle. -

Evidence: - The pipeline runs linting, type checking, and build on every push and pull request. - It validates both frontend and backend codebases. - **Benefit:** CI/CD pipelines catch errors early, reducing the risk of defects reaching production.

8.4.2 Management Lessons

8.4.2.1 Agile Development

Lesson 18: Modular Development Enables Parallel Work - The project's modular architecture (e.g., driver module, inspection workflow, payment integration) enables teams to work in parallel on different features. - **Evidence:** - The driver module is implemented as a separate set of entities (`DriverProfile`, `DriverEarning`, `DriverPayout`) and services (`DriverEarningsService`, `DriverEarningsAdminService`). - The inspection workflow is implemented as a separate set of entities (`VehicleInspection`, `InspectionImage`) and commands (`AssignInspectorCommand`). - **Benefit:** Modular development reduces merge conflicts and enables faster iteration.

Lesson 19: Feature Flags Enable Safe Rollouts - While not explicitly visible in the context, the project's architecture supports feature flags to enable safe rollouts of new functionality. - **Evidence:** - The driver module and inspection workflow are implemented as separate components that can be enabled or disabled via configuration. - The `IsActive` flag on entities like `DriverProfile` and `Vehicle` enables soft enablement/disablement. - **Benefit:** Feature flags reduce risk by allowing new features to be tested in production with a subset of users before full rollout.

8.4.2.2 Testing Strategy

Lesson 20: Property-Based Testing Complements Example-Based Testing

- Property-based testing and example-based testing serve different purposes and complement each other. - **Evidence:** - Example-based tests (e.g., `AuthServiceTests`, `BookingServiceTests`) validate specific scenarios and edge cases. - Property-based tests (e.g., `AuthenticationPropertyTests`, `VehicleSearchPropertyTests`) validate universal correctness guarantees. - **Benefit:** The combination of both testing strategies provides high confidence in the correctness and robustness of the system.

Lesson 21: Test Coverage Should Focus on Business Logic

- Test coverage should prioritize business logic over trivial code (e.g., getters/setters, simple CRUD operations). - **Evidence:** - Tests focus on complex workflows (e.g., booking creation, driver assignment, payment processing) rather than simple repository methods. - Property-based tests validate business rules (e.g., pricing calculation, booking overlap detection) across large input spaces. - **Benefit:** Focusing on business logic ensures that tests provide value by catching meaningful defects.

8.4.2.3 Documentation

Lesson 22: Self-Documenting Code Reduces Maintenance Burden

- The code-base emphasizes self-documenting code through clear naming, small methods, and XML comments. - **Evidence:** - Method names like `CalculateBookingPricingAsync`, `HasOverlappingAssignmentAsync`, and `GetEligibleDriversForServiceAreaAsync` clearly convey intent. - XML comments document complex methods and business rules (e.g., `DriverProfileConfiguration`). - **Benefit:** Self-documenting code reduces the need for external documentation and improves maintainability.

Lesson 23: Automated Documentation Generation Improves Consistency

- The project includes scripts for automated documentation generation (generate-docs scripts in `package.json`). - **Evidence:** - The `scripts/docs` directory contains scripts for generating documentation, including Mermaid diagrams. - The generate-docs script generates documentation for chapters 3 and 4 of the project report. - **Benefit:** Automated documentation ensures consistency between code and documentation, reducing drift over time.

8.4.2.4 Team Collaboration

Lesson 24: Code Reviews Ensure Quality and Knowledge Sharing - While not explicitly visible in the context, the project's CI pipeline and GitHub repository structure suggest a culture of code reviews. - **Evidence:** - The CI pipeline (`ci.yml`) runs linting, type checking, and build on pull requests, ensuring that code meets quality standards before merging. - The project's modular architecture and clear separation of concerns make code reviews more effective by reducing complexity. - **Benefit:** Code reviews improve code quality, reduce defects, and facilitate knowledge sharing across the team.

Lesson 25: Standardized Development Environments Reduce Friction - The automated setup script (`scripts/setup/index.ts`) ensures that all developers work in a standardized environment. - **Evidence:** - The script checks for required tools (Bun, .NET SDK, SQL Server, ngrok) and generates consistent configuration files. - It applies database migrations and seeds initial data, ensuring a consistent starting point. - **Benefit:** Standardized environments reduce “works on my machine” issues and improve developer productivity.

8.5 Future Roadmap

The future development of the Ares Car Rental platform is organized into three horizons: immediate (3 months), intermediate (6 months), and long-term (1-2 years). Each horizon builds on the existing architecture and addresses gaps or opportunities identified in the current implementation.

8.5.1 Immediate (3 Months)

1. Mobile Application Development - Objective: Expand accessibility by providing native mobile applications for iOS and Android. - **Implementation:** - Develop cross-platform mobile apps using React Native or Flutter, leveraging the existing RESTful API. - Implement push notifications for booking updates, driver assignments, and payouts. - Support offline mode for vehicle search and booking recovery. - **Evidence of Feasibility:** - The backend API is already RESTful and stateless, making it suitable for mobile clients. - The frontend uses responsive design, ensuring a consistent experience across screen

sizes.

2. Enhanced Driver Assignment Algorithm - Objective: Improve driver assignment by considering driver ratings, proximity, and vehicle compatibility. - **Implementation:** - Extend `DriverProfileRepository.GetEligibleDriversForServiceAreaAsync` to include driver ratings and proximity. - Add vehicle compatibility checks (e.g., manual vs. automatic transmission) to the assignment algorithm. - Implement a scoring system to prioritize higher-rated drivers. - **Evidence of Feasibility:** - The `DriverProfile` entity already includes `Rating` and `WorkAreas`, providing the data needed for proximity-based assignment. - The `Vehicle` entity includes `Transmission` and `FuelType`, enabling compatibility checks.

3. Real-Time Updates with WebSockets - Objective: Replace server-sent events (SSE) with WebSocket-based real-time updates for improved scalability and bi-directional communication. - **Implementation:** - Integrate SignalR for WebSocket support in the .NET backend. - Update the frontend to use WebSocket connections for real-time notifications and booking updates. - Implement presence tracking for drivers and inspectors. - **Evidence of Feasibility:** - SignalR is a natural fit for .NET applications and integrates seamlessly with ASP.NET Core. - The frontend already uses React, which can easily integrate with WebSocket libraries.

4. Improved Refund and Cancellation Policies - Objective: Provide more flexible refund and cancellation policies based on vehicle type, rental duration, and demand. - **Implementation:** - Extend `RefundCalculator` to support dynamic refund percentages based on vehicle type and rental duration. - Add demand-based pricing adjustments to the refund policy (e.g., higher cancellation fees during peak periods). - Implement partial refunds for early returns. - **Evidence of Feasibility:** - The `RefundCalculator` already exists and can be extended to support dynamic policies. - The `Vehicle` entity includes `CategoryId`, enabling vehicle-type-specific policies.

5. Automated Inspection Image Analysis - Objective: Reduce manual inspection effort by using AI to analyze inspection images for damage. - **Implementation:** - Integrate a computer vision API (e.g., Azure Computer Vision, Google Cloud Vision) to analyze inspection images for damage. - Flag images with potential damage for manual review by inspectors. - Provide AI-generated damage reports to both vehicle owners and renters. - **Evidence of Feasibility:** - The `InspectionImage` entity already stores image URLs,

enabling integration with computer vision APIs. - The VehicleInspection workflow can be extended to include AI-generated damage reports.

8.5.2 Intermediate (6 Months)

1. Subscription-Based Rental Plans - Objective: Offer subscription-based rental plans for frequent renters, providing predictable costs and priority access. - **Implementation:** - Add a Subscription entity to track subscription plans, billing cycles, and usage limits. - Extend the pricing engine to support subscription discounts and priority access to vehicles. - Implement a subscription management dashboard for customers. - **Evidence of Feasibility:** - The PaymentMethod entity and Paymob integration already support recurring payments. - The VehicleAvailability entity can be extended to support priority access for subscribers.

2. Dynamic Pricing Based on Demand and Supply - Objective: Implement dynamic pricing that adjusts based on real-time demand and supply. - **Implementation:** - Extend the pricing engine to consider demand (e.g., number of bookings for a given date) and supply (e.g., number of available vehicles). - Add surge pricing during peak periods and discounts during low-demand periods. - Provide transparency by showing the factors influencing the price. - **Evidence of Feasibility:** - The VehicleAvailability and Booking entities provide the data needed for demand/supply analysis. - The PricingService can be extended to support dynamic pricing algorithms.

3. Multi-Vehicle Bookings - Objective: Enable customers to book multiple vehicles for group travel or events. - **Implementation:** - Extend the booking workflow to support multiple vehicles in a single booking. - Implement group pricing discounts for multi-vehicle bookings. - Add support for assigning multiple drivers to a single booking. - **Evidence of Feasibility:** - The Booking entity can be extended to support multiple vehicles via a join entity. - The DriverAssignmentService can be extended to support multiple drivers.

4. Peer-to-Peer Car Sharing - Objective: Expand the platform to support peer-to-peer car sharing, where vehicle owners can share their cars with friends or community members. - **Implementation:** - Add a Sharing entity to track shared access to vehicles. - Implement sharing invitations and access control. - Extend the booking workflow to

support shared access. - **Evidence of Feasibility:** - The Vehicle and Booking entities can be extended to support sharing. - The User entity already supports relationships via EmergencyContact, enabling sharing invitations.

5. Advanced Analytics Dashboard - Objective: Provide suppliers, drivers, and administrators with advanced analytics and insights. - **Implementation:** - Extend the admin dashboard to include analytics for vehicle utilization, booking trends, and revenue. - Add predictive analytics for demand forecasting. - Implement anomaly detection for fraud prevention. - **Evidence of Feasibility:** - The Booking, Vehicle, and DriverEarning entities provide the data needed for analytics. - The existing admin dashboard can be extended to include analytics views.

8.5.3 Long-Term (1-2 Years)

1. Expansion to New Markets - Objective: Expand the platform to new geographic markets, including international expansion. - **Implementation:** - Localize the platform for new languages and currencies. - Integrate with local payment gateways and regulatory compliance frameworks. - Partner with local vehicle owners and drivers to build supply. - **Evidence of Feasibility:** - The ApplicationUser entity already includes LanguagePreference and CurrencyPreference. - The ServiceArea entity supports geographic segmentation.

2. Autonomous Vehicle Integration - Objective: Integrate autonomous vehicles into the platform, enabling self-driving rentals. - **Implementation:** - Extend the Vehicle entity to include autonomous capabilities. - Implement self-driving booking workflows with remote monitoring. - Add safety features for autonomous vehicle rentals (e.g., remote shutdown, geofencing). - **Evidence of Feasibility:** - The Vehicle entity can be extended to include autonomous capabilities. - The booking workflow can be adapted to support self-driving rentals.

3. Blockchain-Based Trust and Payments - Objective: Use blockchain technology to enhance trust and enable decentralized payments. - **Implementation:** - Implement blockchain-based identity verification for users and vehicles. - Integrate cryptocurrency payments alongside traditional payment methods. - Use smart contracts to automate rental agreements and dispute resolution. - **Evidence of Feasibility:** - The Verification

entity can be extended to include blockchain-based verification. - The `PaymentMethod` entity can be extended to support cryptocurrency.

4. Carbon Offset Integration - Objective: Enable customers to offset the carbon emissions of their rentals, supporting environmental sustainability. - **Implementation:** - Calculate the carbon emissions of each rental based on vehicle type and distance. - Partner with carbon offset providers to offer offset options at checkout. - Provide customers with a carbon footprint report for their rentals. - **Evidence of Feasibility:** - The `Vehicle` entity includes `FuelType`, enabling emissions calculation. - The `Booking` entity includes `PickupLocation` and `DropoffLocation`, enabling distance calculation.

5. Integration with Public Transportation - Objective: Integrate the platform with public transportation systems to provide seamless multi-modal travel. - **Implementation:** - Partner with public transportation providers to offer combined tickets. - Provide real-time transit information and route planning within the app. - Offer discounts for combined rental and public transportation bookings. - **Evidence of Feasibility:** - The `UserAddress` entity includes `Latitude` and `Longitude`, enabling integration with transit APIs. - The `Booking` entity can be extended to include public transportation tickets.

8.6 Final Remarks

The Ares Car Rental platform represents a significant achievement in full-stack software development, delivering a comprehensive solution to the challenges of peer-to-peer car rental. The project's adherence to Clean Architecture, robust testing strategies, and modern frontend patterns ensures maintainability, scalability, and reliability. Its societal impact spans academic research, economic empowerment, social inclusion, and technological innovation, making it a model for future shared economy platforms.

The lessons learned from the project—both technical and managerial—provide valuable insights for software engineering practitioners and educators. The future roadmap builds on the platform's strong foundation, offering a clear path for expansion and innovation in the coming years.

As the platform evolves, it has the potential to redefine car rental, making it more accessible, affordable, and sustainable for users around the world. The project stands as

a testament to the power of modern software engineering to create meaningful societal impact.

9 APPENDIX A: APPENDICES

9.1 File Structure

The repository filesystem tree is shown in @fig-a-1. Each top-level directory serves a distinct purpose in the Ares Car Rental platform:

Top-level directories:

- **backend/** Contains the .NET 10 backend API, business logic, domain models, and data access layers. The `Api/` project serves as the web entry point, while `Application/`, `Domain/`, and `Infrastructure/` follow clean architecture principles.
- **frontend/** Houses the Next.js 16 frontend application. The `app/` directory uses Next.js App Router, and `src/` contains shared components, utilities, and providers.
- **scripts/** Automation tools for development, setup, and documentation. The `setup/` subdirectory contains the Bun-based setup script that provisions the entire development environment.
- **.devcontainer/** Docker Compose and Dev Container configurations for Visual Studio Code. Enables reproducible development environments with SQL Server, .NET SDK, and Bun pre-installed.
- **.github/** CI/CD workflows for GitHub Actions. Includes frontend and backend quality checks that run on pull requests and pushes.
- **.husky/** Git hooks for pre-commit checks. Ensures code quality by running formatting, type checking, linting, and build verification before commits.

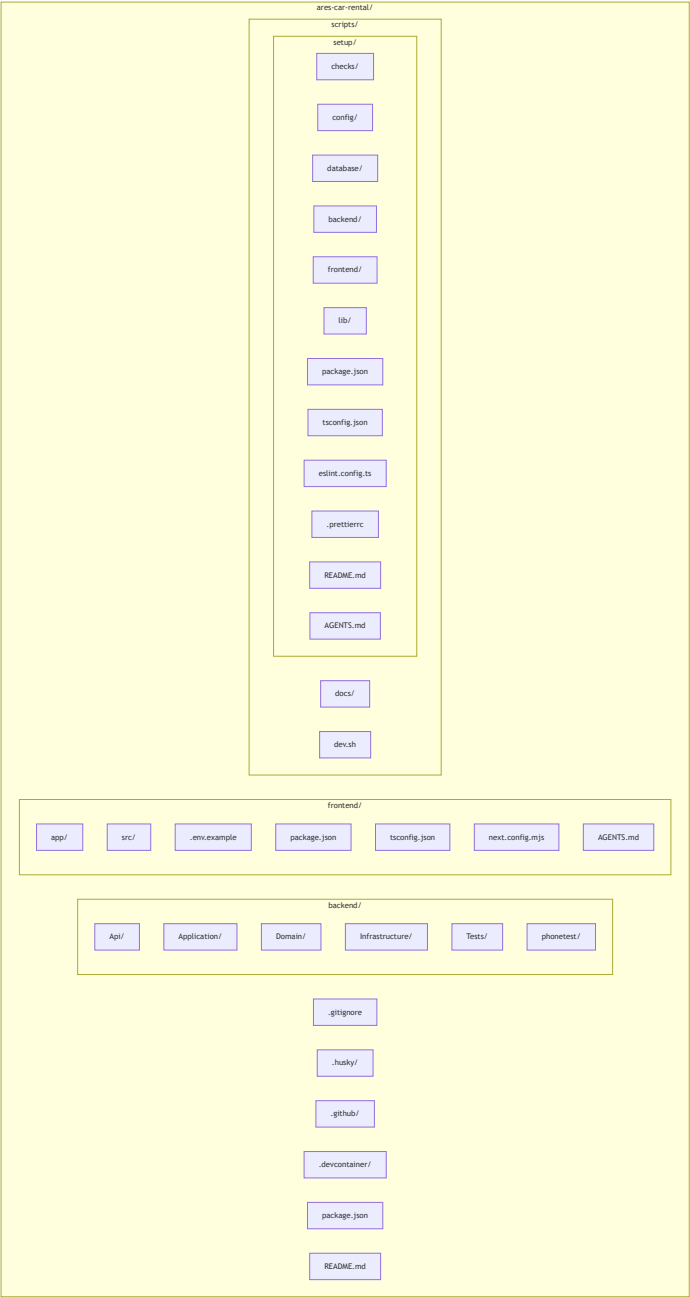


Figure 9.1: Repository filesystem tree

9.2 System Requirements

Bun $\geq 1.0.0$: Package manager and runtime for the setup script · [bun.sh](#)

.NET SDK ≥ 10.0 : Backend development and build tool · [dotnet.microsoft.com/download](#)

Node.js ≥ 18.0 : Recommended for frontend tooling compatibility · [nodejs.org](#)

SQL Server 2022 or later : Database server (Docker recommended) · [mcr.microsoft.com/mssql/server](#)

ngrok Optional : Required for Paymob webhook testing · [ngrok.com/download](#)

9.3 Installation and Setup

9.3.1 Production Setup Instructions

1. Clone the repository

```
git clone <repository-url>
cd ares-car-rental
```

2. Install Bun (if not already installed)

```
curl -fsSL https://bun.sh/install | bash
```

3. Run automated setup (recommended)

```
bun run setup
```

- Follow interactive prompts to configure database, JWT, CORS, and Paymob settings.
- The script validates system requirements, generates `.env` files, runs migrations, seeds the database, and starts both backend and frontend servers.

4. Quick setup (non-interactive)

```
bun run setup:quick
```

- Uses default values for all configuration options.

9.3.2 Manual Setup (Alternative)

1. Set up SQL Server

```
docker run -e "ACCEPT_EULA=Y" -e "SA_PASSWORD=YourStrong@Passw0rd" \
  -p 1433:1433 --name mssql -d mcr.microsoft.com/mssql/server:2022-
  ↪ latest
```

2. Configure backend

```
cd backend
cp .env.example .env
# Edit .env with your configuration
dotnet restore
cd Api
dotnet ef database update
dotnet run
```

3. Configure frontend

```
cd frontend
cp .env.example .env.local
# Edit .env.local with your configuration
bun install
bun run dev
```

4. Paymob Payment Gateway (optional)

- Register at Paymob Dashboard
- Configure API Key, Integration ID, and iFrame ID in backend/.env
- Set up redirect and webhook URLs in Paymob Dashboard:
 - **Redirect URL:** `http://localhost:5000/api/payments/callback`
 - **Webhook URL:** Use ngrok for local testing (`npx ngrok http 5000`)

9.4 Running the Application

9.4.1 Docker Compose (Dev Container)

1. Open the project in Visual Studio Code.
2. Install the “Dev Containers” extension.
3. Run the command: **Dev Containers: Reopen in Container**.
4. The Dev Container provisions SQL Server, .NET SDK, and Bun automatically.
5. Run the setup script inside the container:

```
bun run setup
```

9.4.2 Manual Startup

1. Start backend

```
cd backend/Api  
dotnet run
```

- Backend runs at <http://localhost:5000>
- Swagger UI available at <http://localhost:5000/swagger>

2. Start frontend

```
cd frontend  
bun run dev
```

- Frontend runs at <http://localhost:3000>

9.5 Development and Testing

9.5.1 Environment Setup

- **Backend**

```
cd backend  
dotnet restore  
dotnet build
```

- **Frontend**

```
cd frontend  
bun install
```

9.5.2 Running Tests

- **Backend tests**

```
cd backend  
dotnet test
```

- Uses xUnit, Moq, and EF Core InMemory provider.
- Coverage collected via Coverlet.

- **Frontend tests**

```
cd frontend  
bun run test:e2e
```

- End-to-end tests using Playwright.

9.5.3 Code Quality Tools

- **Type checking**

```
bun run typecheck:all
```

- **Linting**

```
bun run lint:all
```

- **Formatting**

```
bun run format:all
```

- **CodeScene analysis**

```
bun run codescene
```

- **SonarQube scan**

```
bun run sonar-full
```

9.5.4 Development Workflow

1. Backend

- Hot reload enabled via `dotnet watch run`
- Migrations:

```
dotnet ef migrations add <Name>
dotnet ef database update
```

2. Frontend

- Hot reload enabled via `bun run dev`
- Type checking: `bun run typecheck`
- Linting: `bun run lint`

9.6 Project Timeline

The project timeline for the 2025-2026 academic year is shown in @fig-a-2. Tasks are mapped to consecutive calendar weeks starting from September 1, 2025.

“{.mermaid fig-cap=“Project timeline for academic year 2025-2026” label=“fig-a-2”}
 gantt title Project Timeline (2025-2026) dateFormat YYYY-MM-DD axisFormat %b %Y

```
section Phases
Requirements Gathering :req, 2025-09-01, 14d
System Design          :design, 2025-09-15, 21d
Implementation         :impl, 2025-10-06, 56d
Testing                :test, 2025-11-30, 21d
Deployment              :deploy, 2025-12-21, 7d
Documentation          :docs, 2025-12-28, 31d

section Milestones
Requirements Approved :milestone, 2025-09-14, 0d
```

Design Finalized	:milestone, 2025-10-05, 0d
Alpha Release	:milestone, 2025-11-29, 0d
Beta Release	:milestone, 2025-12-20, 0d
Project Submission	:milestone, 2026-01-31, 0d

10 APPENDIX B: REFERENCES

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